Lecture Notes on Automatic Control

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Chapter 1

Algebra Review

1.1 Why Matrix Mathematics is Essential for Systems and Control

Modern dynamical systems and control problems are most naturally and powerfully expressed using vectors and matrices. The state of a system, whether a UAV, robotic manipulator, power network, or mechanical structure, is represented as a state vector collecting positions, velocities, orientations, and other relevant variables. The coupled differential (or difference) equations governing the system dynamics are compactly written in matrix form, revealing the structure of interactions among states, inputs, and disturbances.

Coordinate transformations between reference frames (e.g., inertial, body-fixed, and wind frames in UAVs) are expressed using rotation matrices, while physical properties such as mass and inertia are encoded in structured matrices (e.g., inertia and mass matrices). Linearization of nonlinear dynamics about an operating point leads to state-space models whose system matrices directly determine stability, transient response, and sensitivity to disturbances.

Core analysis and design tools in systems and control are inherently matrix-based. Stability and modal behavior are characterized by the eigenvalues and eigenvectors of the system matrices, which reveal oscillatory modes, damping ratios, and unstable dynamics. Controllability and observability are assessed by rank conditions on matrix pairs, which determine whether a system can be influenced and estimated from available inputs and measurements. Modern controllers and estimators, including state feedback, observers, LQR, Kalman filters, and model predictive control, are formulated as matrix equations and optimization problems.

Consequently, matrix mathematics is not merely a convenient notation but the fundamental language of systems and control. A strong foundation in linear algebra enables engineers to model complex multi-variable systems, reason about stability and performance, implement algorithms efficiently in software, and design controllers and estimators with mathematical rigor and confidence.

1.2 Notation

The set of natural numbers is denoted by $\mathbb{N}$. The set of whole numbers is denoted by $\mathbb{W}$. The set of integers is denoted by $\mathbb{Z}$. The set of real numbers is denoted by $\mathbb{R}$. The set of complex numbers is denoted by $\mathbb{C}$. The equation $a \triangleq b$ implies that a is defined as the object b.

Fact 1.2.1. An element of $\mathbb{N}, \mathbb{W}, \mathbb{Z}, \mathbb{R}$, or $\mathbb{C}$ is a **scalar**.

Remark 1.2.1. Scalars satisfy the usual addition and multiplication operations.

Definition 1.2.1. The **absolute value** of $\alpha \in \mathbb{R}$, denoted by $|\alpha| \geq 0$, is

$$|\alpha| \triangleq \begin{cases} -\alpha, & \alpha < 0, \\ \alpha, & \alpha \geq 0. \end{cases} \tag{1.1}$$

1.3 Complex Numbers

Definition 1.3.1. A **complex number** z is $z = \alpha + \jmath\beta$ where $\alpha, \beta \in \mathbb{R}$ and $\jmath \triangleq \sqrt{-1}$. The set of all complex numbers is denoted by $\mathbb{C}$.

Definition 1.3.2. The **complex conjugate** of z, denoted by $\overline{z}$, is $\overline{z} = \alpha - \jmath\beta$.

Definition 1.3.3. The **absolute value, modulus, or magnitude** of z, denoted by $|z| \geq 0$, is

$$|z| = \sqrt{\alpha^2 + \beta^2}. \tag{1.2}$$

Fact 1.3.1. Let $z_1, z_2 \in \mathbb{C}$. Then,

$$z_1 + z_2 \triangleq \alpha_1 + \alpha_2 + \jmath(\beta_1 + \beta_2), \tag{1.3}$$

$$z_1 z_2 \triangleq \alpha_1\alpha_2 - \beta_1\beta_2 + \jmath(\alpha_1\beta_2 + \alpha_2\beta_1), \tag{1.4}$$

$$\frac{z_1}{z_2} \triangleq \frac{z_1\overline{z}_2}{z_2\overline{z}_2} = \frac{\alpha_1\alpha_2 + \beta_1\beta_2 + \jmath(\alpha_1\beta_1 - \alpha_2\beta_2)}{\alpha_2^2 + \beta_2^2}. \tag{1.5}$$

1.3.1 Problems

Problem 1.3.1. Let $z_1 = 1 + 2\jmath$ and $z_2 = -2 + 1\jmath$. Compute $z_1 + z_2$, $z_1 z_2$, and z_1/z_2.

1.4 Vectors

Definition 1.4.1. An n-dimensional **vector** x is a column of n scalars $x_1, x_2, \ldots, x_n$, that is

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}. \tag{1.6}$$

Remark 1.4.1. For all $i \in \{1, 2, \ldots, n\}$, if $x_i \in \mathbb{R}$, then x is an n-dimensional real vector. For any $i \in \{1, 2, \ldots, n\}$, if $x_i \in \mathbb{C}$, then x is an n-dimensional complex vector.

Remark 1.4.2. A vector is denoted by lowercase letters.

Remark 1.4.3. The ith element of x is denoted by x_i.

Definition 1.4.2. The **dimension** of x is the number of scalar elements in the vector.

Remark 1.4.4. The set of n-dimensional real vectors is denoted by $\mathbb{R}^n$. The set of n-dimensional complex vectors is denoted by $\mathbb{C}^n$.

Fact 1.4.1. Two vectors can be added if and only if they have the same dimension. Let $x, y \in \mathbb{R}^n$. Then,

$$z \triangleq x + y = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}. \tag{1.7}$$

Fact 1.4.2. A vector can be multiplied by a scalar, which results in another vector. Let $\alpha \in \mathbb{R}$ and $x \in \mathbb{R}^n$.

Then,

$$\alpha x \triangleq \begin{bmatrix} \alpha x_1 \\ \alpha x_2 \\ \vdots \\ \alpha x_n \end{bmatrix}. \tag{1.8}$$

Fact 1.4.3. The dot product of two vectors $x, y \in \mathbb{R}^n$, denoted by $x \bullet y \in \mathbb{R}$, is

$$x \bullet y \triangleq \sum_{i=1}^{n} x_i y_i = x_1 y_1 + x_2 y_2 + \ldots + x_n y_n. \tag{1.9}$$

Note that $x \bullet y = y \bullet x$.

Definition 1.4.3. The **length, magnitude, or norm** of the vector x, denoted by $\|x\| \in [0, \infty]$, is

$$\|x\| \triangleq \sqrt{x_1^2 + x_2^2 + \ldots + x_n^2}. \tag{1.10}$$

Definition 1.4.4. Let $x \in \mathbb{R}^n$. If $\|x\| = 1$, then x is called a **unit vector**.

Definition 1.4.5. The **angle** between two vectors $x, y \in \mathbb{R}^n$, denoted by $\theta_{xy} \in \mathbb{R}$, is

$$\theta_{xy} \triangleq \cos^{-1}\left(\frac{x \bullet y}{\|x\|\|y\|}\right). \tag{1.11}$$

Note that $\theta_{xy} = \theta_{yx}$.

Fact 1.4.4. Let $x \in \mathbb{R}^n$. Then, $\theta_{xx} = 0$.

Definition 1.4.6. Two vectors $x, y \in \mathbb{R}^n$ are **orthogonal** if $\theta_{xy} = \pi/2$.

Example 1.4.1. Let $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $y = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. Then, $\theta_{xy} = \pi/2$.

Remark 1.4.5. Let $x, y \in \mathbb{R}^n$ be orthogonal. Then, x is orthogonal to y. Similarly, y is orthogonal to x.

Fact 1.4.5. Let $x, y \in \mathbb{R}^n$ be orthogonal. Then, $x \bullet y = 0$.

Definition 1.4.7. The **cross-product** of two vectors $x, y \in \mathbb{R}^3$, denoted by $x \times y \in \mathbb{R}^3$, is

$$x \times y \triangleq \begin{bmatrix} x_2 y_3 - x_3 y_2 \\ x_3 y_1 - x_1 y_3 \\ x_1 y_2 - x_2 y_1 \end{bmatrix}. \tag{1.12}$$

Fact 1.4.6. For all $x, y \in \mathbb{R}^3$, $(x \times y) \bullet x = (x \times y) \bullet y = 0$.

Fact 1.4.7. For all $x, y \in \mathbb{R}^3$, $\|x \times y\| = \|x\|\|y\||\sin\theta_{xy}|$.

1.4.1 Useful Matlab Functions

- `abs`: Absolute value of real or complex number.
- `length`: Can be used to calculate the dimension of a vector.
- `dot`: Dot product of two vectors.
- `norm`: Norm of a vector.
- `acos`: Can be used to calculate $\cos^{-1}$.
- `cross`: Cross product of 3-dimensional vectors.

1.4.2 Problems

Problem 1.4.1. Let $x = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $y = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$.

1. What is the dimension of x and y ?
2. Compute $x + y$ and $x - y$ and draw them on a 2D plane.
3. Compute $x \bullet y$ and $y \bullet x$.
4. Compute the length of x and y.
5. Compute the angle θ_{xy}.
6. Generate a vector u orthogonal to x. Compute θ_{ux}.

Problem 1.4.2. Consider a unit cube with one corner at the point $(0, 0, 0)$, and the three edges along the $x-$, $y-$, and $z-$axis. Describe all corners of the cube using 3-dimensional vectors.

1.5 Matrices

Definition 1.5.1. A $n \times m$ matrix is an array of n rows and m columns of scalars, that is,

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1m} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nm} \end{bmatrix}. \tag{1.13}$$

The matrix A above has n rows and m columns.

Definition 1.5.2. The dimension of a matrix is the number of rows times the number of columns. The dimension of A above is $n \times m$. If $n > m$, then A is a tall matrix. If $n = m$, then A is a square matrix. If $n < m$, then A is a wide matrix.

Example 1.5.1. Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}. \tag{1.14}$$

Then, A is a **wide** matrix, B is a **square** matrix, and C is a **tall** matrix.

Remark 1.5.1. The set of $n \times m$ real matrices is denoted by $\mathbb{R}^{n \times m}$. The set of $n \times m$ complex matrices is denoted by $\mathbb{C}^{n \times m}$.

Remark 1.5.2. A matrix is denoted by uppercase letters.

Remark 1.5.3. The element in the ith row and the $j$the column of A is denoted by a_{ij}. The element a_{ij} is also denoted by the (i, j)th element of A and $A(i, j)$.

Fact 1.5.1. Two matrices can be added if and only if they have the same dimension. Let $A, B \in \mathbb{R}^{n \times m}$. Then,

$$C \triangleq A + B = \begin{bmatrix} a_{11} + b_{11} & \cdots & a_{1m} + b_{1m} \\ \vdots & \ddots & \vdots \\ a_{n1} + b_{n1} & \cdots & a_{nm} + b_{nm} \end{bmatrix}. \tag{1.15}$$

Fact 1.5.2. A matrix can be multiplied by a scalar. Let $\alpha \in \mathbb{R}$ and $A \in \mathbb{R}^{n\times m}$. Then,

$$\alpha A \triangleq \begin{bmatrix} \alpha a_{11} & \cdots & \alpha a_{1m} \\ \vdots & \ddots & \vdots \\ \alpha a_{n1} & \cdots & \alpha a_{nm} \end{bmatrix}. \tag{1.16}$$

Fact 1.5.3. Two matrices can be multiplied if and only if the number of columns of the first matrix equals the number of rows of the second matrix. Let $A \in \mathbb{R}^{n\times m}$ and $B \in \mathbb{R}^{p\times q}$ be two matrices. Then, the matrix product AB is defined if and only if $m = p$.

Fact 1.5.4. Let $A \in \mathbb{R}^{n\times m}$ and $B \in \mathbb{R}^{m\times q}$. Define $C \triangleq AB \in \mathbb{R}^{n\times q}$. Then,

$$c_{ij} \triangleq \sum_{k=1}^{m} a_{ik} b_{kj}. \tag{1.17}$$

Remark 1.5.4. Let $A, B \in \mathbb{R}^{n\times n}$. In general, $AB \neq BA$.

Example 1.5.2. Let

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 2 \\ -2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}. \tag{1.18}$$

Then,

$$AB = \begin{bmatrix} -2 & -1 & 0 \\ 9 & 12 & 15 \\ 2 & 1 & 0 \end{bmatrix}. \tag{1.19}$$

Definition 1.5.3. The $n \times n$ identity matrix I_n is

$$I_n(i,j) = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases} \tag{1.20}$$

Example 1.5.3. The 2×2 identity matrix I_2 is

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \tag{1.21}$$

Fact 1.5.5. Let $A \in \mathbb{R}^{n\times n}$. The $n \times n$ identity matrix I_n satisfies $AI_n = I_n A = A$.

Fact 1.5.6. A matrix and a vector can be multiplied if and only if the number of columns of the matrix equals the dimension of the vector. Let $A \in \mathbb{R}^{n\times m}$ and $x \in \mathbb{R}^p$. Then, Ax is defined if and only if $m = p$.

Fact 1.5.7. Let $A \in \mathbb{R}^{n\times m}$ and $x \in \mathbb{R}^m$. Define $y \triangleq Ax \in \mathbb{R}^n$. Then,

$$y_i \triangleq \sum_{k=1}^{m} a_{ik} x_k. \tag{1.22}$$

Fact 1.5.8. Let $x \in \mathbb{R}^n$. The $n \times n$ identity matrix I_n satisfies $I_n x = x$.

Definition 1.5.4. Let $A \in \mathbb{R}^{n\times n}$. Then,

$$A^p \triangleq \underbrace{AA\cdots A}_{p \text{ times}}. \tag{1.23}$$

Definition 1.5.5. Let $A \in \mathbb{R}^{n\times m}$. Then, the transpose of A is an $m \times n$ matrix such that the (i,j)th element of the transpose of A is a_{ji}. The transpose of A is denoted by A^{T}. Note that $A^{\mathrm{T}} \in \mathbb{R}^{m\times n}$.

Example 1.5.4. Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}. \tag{1.24}$$

Then,

$$A^{\mathrm{T}} = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}. \tag{1.25}$$

Fact 1.5.9. Let $A \in \mathbb{R}^{n\times m}$ and $B \in \mathbb{R}^{m\times p}$. Then, $(AB)^{\mathrm{T}} = B^{\mathrm{T}}A^{\mathrm{T}} \in \mathbb{R}^{p\times n}$.

Definition 1.5.6. Let $x \in \mathbb{R}^n$. Then, $x^{\mathrm{T}} \in \mathbb{R}^{1\times n}$ is called a **row vector**.

Fact 1.5.10. Let $x, y \in \mathbb{R}^n$. Then,

$$x \bullet y = x^{\mathrm{T}}y. \tag{1.26}$$

Fact 1.5.11. Let $x, y \in \mathbb{R}^n$. Then, $x^{\mathrm{T}}y \in \mathbb{R}$ and $xy^{\mathrm{T}} \in \mathbb{R}^{n\times n}$.

Definition 1.5.7. The **trace** of $A \in \mathbb{R}^{n\times n}$, denoted by $\operatorname{tr} A$, is the sum of its diagonal elements.

$$\operatorname{tr} A \triangleq \sum_{i=1}^{n} a_{ii}. \tag{1.27}$$

Definition 1.5.8. The **determinant** of $A \in \mathbb{R}^{n\times n}$, denoted by $\det(A) \in \mathbb{R}$, is given by the recursive expression

$$\det(A) \triangleq (-1)^{i-1}\sum_{j=1}^{n}(-1)^{j-1}a_{ij}\det(M_{ij}), \tag{1.28}$$

where $M_{ij} \in \mathbb{R}^{(n-1)\times(n-1)}$ is the $(n-1)\times(n-1)$ matrix formed by removing the ith row and jth column of A. For simplicity, we choose $i = 1$.

Fact 1.5.12. Let $A \in \mathbb{R}^{1\times 1}$. The determinant of A is given by $\det(A) = A$.

Fact 1.5.13. Let $A \in \mathbb{R}^{2\times 2}$. The determinant of A is given by

$$\det(A) = a_{11}a_{22} - a_{12}a_{21}. \tag{1.29}$$

Fact 1.5.14. Let $A \in \mathbb{R}^{3\times 3}$. The determinant of A is given by

$$\begin{aligned} \det(A) &= a_{11}\det\left(\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}\right) - a_{12}\det\left(\begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix}\right) + a_{13}\det\left(\begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}\right) \\ &= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}). \end{aligned} \tag{1.30}$$

Definition 1.5.9. Let $A, B \in \mathbb{R}^{n\times n}$ be two square matrices. Then, B is an **inverse** of A if and only if

$$AB = BA = I_n. \tag{1.31}$$

The inverse of A, if it exists, is denoted by A^{-1}.

Remark 1.5.5. All square matrices are not invertible.

Remark 1.5.6. In general, computation of the inverse of a matrix is a difficult process.

Fact 1.5.15. The square matrix A is invertible if and only if $\det(A) \neq 0$.

Definition 1.5.10. Let $A \in \mathbb{R}^{n\times n}$ and suppose that A^p is invertible. Then, $A^{-p} \triangleq (A^p)^{-1}$.

Fact 1.5.16. Let $A \in \mathbb{R}^{2\times 2}$. In the case where $\det(A) = a_{11}a_{22} - a_{12}a_{21} \neq 0$, the inverse of A is given by

$$A^{-1} = \frac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}. \tag{1.32}$$

Fact 1.5.17. Let $x, y \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n\times n}$ such that $y = Ax$. If A is invertible, then $x = A^{-1}y$.

Definition 1.5.11. Let $A \in \mathbb{R}^{n\times n}$. If each column of A is unit-length and orthogonal to every other column of A, then A is an **orthogonal** matrix.

Fact 1.5.18. Let $A \in \mathbb{R}^{n\times n}$ be orthogonal. Then, $A^{-1} = A^{\mathrm{T}}$. Consequently, $AA^{\mathrm{T}} = A^{\mathrm{T}}A = I_n$.

Remark 1.5.7. Orthogonal matrices preserve vector length and represent rigid body rotations.

Example 1.5.5. Let

$$R = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}. \tag{1.33}$$

Then, R is an orthogonal matrix. Note that

$$R^{-1} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = R^{\mathrm{T}}. \tag{1.34}$$

Definition 1.5.12. Let $A \in \mathbb{R}^{n\times n}$. Then, $v \in \mathbb{C}^n$ is an eigenvector of A and $\lambda \in \mathbb{C}$ is an eigenvalue of A if

$$Av = \lambda v. \tag{1.35}$$

Fact 1.5.19. Let λ be an eigenvalue and v be an eigenvector of $A \in \mathbb{R}^{n\times n}$. Then,

$$\det(\lambda I_n - A) = 0. \tag{1.36}$$

1.5.1 Useful Matlab Functions

- `size`: Can be used to calculate the dimension of a matrix.
- `eye`: Creates identity matrix of chosen dimensions.
- `transpose`: Transpose of a matrix.
- `trace`: Trace of a square matrix.
- `det`: Determinant of a square matrix.
- `inv`: Inverse of an invertible matrix.
- `eig`: Obtains eigenvalues and eigenvectors of a matrix.
- `ones`: Creates a matrix whose elements are all 1.
- `zeros`: Creates a matrix whose elements are all 0.

1.5.2 Problems

Problem 1.5.1. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 2 \\ 3 & -4 \end{bmatrix}$.

1. Compute $A + B$ and $A - B$.
2. Compute AB and BA. Are they equal?
3. Compute $A + A^{\mathrm{T}}$ and $B + B^{\mathrm{T}}$.
4. Compute $\det(A)$ and $\det(B)$.
5. Are the matrices A and B orthogonal?
6. Using Matlab, compute A^{-1} and B^{-1}. Verify that $AA^{-1} = BB^{-1} = I_2$.

Chapter 2

Mathematical Representation of Physical Systems

A physical system is a real system, such as a car, a bank, an aircraft, or the climate. A **mathematical system** is a mathematical description of a physical system. The process of describing a physical system using mathematics is called **modeling**, and the resulting mathematical description is called the **system model**.

Example 2.0.1. Consider a mass-damper-spring (MCK) system. Using Newton's Second Law, the system can be described by

$$m\ddot{q} + c\dot{q} + kq = f, \tag{2.1}$$

where q is the displacement, m is the mass, c is the damper coefficient, k is the spring constant, and f is the force applied to the system. The differential equation (2.1) is a model of the physical MCK system.

A physical system and its model do not necessarily behave identically. However, a model can approximate the physical system under appropriately chosen conditions and parameter values. For example, the model (2.1) will behave similarly to an actual MCK system when its parameters match the physical mass, damping, and stiffness. Importantly, a mathematical model can also be explored beyond physically realizable parameter values. For instance, a physical MCK system cannot have negative damping, but the model can be simulated with a negative damping coefficient to study how instability arises. Therefore, models are valuable for parametric studies, design exploration, and simulation of scenarios that may be difficult (or impossible) to reproduce experimentally.

Physical systems can often be described by differential equations. Differential equations can be ordinary (ODE) or partial (PDE). Ordinary differential equations have only one independent variable, whereas partial differential equations have more than one independent variable. For example, (2.1) is an ODE since time is the only independent variable. In contrast, the differential equation

$$\mu\frac{\partial^2 w}{\partial t^2} + \frac{\partial^2}{\partial x^2}EI\frac{\partial^2 q}{\partial x^2} = q(x), \tag{2.2}$$

is a linear PDE since the independent variables are both time and space.

In general, the differential equation derived from physical principles to model a physical system is nonlinear. However, mathematical tools to analyze nonlinear systems are often complex and limited. On the other hand, far simpler tools exist to analyze linear systems. The objective of this chapter is therefore to extract a linear or linearized system from a nonlinear system that can be analyzed and, more importantly, used to design control systems for nonlinear systems.

Locally, the dynamics of a nonlinear system can be approximated by a linear system obtained via first-order Taylor expansion about a trajectory or an equilibrium. This approximation is valid only within a neighborhood where higher-order terms are negligible.

2.1 Nonlinear Differential Equations

A general nth order nonlinear ODE is

$$y^{(n)} = \mathcal{F}(y^{(n-1)}, y^{(n-2)}, \dots, y^{(1)}, y, u^{(m)}, u^{(m-1)}, \dots, u^{(1)}, u) \tag{2.3}$$

where

$$y^{(i)} \triangleq \frac{\mathrm{d}^i y}{\mathrm{d}t^i}. \tag{2.4}$$

Remark 2.1.1. The order of the ODE (2.3) is n.

Remark 2.1.2. The ODE is **linear** if the function $\mathcal{F}$ is linear in each argument, that is,

$$\mathcal{F}(y^{(n-1)}, y^{(n-2)}, \dots, y^{(1)}, y, u^{(m)}, u^{(m-1)}, \dots, u^{(1)}, u) = \sum_{i=1}^{n} \alpha_i y^{(n-i)} + \sum_{j=0}^{m} \beta_j u^{(m-i)}. \tag{2.5}$$

Remark 2.1.3. The ODE is **strictly causal/strictly proper** if $n > m$.

Remark 2.1.4. The ODE is **exactly causal/exactly proper** if $n = m$.

Remark 2.1.5. The ODE is **noncausal/improper** if $n < m$.

Example 2.1.1. Van der Pol system

$$\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} = \mu(1 - y^2)\frac{\mathrm{d}y}{\mathrm{d}t} - y. \tag{2.6}$$

Example 2.1.2. Bellman's equation

$$\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} = k t^a y^b. \tag{2.7}$$

Example 2.1.3. Bernoulli's equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)y^n. \tag{2.8}$$

2.2 State Space Form

A general nonlinear nth-order ODE is often difficult to solve directly. However, first-order ODEs are typically easier to analyze, and many tools in systems and control are naturally expressed in first-order vector form. Using algebraic manipulation, an nth-order ODE can be converted into a set of coupled first-order ODEs. These first-order equations can then be written compactly as a single first-order vector differential equation. This first-order vector differential form is called the **state-space form** of the ODE.

A general nth order ODE in the state-space form is

$$\dot{x} = f(x, u, t), \tag{2.9}$$

$$y = h(x, u, t), \tag{2.10}$$

where $x \in \mathbb{R}^n$, $u \in \mathbb{R}^m$, $y \in \mathbb{R}^p$, $f : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R} \to \mathbb{R}^n$ and $h : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R} \to \mathbb{R}^p$ are vector functions. Note that (2.9) is a **first-order ODE** and (2.10) is an **algebraic equation**. The system of equations (2.9), (2.10) is said to be in **state-space form**, where x is the **state** of the system, u is the **input** to the system, y is the **output** of the system, f is the **dynamics** of the system, and h is the **output** of the system.

Remark 2.2.1. A **state**, at an instant t_0, is a mathematical object that contains all the necessary information about the system so that the output of the system at any time $t \geq t_0$ can be precisely determined if the input $u(t)$ for $[t_0, t]$ is known.

Remark 2.2.2. If the system can be represented as

$$\dot{x} = f(x, u, t) = A(t)x + B(t)u, \tag{2.11}$$
$$y = h(x, u, t) = C(t)x + D(t)u, \tag{2.12}$$

where $A(t) \in \mathbb{R}^{n\times n}$, $B(t) \in \mathbb{R}^{n\times m}$, $C(t) \in \mathbb{R}^{p\times n}$, $D(t) \in \mathbb{R}^{p\times m}$, then the system is **linear**.

Remark 2.2.3. If the functions f and h do not explicitly depend on time t, that is,

$$\dot{x} = f(x, u), \tag{2.13}$$
$$y = h(x, u), \tag{2.14}$$

then the system is **time-invariant**.

Remark 2.2.4. If the system can be represented as

$$\dot{x} = f(x, u, t) = Ax + Bu, \tag{2.15}$$
$$y = h(x, u, t) = Cx + Du, \tag{2.16}$$

where $A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{n\times m}$, $C \in \mathbb{R}^{p\times n}$, $D \in \mathbb{R}^{p\times m}$, then the system is **linear time-invariant (LTI)**.

Example 2.2.1. Consider the Van der Pol system

$$\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} = \mu(1 - y^2)\frac{\mathrm{d}y}{\mathrm{d}t} - y + u. \tag{2.17}$$

Define

$$x_1 \triangleq y, \tag{2.18}$$
$$x_2 \triangleq \dot{y}. \tag{2.19}$$

Then

$$\dot{x}_1 = x_2, \tag{2.20}$$
$$\dot{x}_2 = \mu(1 - x_1^2)x_2 - x_1 + u. \tag{2.21}$$

Define

$$x \triangleq \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}. \tag{2.22}$$

Then the system can be written as

$$\dot{x} = f(x, u), \tag{2.23}$$

where

$$f(x, u) \triangleq \begin{bmatrix} x_2 \\ \mu(1 - x_1^2)x_2 - x_1 + u \end{bmatrix}. \tag{2.24}$$

Thus, $x \in \mathbb{R}^2$ is the state and $f\colon \mathbb{R}^2 \times \mathbb{R} \to \mathbb{R}^2$ is the dynamics.

Example 2.2.2. Consider the nth order linear ODE

$$q^{(n)} + a_{n-1}q^{(n-1)} + \cdots + a_1 q^{(1)} + a_0 q = b_0 v. \tag{2.25}$$

Define

$$x \triangleq \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \qquad x_1 \triangleq q,\ x_2 \triangleq q^{(1)},\ \ldots,\ x_n \triangleq q^{(n-1)}. \tag{2.26}$$

Then

$$\dot{x}_1 = x_2, \tag{2.27}$$
$$\dot{x}_2 = x_3, \tag{2.28}$$
$$\vdots \tag{2.29}$$
$$\dot{x}_n = -a_{n-1}x_n - \cdots - a_1 x_2 - a_0 x_1 + b_0 v, \tag{2.30}$$

and hence

$$\dot{x} = Ax + Bu, \tag{2.31}$$

where

$$A \triangleq \begin{bmatrix} 0 & 1 & 0 & \ldots & 0 \\ 0 & 0 & 1 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & 1 \\ -a_0 & -a_1 & \cdots & a_{n-2} & a_{n-1} \end{bmatrix}, \qquad B \triangleq \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ b_0 \end{bmatrix}. \tag{2.32}$$

Example 2.2.3. Consider a mass-spring-damper system with dynamics

$$m\ddot{q} + c\dot{q} + kq = f, \tag{2.33}$$

where q is the extension and f is the applied force. Define

$$x_1 \triangleq q, \tag{2.34}$$
$$x_2 \triangleq \dot{q}. \tag{2.35}$$

Then

$$\dot{x}_1 = x_2, \tag{2.36}$$
$$\dot{x}_2 = \frac{1}{m}\left(u - kx_1 - cx_2\right). \tag{2.37}$$

Define

$$x \triangleq \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}. \tag{2.38}$$

Then (2.36)-(2.37) can be written as

$$\dot{x} = f(x,u),$$
$$y = g(x,u),$$

where

$$f(x,u) \triangleq \begin{bmatrix} x_2 \\ \frac{1}{m}\left(u - kx_1 - cx_2\right) \end{bmatrix}, \qquad g(x,u) \triangleq x_1. \tag{2.39}$$

Furthermore,

$$f(x,u) = Ax + Bu, \tag{2.40}$$
$$g(x,u) = Cx + Du, \tag{2.41}$$

where

$$A \triangleq \begin{bmatrix} 0 & 1 \\ -\dfrac{k}{m} & -\dfrac{c}{m} \end{bmatrix}, \quad B \triangleq \begin{bmatrix} 0 \\ \dfrac{1}{m} \end{bmatrix}, \quad C \triangleq \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D \triangleq 0, \tag{2.42}$$

which implies that the system is linear.

Example 2.2.4. Consider the nth-order ODE

$$y^{(n)} = \mathcal{F}(y^{(n-1)}, y^{(n-2)}, \ldots, y^{(1)}, y, u). \tag{2.43}$$

Define

$$x_1 \triangleq y, \tag{2.44}$$
$$x_2 \triangleq y^{(1)}, \tag{2.45}$$
$$\vdots \tag{2.46}$$
$$x_n \triangleq y^{(n-1)}. \tag{2.47}$$

Then

$$\dot{x}_1 = x_2, \tag{2.48}$$
$$\dot{x}_2 = x_3, \tag{2.49}$$
$$\vdots \tag{2.50}$$
$$\dot{x}_{n-1} = x_n, \tag{2.51}$$
$$\dot{x}_n = \mathcal{F}(x_n, x_{n-1}, \ldots, x_2, x_1, u). \tag{2.52}$$

Define

$$x \triangleq \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n. \tag{2.53}$$

Then the state-space form is

$$\dot{x} = \begin{bmatrix} x_2 \\ \vdots \\ \mathcal{F}(x_n, x_{n-1}, \ldots, x_2, x_1, u) \end{bmatrix} = f(x,u). \tag{2.54}$$

Equation (2.54) is the state-space form of (2.43), where $f : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^n$.

2.2.1 Matrix Form of Differential Equations

Many coupled mechanical systems can be compactly written in second-order matrix form as

$$\mathcal{M}(\xi)\ddot{\xi} + \mathcal{G}(\xi, \dot{\xi}) + \mathcal{K}(\xi) = \mathcal{F}, \tag{2.55}$$

where $\xi \in \mathbb{R}^n$ is the generalized coordinate vector, $\mathcal{M}(\xi) \in \mathbb{R}^{n \times n}$ is the **mass matrix**, and $\mathcal{G}(\xi, \dot{\xi}), \mathcal{K}(\xi), \mathcal{F} \in \mathbb{R}^n$ are vector-valued functions constructed from the equations of motion. The term $\mathcal{G}(\xi, \dot{\xi})$ collects Coriolis and gyroscopic effects, $\mathcal{K}(\xi)$ represents conservative forces (e.g., gravity and elastic restoring forces), and $\mathcal{F}$ denotes generalized external inputs/forces.

To obtain a first-order state-space representation, define the state

$$\chi \triangleq \begin{bmatrix} \xi \\ \dot{\xi} \end{bmatrix} \in \mathbb{R}^{2n}. \tag{2.56}$$

Then, the dynamics can be written in state-space form as

$$\dot{\chi} = \begin{bmatrix} \dot{\xi} \\ \mathcal{M}(\xi)^{-1}\big(\mathcal{F} - \mathcal{G}(\xi, \dot{\xi}) - \mathcal{K}(\xi)\big) \end{bmatrix}. \tag{2.57}$$

It is often convenient to define $\mathcal{M}, \mathcal{G}, \mathcal{K}$, and $\mathcal{F}$ symbolically in MATLAB. This enables automated generation of the state-space model and its linearization via symbolic Jacobians. In particular, the script `Automatic_Linearization.m` can be used to compute the linearized dynamics, described in the next section, about a chosen equilibrium or trajectory with minimal algebraic effort.

Example 2.2.5. Consider the coupled equations of motion

$$(m + M)\ddot{x} + ml\ddot{\theta}\cos\theta - ml\dot{\theta}^2 \sin\theta = F, \tag{2.58}$$

$$(I + ml^2)\ddot{\theta} + ml\ddot{x}\cos\theta - mgl\sin\theta = 0. \tag{2.59}$$

The objective is to write the coupled differential equation in the state-space form.

Define the generalized coordinates

$$\xi \triangleq \begin{bmatrix} x \\ \theta \end{bmatrix}, \tag{2.60}$$

which implies that

$$\dot{\xi} = \begin{bmatrix} \dot{x} \\ \dot{\theta} \end{bmatrix}, \qquad \ddot{\xi} = \begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix}. \tag{2.61}$$

Collecting the acceleration terms in (2.58)–(2.59) yields the **mass matrix**

$$\mathcal{M}(\xi) = \begin{bmatrix} m + M & ml\cos\theta \\ ml\cos\theta & I + ml^2 \end{bmatrix}. \tag{2.62}$$

Similarly, collecting the velocity-dependent terms yields the **gyroscopic vector**

$$\mathcal{G}(\xi, \dot{\xi}) = \begin{bmatrix} -ml\dot{\theta}^2 \sin\theta \\ 0 \end{bmatrix}, \tag{2.63}$$

collecting the conservative terms (gravity, spring, etc.) yields

$$\mathcal{K}(\xi) = \begin{bmatrix} 0 \\ -mgl\sin\theta \end{bmatrix}, \tag{2.64}$$

and collecting the external inputs yields the **generalized forces vector**

$$\mathcal{F} = \begin{bmatrix} F \\ 0 \end{bmatrix}. \tag{2.65}$$

The equations of motion can then be written compactly as

$$\mathcal{M}(\xi)\ddot{\xi} + \mathcal{G}(\xi, \dot{\xi}) + \mathcal{K}(\xi) = \mathcal{F}. \tag{2.66}$$

Next, define the state

$$\chi \triangleq \begin{bmatrix} \xi \\ \dot{\xi} \end{bmatrix} = \begin{bmatrix} x \\ \theta \\ \dot{x} \\ \dot{\theta} \end{bmatrix}. \tag{2.67}$$

The dynamics then can be written as

$$\dot{\chi} = \begin{bmatrix} \dot{\xi} \\ \ddot{\xi} \end{bmatrix} = \begin{bmatrix} \dot{x} \\ \dot{\theta} \\ \ddot{x} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} \dot{\xi} \\ \mathcal{M}(\xi)^{-1}\big(\mathcal{F} - \mathcal{G}(\xi, \dot{\xi}) - \mathcal{K}(\xi)\big) \end{bmatrix}. \tag{2.68}$$

Listing 2.1: Symbolic definition of M, G, K, and F in MATLAB

```matlab
% Cart pendulum: build M(xi), G(xi,xid), K(xi), and F(u) symbolically
% xi = [x; th], xid = [dx; dth], xidd = [ddx; ddth]

clear; clc; close all
syms x th dx dth F m M l I g

% Generalized coordinates
xi = [x; th];
xid = [dx; dth];

% Mass matrix M(xi)
Mmat = [m+M, m*l*cos(th);
       m*l*cos(th), I + m*l^2];

% Coriolis/gyroscopic vector G(xi,xid)
Gvec = [-m*l*dth^2*sin(th);
        0];

% Conservative forces vector K(xi)
Kvec = [0;
       -m*g*l*sin(th)];

% Generalized input/force vector F
Fvec = [F;
       0];

syms chi dchi
chi = [xi; xid]; % State definition
dchi = [xid; inv(Mmat)*(Fvec - Kvec - Gvec)] % dynamics map
```

2.3 Perturbation Dynamics

Definition 2.3.1. Consider the system (2.13), (2.14). A state $x_\mathrm{e} \in \mathbb{R}^n$ and an input $u_\mathrm{e} \in \mathbb{R}^m$ is an **equilibrium state-input pair** if

$$f(x_\mathrm{e}, u_\mathrm{e}) = 0. \tag{2.69}$$

Definition 2.3.2. Consider the system (2.13), (2.14). A state trajectory $x_\mathrm{s} : [t_0, \infty) \to \mathbb{R}^n$ and an input trajectory $u_\mathrm{s} : [t_0, \infty) \to \mathbb{R}^m$ is a **solution** of (2.13) if, for all $t \geq t_0$,

$$\dot{x}_\mathrm{s}(t) = f(x_\mathrm{s}(t), u_\mathrm{s}(t)). \tag{2.70}$$

In particular, an equilibrium pair $(x_\mathrm{e}, u_\mathrm{e})$ is a solution of (2.13), (2.14).

Let $(x_s(t), u_s(t))$ be a solution of (2.13), (2.14), that is,

$$\dot{x}_s = f(x_s, u_s). \tag{2.71}$$

Define the perturbed trajectories

$$x(t) = x_s(t) + \xi(t), \qquad u(t) = u_s(t) + v(t), \tag{2.72}$$

where $\xi(t)$ and $v(t)$ are the **state and input perturbations**. Substituting into (2.13) yields

$$\dot{x}_s + \dot{\xi} = f(x_s + \xi, u_s + v). \tag{2.73}$$

Using first-order Taylor expansion,

$$f(x_s + \xi, u_s + v) \approx f(x_s, u_s) + \left.\frac{\partial f}{\partial x}\right|_{(x_s, u_s)} \xi + \left.\frac{\partial f}{\partial u}\right|_{(x_s, u_s)} v. \tag{2.74}$$

Subtracting (2.71) from (2.73) gives the **linearized perturbation dynamics**

$$\dot{\xi} = A\xi + Bv, \tag{2.75}$$

where

$$A \triangleq \left.\frac{\partial f}{\partial x}\right|_{(x_s, u_s)}, \qquad B \triangleq \left.\frac{\partial f}{\partial u}\right|_{(x_s, u_s)}. \tag{2.76}$$

Example 2.3.1. Consider Example 2.2.1. The trajectory $x_s(t) = 0$ and $u_s(t) = 0$ is a solution for all $t \geq 0$. The linearized perturbation dynamics is

$$\dot{\xi} = A\xi + Bv, \tag{2.77}$$

where

$$A \triangleq \left.\frac{\partial f}{\partial x}\right|_{(0,0)} = \left.\begin{bmatrix} 0 & 1 \\ \mu x_2(-2x_1) - 1 & \mu(1 - x_1^2) \end{bmatrix}\right|_{(0,0)} = \begin{bmatrix} 0 & 1 \\ -1 & \mu \end{bmatrix}, \tag{2.78}$$

$$B \triangleq \left.\frac{\partial f}{\partial u}\right|_{(0,0)} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \tag{2.79}$$

Example 2.3.2. Consider the ball and beam setup in Figure 2.1. The equations of motion are

$$\ddot{x} + g\sin\theta - x\dot{\theta}^2 = 0, \tag{2.80}$$

$$(mx^2 + J)\ddot{\theta} + 2mx\dot{x}\dot{\theta} + mgx\cos\theta = \tau, \tag{2.81}$$

where m is the mass of the ball, J is the moment of inertia of the beam, x is the distance between the point of rotation and the ball, and θ is the angle between the beam and horizontal.

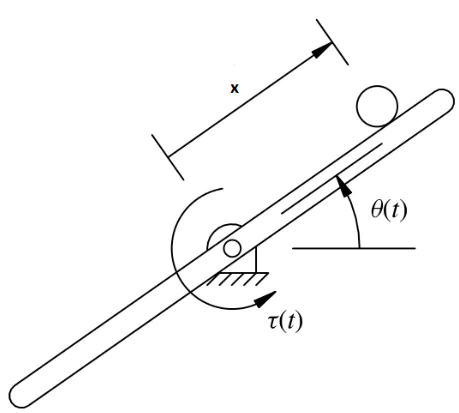

Figure 2.1: Ball and beam setup.

Define $x_1 = x$, $x_2 = \dot{x}$, $x_3 = \theta$, $x_4 = \dot{\theta}$, and $u = \tau$. Then

$$\dot{x}_1 = x_2, \tag{2.82}$$
$$\dot{x}_2 = -g \sin x_3 + x_1 x_4^2, \tag{2.83}$$
$$\dot{x}_3 = x_4, \tag{2.84}$$
$$\dot{x}_4 = \frac{-2mx_1x_2x_4 - mgx_1 \cos x_3 + u}{mx_1^2 + J}. \tag{2.85}$$

For any $x_0 \in \mathbb{R}$, the pair

$$x_{\mathrm{s}}(t) = \begin{bmatrix} x_0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \qquad u_{\mathrm{s}}(t) = mgx_0 \tag{2.86}$$

is a solution. The linearized dynamics is (2.75), where

$$A(t) \triangleq \left.\frac{\partial f}{\partial x}\right|_{x_{\mathrm{s}}(t),u_{\mathrm{s}}(t)} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ x_4^2 & 0 & -g\cos x_3 & 2x_1x_4 \\ 0 & 0 & 0 & 1 \\ M & \dfrac{-2mx_1x_4}{J+mx_1^2} & \dfrac{-mgx_1\sin x_3}{J+mx_1^2} & \dfrac{-2mx_1x_2}{J+mx_1^2} \end{bmatrix}_{x_{\mathrm{s}}(t),u_{\mathrm{s}}(t)} \tag{2.87}$$

$$= \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -g & 0 \\ 0 & 0 & 0 & 1 \\ \dfrac{-mg}{mx_0^2+J} & 0 & 0 & 0 \end{bmatrix}, \tag{2.88}$$

$$B(t) \triangleq \left.\frac{\partial f}{\partial u}\right|_{x_{\mathrm{s}}(t),u_{\mathrm{s}}(t)} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \dfrac{1}{J+mx_1^2} \end{bmatrix}_{x_{\mathrm{s}}(t),u_{\mathrm{s}}(t)} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \dfrac{1}{J+mx_0^2} \end{bmatrix}, \tag{2.89}$$

where M denotes higher-order coupling terms omitted in the linearization.

Alternatively, define

$$\xi \triangleq \begin{bmatrix} x \\ \theta \end{bmatrix}. \tag{2.90}$$

Then (2.80)–(2.81) can be written in the form

$$\mathcal{M}(\xi)\ddot{\xi} + \mathcal{G}(\xi,\dot{\xi}) + \mathcal{K}(\xi) = \mathcal{F}, \tag{2.91}$$

where

$$\mathcal{M}(\xi) \triangleq \begin{bmatrix} 1 & 0 \\ 0 & mx^2 + J \end{bmatrix}, \quad \mathcal{G}(\xi,\dot{\xi}) \triangleq \begin{bmatrix} -x\dot{\theta}^2 \\ 2mx\dot{x}\dot{\theta} \end{bmatrix}, \quad \mathcal{K}(\xi) \triangleq \begin{bmatrix} g\sin\theta \\ mgx\cos\theta \end{bmatrix}, \quad \mathcal{F} \triangleq \begin{bmatrix} 0 \\ \tau \end{bmatrix}. \tag{2.92}$$

Next, define the state

$$\chi \triangleq \begin{bmatrix} \xi \\ \dot{\xi} \end{bmatrix} = \begin{bmatrix} x \\ \theta \\ \dot{x} \\ \dot{\theta} \end{bmatrix}. \tag{2.93}$$

Then the state-space form is

$$\dot{\chi} = \begin{bmatrix} \dot{\xi} \\ \mathcal{M}(\xi)^{-1}\Big(\mathcal{F} - \mathcal{G}(\xi,\dot{\xi}) - \mathcal{K}(\xi)\Big) \end{bmatrix} = \begin{bmatrix} \dot{x} \\ \dot{\theta} \\ -g\sin\theta + x\dot{\theta}^2 \\ \dfrac{\tau - 2mx\dot{x}\dot{\theta} - mgx\cos\theta}{mx^2 + J} \end{bmatrix}. \tag{2.94}$$

For any $x_0 \in \mathbb{R}$, the pair

$$\xi_{\rm e} = \begin{bmatrix} x_0 \\ 0 \end{bmatrix}, \qquad \dot{\xi}_{\rm e} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \tau_{\rm e} = mgx_0 \tag{2.95}$$

satisfies $\mathcal{F} - \mathcal{G}(\xi_{\rm e},\dot{\xi}_{\rm e}) - \mathcal{K}(\xi_{\rm e}) = 0$, and hence defines an equilibrium of (2.94). Let

$$\chi_{\rm e} \triangleq \begin{bmatrix} \xi_{\rm e} \\ \dot{\xi}_{\rm e} \end{bmatrix} = \begin{bmatrix} x_0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \qquad u_{\rm e} \triangleq \tau_{\rm e}. \tag{2.96}$$

Define perturbations $\chi = \chi_{\rm e} + \zeta$ and $u = u_{\rm e} + \nu$. Then the linearized perturbation dynamics is

$$\dot{\zeta} = A\zeta + B\nu, \tag{2.97}$$

where $A = \left.\dfrac{\partial f}{\partial \chi}\right|_{(\chi_{\rm e},u_{\rm e})}$ and $B = \left.\dfrac{\partial f}{\partial u}\right|_{(\chi_{\rm e},u_{\rm e})}$, with $f(\chi,u)$ defined by (2.94). Carrying out the partial derivatives and evaluating at $(\chi_{\rm e},u_{\rm e})$ yields

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -g & 0 & 0 \\ -\dfrac{mg}{mx_0^2 + J} & 0 & 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \dfrac{1}{mx_0^2 + J} \end{bmatrix}. \tag{2.98}$$

Example 2.3.3. Consider the equations of motion of a satellite in a fixed plane.

$$m\ddot{r} - mr\dot{\theta}^2 = u_r - \frac{mk}{r^2}, \tag{2.99}$$

$$mr^2\ddot{\theta} + 2mr\dot{r}\dot{\theta} = ru_\theta, \tag{2.100}$$

where m is the mass of the satellite, θ is the true anomaly, r is the distance of the satellite from the center of the planet, u_r is the radial thrust, u_θ is the tangential thrust, and k is a parameter.

Define $x_1 = r$, $x_2 = \dot{r}$, $x_3 = \theta$, $x_4 = \dot{\theta}$, $u_1 = u_r$, and $u_2 = u_\theta$. Then

$$\dot{x}_1 = x_2, \tag{2.101}$$

$$\dot{x}_2 = x_1x_4^2 - \frac{k}{x_1^2} + \frac{u_1}{m}, \tag{2.102}$$

$$\dot{x}_3 = x_4, \tag{2.103}$$

$$\dot{x}_4 = \frac{u_2}{mx_1} - \frac{2x_2x_4}{x_1}. \tag{2.104}$$

The trajectory

$$x_{\rm s}(t) = \begin{bmatrix} r_0 \\ 0 \\ \omega t \\ \omega \end{bmatrix}, \qquad u_{\rm s} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{2.105}$$

is a solution corresponding to a circular orbit. The linearized dynamics is (2.75), where

$$A(t) \triangleq \left.\frac{\partial f}{\partial x}\right|_{x_{\rm s}(t),u_{\rm s}(t)} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ x_4^2 + \dfrac{2k}{x_1^3} & 0 & 0 & 2x_1x_4 \\ 0 & 0 & 0 & 1 \\ \dfrac{2x_2x_4}{x_1^2} & \dfrac{-2x_4}{x_1} & 0 & \dfrac{-2x_2}{x_1} \end{bmatrix}_{x_{\rm s}(t),u_{\rm s}(t)} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 3\omega^2 & 0 & 0 & 2r_0\omega \\ 0 & 0 & 0 & 1 \\ 0 & \dfrac{-2\omega}{r_0} & 0 & 0 \end{bmatrix}, \tag{2.106}$$

$$B(t) \triangleq \left.\frac{\partial f}{\partial u}\right|_{x_{\rm s}(t),u_{\rm s}(t)} = \begin{bmatrix} 0 & 0 \\ \dfrac{1}{m} & 0 \\ 0 & 0 \\ 0 & \dfrac{1}{mx_1} \end{bmatrix}_{x_{\rm s}(t),u_{\rm s}(t)} = \begin{bmatrix} 0 & 0 \\ \dfrac{1}{m} & 0 \\ 0 & 0 \\ 0 & \dfrac{1}{mr_0} \end{bmatrix}. \tag{2.107}$$

Note that, in a circular orbit, $\omega = \sqrt{\frac{\mu}{r^3}}$, where μ is the standard gravitational parameter.

Alternatively, define

$$\xi \triangleq \begin{bmatrix} r \\ \theta \end{bmatrix}, \qquad u \triangleq \begin{bmatrix} u_r \\ u_\theta \end{bmatrix}. \tag{2.108}$$

Then (2.99)–(2.100) can be written as

$$\mathcal{M}(\xi)\ddot{\xi} + \mathcal{G}(\xi,\dot{\xi}) + \mathcal{K}(\xi) = \mathcal{F}, \tag{2.109}$$

where

$$\mathcal{M}(\xi) \triangleq \begin{bmatrix} m & 0 \\ 0 & mr^2 \end{bmatrix}, \quad \mathcal{G}(\xi,\dot{\xi}) \triangleq \begin{bmatrix} -mr\dot{\theta}^2 \\ 2mr\dot{r}\dot{\theta} \end{bmatrix}, \quad \mathcal{K}(\xi) \triangleq \begin{bmatrix} \dfrac{mk}{r^2} \\ 0 \end{bmatrix}, \quad \mathcal{F} \triangleq \begin{bmatrix} u_r \\ ru_\theta \end{bmatrix}. \tag{2.110}$$

Next, define the state

$$\chi \triangleq \begin{bmatrix} \xi \\ \dot{\xi} \end{bmatrix} = \begin{bmatrix} r \\ \theta \\ \dot{r} \\ \dot{\theta} \end{bmatrix}. \tag{2.111}$$

Then, the state-space form is

$$\dot{\chi} = \begin{bmatrix} \dot{\xi} \\ \mathcal{M}(\xi)^{-1}\Big(\mathcal{F} - \mathcal{G}(\xi,\dot{\xi}) - \mathcal{K}(\xi)\Big) \end{bmatrix} = \begin{bmatrix} \dot{r} \\ \dot{\theta} \\ r\dot{\theta}^2 - \dfrac{k}{r^2} + \dfrac{u_r}{m} \\ \dfrac{u_\theta}{mr} - \dfrac{2\dot{r}\dot{\theta}}{r} \end{bmatrix}. \tag{2.112}$$

Consider the circular-orbit solution

$$\xi_{\rm s}(t) = \begin{bmatrix} r_0 \\ \omega t \end{bmatrix}, \qquad \dot{\xi}_{\rm s}(t) = \begin{bmatrix} 0 \\ \omega \end{bmatrix}, \qquad u_{\rm s}(t) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \tag{2.113}$$

Equivalently,

$$\chi_{\rm s}(t) = \begin{bmatrix} r_0 \\ \omega t \\ 0 \\ \omega \end{bmatrix}, \qquad u_{\rm s}(t) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \tag{2.114}$$

Define perturbations $\chi = \chi_{\rm s} + \zeta$ and $u = u_{\rm s} + \nu$. Then the linearized perturbation dynamics is

$$\dot{\zeta} = A(t)\zeta + B(t)\nu, \tag{2.115}$$

where $A(t) = \left.\dfrac{\partial f}{\partial \chi}\right|_{(\chi_{\rm s}(t), u_{\rm s}(t))}$ and $B(t) = \left.\dfrac{\partial f}{\partial u}\right|_{(\chi_{\rm s}(t), u_{\rm s}(t))}$, with $f(\chi, u)$ defined by (2.112). Evaluating the Jacobians yields

$$A(t) = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \omega^2 + \dfrac{2k}{r_0^3} & 0 & 0 & 2r_0\omega \\ 0 & 0 & \dfrac{-2\omega}{r_0} & 0 \end{bmatrix}, \qquad B(t) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \dfrac{1}{m} & 0 \\ 0 & \dfrac{1}{mr_0} \end{bmatrix}. \tag{2.116}$$

2.4 Linear Systems

A large class of physical and engineered systems can be approximated, either exactly or locally, by **linear dynamical models**. Linear models play a central role in systems and control because they admit powerful analysis and design tools (stability, controllability, observers, LQR, Kalman filtering, frequency response, etc.) that are far less tractable for general nonlinear systems.

A **linear (state-space) dynamical system** is described by

$$\dot{x} = Ax + Bu, \tag{2.117}$$

$$y = Cx + Du, \tag{2.118}$$

where $x \in \mathbb{R}^n$ is the **state**, $u \in \mathbb{R}^m$ is the **input**, and $y \in \mathbb{R}^p$ is the **output**.

Recall that the state x is a compact mathematical representation of the system's internal configuration (e.g., positions, velocities, currents, temperatures), while the input u represents external signals or actuations applied to the system. The output y represents the quantities that are measured or of interest for performance objectives. The matrices $A \in \mathbb{R}^{n\times n}, B \in \mathbb{R}^{n\times m}, C \in \mathbb{R}^{p\times n}$, and $D \in \mathbb{R}^{p\times m}$ encode how the state evolves in time and how the inputs influence both the state and the output. In particular, the matrix A determines the **internal dynamics** of the system, while B captures how control inputs affect the evolution of the state.

Remark 2.4.1. If $u(t) = 0$, the system is called **unforced**. The resulting dynamics

$$\dot{x} = Ax \tag{2.119}$$

describe the **natural behavior** of the system in the absence of any external actuation. The eigenvalues of A determine whether the system's natural response grows, decays, or oscillates.

2.4.1 Solution of Linear State Equations

With the initial condition $x(0) = x_0$, the solution $x(t)$ of (2.117) is given by

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-\tau)}Bu(\tau)\,\mathrm{d}\tau. \tag{2.120}$$

The first term $e^{At}x_0$ is called the **homogeneous solution/natural response/free response/zero-input response**, while the integral term is the **particular solution/ forced response/zero-state response**.

Remark 2.4.2. The forced response term

$$\int_0^t e^{A(t-\tau)}Bu(\tau)\,\mathrm{d}\tau \tag{2.121}$$

has the structure of a **convolution integral**. In particular, the forced response is the convolution of the matrix-valued function $e^{At}B$ and the input signal $u(t)$. See Appendix A.11.

2.4.2 Four Canonical Mechanical Systems

This section introduces four simple second-order mechanical systems. Despite their simplicity, these models capture fundamental behaviors, such as drift, damping, and oscillation, that arise in much more complex systems, including robotic joints, flexible structures, and aerospace vehicles. As such, these four systems serve as **canonical building blocks** for modeling and analysis. For example, multi-degree-of-freedom mechanical systems can often be interpreted as combinations of rigid-body modes and oscillatory modes.

Undamped Rigid Body (URB)

A URB is a point mass m with a force f applied to it. The equation of motion is

$$m\ddot{q} = f. \tag{2.122}$$

Defining the state and input

$$x \triangleq \begin{bmatrix} q \\ \dot{q} \end{bmatrix}, \qquad u \triangleq f, \tag{2.123}$$

the state-space form of URB dynamics is

$$\dot{x} = \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}_{A} x + \underbrace{\begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}}_{B} u. \tag{2.124}$$

Damped Rigid Body (DRB)

A DRB is a point mass m with viscous damping c and an applied force f. The equation of motion is

$$m\ddot{q} + c\dot{q} = f. \tag{2.125}$$

Using the state and input defined by (2.123), the state-space form of DRB dynamics is

$$\dot{x} = \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & -\frac{c}{m} \end{bmatrix}}_{A} x + \underbrace{\begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}}_{B} u. \tag{2.126}$$

Undamped Oscillator (UO)

A UO is a point mass m attached to a linear spring with stiffness k, with an applied force f. The equation of motion is

$$m\ddot{q} + kq = f. \tag{2.127}$$

Using the state and input defined by (2.123), the state-space form of UO dynamics is

$$\dot{x} = \underbrace{\begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & 0 \end{bmatrix}}_{A} x + \underbrace{\begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}}_{B} u. \tag{2.128}$$

Damped Oscillator (DO)

A DO is a point mass m attached to a linear spring with stiffness k and a viscous damper with coefficient c, with an applied force f. The equation of motion is

$$m\ddot{q} + c\dot{q} + kq = f. \tag{2.129}$$

Using the state and input defined by (2.123), the state-space form of DO dynamics is

$$\dot{x} = \underbrace{\begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix}}_{A} x + \underbrace{\begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}}_{B} u. \tag{2.130}$$

Qualitative Position and Velocity Response to a Short External Force

Starting from zero initial conditions, suppose each system is subjected to a short, brief "kick" by a hammer (i.e., a large force applied over a very small time interval and then removed). The resulting qualitative behavior of the position and velocity is as follows.

- **URB:** The short kick sets the mass into motion. After the force is removed, the velocity remains constant and the position grows linearly in time.
- **DRB:** The short kick sets the mass into motion, but damping causes the velocity to decay to zero. The position approaches a constant offset.
- **UO:** The short kick excites oscillatory motion. Both position and velocity oscillate with constant amplitude.
- **DO:** The short kick excites oscillatory motion, but damping causes the oscillations in both position and velocity to decay to zero.

Figure 2.2 shows the qualitative position responses $q(t)$ to a short hammer "kick" starting from zero initial condition.

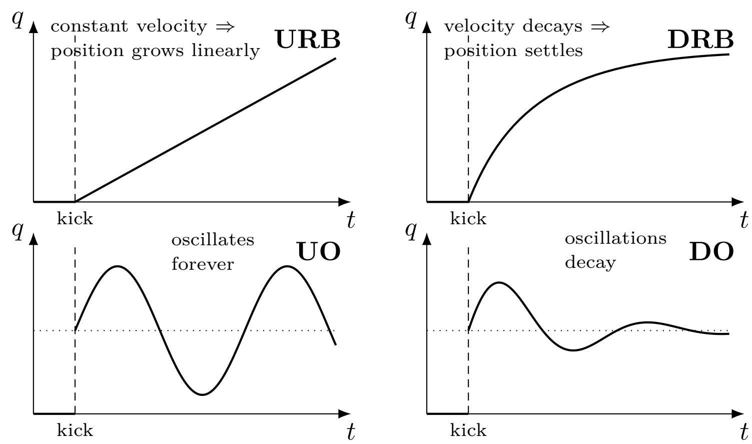

Figure 2.2: Qualitative position responses $q(t)$ to a short "kick" starting from zero initial conditions.

Chapter 3

Signals

3.1 Common Input Signals

This section summarizes several canonical input signals that are frequently used to probe and analyze the behavior of linear dynamical systems. These signals serve as basic building blocks for constructing more general inputs.

3.1.1 Unit Impulse

The unit impulse $\delta(t)$ is defined as

$$\delta(t) \triangleq \begin{cases} 0, & t \neq 0, \\ \infty, & t = 0, \end{cases} \tag{3.1}$$

and satisfies

$$\int_{-\infty}^{\infty} \delta(t)\, dt = 1, \tag{3.2}$$

and

$$\int_{-\infty}^{\infty} g(t)\, \delta(t - t_0)\, dt = g(t_0), \tag{3.3}$$

where $g : \mathbb{R} \to \mathbb{R}$. The property above is called the **sifting property.**

Although $\delta(t)$ is not a function in the classical sense, it is a useful idealization that models an impulsive input applied over an infinitesimal time interval with finite magnitude.

Impulse Response. Consider the linear system

$$\dot{x}(t) = Ax(t) + Bu(t). \tag{3.4}$$

Consider an impulse input, that is, $u(t) = \delta(t)u_0$. The state response is

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-\tau)}B\delta(\tau)u_0\, d\tau = e^{At}x_0 + e^{At}Bu_0. \tag{3.5}$$

Thus, an impulse input produces an instantaneous change in the state that is equivalent to modifying the initial condition from x_0 to $x_0 + Bu_0$.

3.1.2 Step Input

The unit step function is defined as

$$1(t) \triangleq \begin{cases} 0, & t \leq 0, \\ 1, & t > 0. \end{cases} \tag{3.6}$$

The step input represents the sudden application of a constant input at $t = 0$.

Note that the step function is the integral of the impulse function, that is,

$$1(t) = \int_{-\infty}^{t} \delta(\tau)\, d\tau. \tag{3.7}$$

Applying a constant step input $u(t) = 1(t)u_0$ yields

$$x(t) = e^{At}x_0 + \int_{0}^{t} e^{A(t-\tau)} B u_0 \, d\tau. \tag{3.8}$$

Step responses are commonly used to characterize transient behavior, steady-state response, and time constants of linear systems.

3.1.3 Ramp and Quadratic Inputs

The ramp input is defined as the integral of the step, that is,

$$\text{ramp}(t) = \int_{-\infty}^{t} 1(\tau)\, d\tau, \tag{3.9}$$

which grows linearly with time for $t > 0$.

Similarly, the quadratic input is obtained by integrating the ramp, that is,

$$\text{quadratic}(t) = \int_{-\infty}^{t} \text{ramp}(\tau)\, d\tau. \tag{3.10}$$

Ramp and polynomial inputs are useful for studying tracking performance and steady-state error properties of control systems.

3.1.4 Sinusoidal Input

A sinusoidal input is given by

$$u(t) = A \sin(\omega t + \phi), \tag{3.11}$$

where A is the amplitude, ω is the angular frequency, and ϕ is the phase. Sinusoidal inputs are central to frequency-response analysis and provide insight into resonance and steady-state oscillatory behavior.

3.2 Noise

In practice, measured signals are corrupted by noise, most commonly due to sensor imperfections and environmental disturbances. A standard modeling assumption is additive noise:

$$y_{\text{meas}}(t) = y(t) + w(t). \tag{3.12}$$

A commonly used idealized model is *white noise*, which (informally) contains all frequencies with equal power. One heuristic representation is

$$w(t) = \sum_{i=0}^{\infty} a_i \sin(\omega_i t + \phi_i), \tag{3.13}$$

although this representation is not physically realizable.

A stochastic process $w(t)$ is said to be white noise if

$$\mathbb{E}[w(t)w(t+\tau)] = 0, \qquad \tau \neq 0, \tag{3.14}$$

i.e., samples at different times are uncorrelated.

Chapter 4

Laplace Transform

The Laplace transform converts linear differential equations into algebraic equations, enabling efficient analysis and solution of linear dynamical systems.

Definition 4.0.1. Consider a function

$$f\colon [0,\infty) \to \mathbb{R}. \tag{4.1}$$

The **Laplace transform** of the function $f(t)$, denoted by $F(s)$, is the function

$$F(s) = \mathcal{L}[f(t)] \triangleq \int_0^\infty e^{-st} f(t)\mathrm{d}t, \tag{4.2}$$

where $s \in \mathbb{C}$. Note that the Laplace transform of a function is a complex-valued function.

4.1 Laplace Transform of Functions

The Laplace transform of simple functions can be computed explicitly by evaluating (4.2).

Example 4.1.1. Let $f(t) = \delta(t)$. Then, $F(s) = \int_0^\infty e^{-st} f(t)\mathrm{d}t = \int_0^\infty e^{-st}\delta(t)\mathrm{d}t = e^{-s0} = 1$.

Example 4.1.2. Let $f(t) = 1(t)$. Then, $F(s) = \int_0^\infty e^{-st} f(t)\mathrm{d}t = \int_0^\infty e^{-st}1(t)\mathrm{d}t = \frac{1}{-s}e^{-st}\Big|_0^\infty = \frac{1}{s}$.

Example 4.1.3. Let $f(t) = e^{at}$. Then, $F(s) = \int_0^\infty e^{-st} f(t)\mathrm{d}t = \int_0^\infty e^{-st}e^{at}\mathrm{d}t = \frac{1}{-(s-a)}e^{-(s-a)t}\Big|_0^\infty = \frac{1}{s-a}$.

Table 4.1 shows the Laplace transforms of various functions.

Table 4.1: Common Laplace Transform Pairs

$f(t)$	$F(s)$	$f(t)$	$F(s)$
$\delta(t)$	1	$1(t)$	$\frac{1}{s}$
e^{at}	$\frac{1}{s-a}$	$t^n,\ n = 1, 2, 3, \ldots$	$\frac{n!}{s^{n+1}}$
$t^p,\ p > -1$	$\frac{\Gamma(p+1)}{s^{p+1}}$	$\sqrt{t}$	$\frac{\sqrt{\pi}}{2s^{3/2}}$
$t^{n+\frac{1}{2}},\ n = 0, 1, 2, \ldots$	$\frac{1 \cdot 3 \cdot 5 \cdots (2n+1)\sqrt{\pi}}{2^{n+1}s^{n+3/2}}$	$\sin(at)$	$\frac{a}{s^2+a^2}$

Continued on next page

Table 4.1 (continued)

$f(t)$	$F(s)$	$f(t)$	$F(s)$
$\cos(at)$	$\dfrac{s}{s^2+a^2}$	$t\sin(at)$	$\dfrac{2as}{(s^2+a^2)^2}$
$t\cos(at)$	$\dfrac{s^2-a^2}{(s^2+a^2)^2}$	$\sin(at)-at\cos(at)$	$\dfrac{2a^3}{(s^2+a^2)^2}$
$\sin(at)+at\cos(at)$	$\dfrac{2as^2}{(s^2+a^2)^2}$	$\cos(at)-at\sin(at)$	$\dfrac{s(s^2-a^2)}{(s^2+a^2)^2}$
$\cos(at)+at\sin(at)$	$\dfrac{s(s^2+3a^2)}{(s^2+a^2)^2}$	$\sin(at+b)$	$\dfrac{s\sin b+a\cos b}{s^2+a^2}$
$\cos(at+b)$	$\dfrac{s\cos b-a\sin b}{s^2+a^2}$	$\sinh(at)$	$\dfrac{a}{s^2-a^2}$
$\cosh(at)$	$\dfrac{s}{s^2-a^2}$	$e^{at}\sin(bt)$	$\dfrac{b}{(s-a)^2+b^2}$
$e^{at}\cos(bt)$	$\dfrac{s-a}{(s-a)^2+b^2}$	$e^{at}\sinh(bt)$	$\dfrac{b}{(s-a)^2-b^2}$
$e^{at}\cosh(bt)$	$\dfrac{s-a}{(s-a)^2-b^2}$	$t^n e^{at},\ n=0,1,2,\ldots$	$\dfrac{n!}{(s-a)^{n+1}}$

4.2 Laplace Operator Properties

The Laplace transform of many standard functions can be computed directly from the definition. However, as the functional form becomes more complicated, evaluating the integral quickly becomes tedious and error-prone. In such cases, we use the algebraic properties of the Laplace transform to reduce the computation of transforms of complex signals to combinations of simpler, known transforms.

Fact 4.2.1. The Laplace transform is linear, that is, $\mathcal{L}[\alpha_1 f_1(t)+\alpha_2 f_2(t)] = \alpha_1 F_1(s)+\alpha_2 F_2(s)$.

Proof. Note that

$$\begin{aligned}\mathcal{L}[\alpha_1 f_1(t)+\alpha_2 f_2(t)] &= \int_0^\infty \big(\alpha_1 f_1(t)+\alpha_2 f_2(t)\big)e^{-st}\,dt\\ &= \alpha_1\int_0^\infty f_1(t)e^{-st}\,dt+\alpha_2\int_0^\infty f_2(t)e^{-st}\,dt\\ &= \alpha_1 F_1(s)+\alpha_2 F_2(s),\end{aligned}\tag{4.3}$$

which proves the linearity of the Laplace transform. □

Fact 4.2.2. The Laplace transform of the convolution of two functions is the product of the Laplace transforms of both functions, that is,

$$\mathcal{L}\big[(f*g)(t)\big] = F(s)G(s),\tag{4.4}$$

where $F(s)=\mathcal{L}[f(t)]$ and $G(s)=\mathcal{L}[g(t)]$.

Proof. Recall that

$$\mathcal{L}\big[(f*g)(t)\big] = \int_0^\infty \left(\int_0^t f(\tau)\,g(t-\tau)\,d\tau\right)e^{-st}\,dt.$$

Assuming f and g are such that the integrals converge and we can interchange the order of integration, we rewrite the domain $\{(t,\tau): 0\le\tau\le t<\infty\}$ by swapping the order. Thus

$$\mathcal{L}\big[(f*g)(t)\big] = \int_0^\infty\int_\tau^\infty f(\tau)\,g(t-\tau)\,e^{-st}\,dt\,d\tau.$$

For fixed τ, let $\sigma \triangleq t-\tau$. Note that as t goes from τ to ∞, σ goes from 0 to ∞, and thus

$$\begin{aligned}
\mathcal{L}\big[(f*g)(t)\big] &= \int_0^\infty \int_0^\infty f(\tau)\, g(\sigma)\, e^{-s(\sigma+\tau)}\, d\sigma\, d\tau \\
&= \int_0^\infty \int_0^\infty \big(f(\tau)e^{-s\tau}\big)\big(g(\sigma)e^{-s\sigma}\big)\, d\sigma\, d\tau. \\
&= \left(\int_0^\infty f(\tau)e^{-s\tau}\, d\tau\right)\left(\int_0^\infty g(\sigma)e^{-s\sigma}\, d\sigma\right) \\
&= F(s)\, G(s),
\end{aligned}$$

which proves (4.4). □

Thus, convolution in the time domain corresponds to multiplication in the frequency domain, which is a central reason for the utility of Laplace methods in control and systems analysis.

Fact 4.2.3. If f is differentiable, then

$$\mathcal{L}\big[f'(t)\big] = sF(s) - f(0). \tag{4.5}$$

Proof. Note that

$$\mathcal{L}\big[f'(t)\big] = \int_0^\infty f'(t)e^{-st}\, dt = \Big[f(t)e^{-st}\Big]_0^\infty + s\int_0^\infty f(t)e^{-st}\, dt = -f(0) + sF(s), \tag{4.6}$$

which proves (4.5). □

Fact 4.2.4. Let $F(s) = \mathcal{L}[f(t)]$. Then, $\mathcal{L}[e^{at}f(t)] = F(s-a)$ and $\mathcal{L}[f(t-T)] = e^{-sT}F(s)$.

Proof. Note that

$$\mathcal{L}[e^{at}f(t)] = \int_0^\infty e^{-st}e^{at}f(t)\mathrm{d}t = \int_0^\infty e^{-(s-a)t}f(t)\mathrm{d}t = F(s-a), \tag{4.7}$$

and

$$\mathcal{L}[f(t-T)] = \int_0^\infty e^{-st}f(t-T)\mathrm{d}t = \int_{-T}^\infty e^{-s(t+\tau)}f(\tau)\mathrm{d}\tau = e^{-sT}\int_0^\infty e^{-st}f(\tau)\mathrm{d}\tau = e^{-sT}F(s). \tag{4.8}$$

□

Table 4.2 shows several properties of the Laplace transforms.

Table 4.2: Common Properties and Theorems of the Laplace Transform

Item	Theorem	Name
1.	$\mathcal{L}[f(t)] = F(s) = \int_0^\infty f(t)e^{-st}\, dt$	Definition
2.	$\mathcal{L}[k_1f_1(t) + k_2f_2(t)] = k_1F_1(s) + k_2F_2(s)$	Linearity
3.	$\mathcal{L}[e^{-at}f(t)] = F(s+a)$	Frequency shift
4.	$\mathcal{L}[f(t-T)] = e^{-sT}F(s)$	Time shift
5.	$\mathcal{L}[f(at)] = \frac{1}{a}F\left(\frac{s}{a}\right)$	Scaling
6.	$\mathcal{L}[t^nf(t)] = (-1)^nF^{(n)}(s)$	Time multiplication

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Table 4.2 (continued)

Item	Theorem	Name
7.	$\mathcal{L}\left[\dfrac{df}{dt}\right] = sF(s) - f(0)$	Differentiation
8.	$\mathcal{L}\left[\dfrac{d^2 f}{dt^2}\right] = s^2 F(s) - sf(0) - f'(0)$	Differentiation
9.	$\mathcal{L}\left[\dfrac{d^n f}{dt^n}\right] = s^n F(s) - \sum_{k=1}^{n} s^{n-k} f^{(k-1)}(0)$	Differentiation
10.	$\mathcal{L}\left[\int_0^t f(\tau)\, d\tau\right] = \dfrac{F(s)}{s}$	Integration
11.	$f(\infty) = \lim_{s\to 0} sF(s)$	Final value theorem
12.	$f(0^+) = \lim_{s\to\infty} sF(s)$	Initial value theorem

4.3 Laplace Transform Practice Problems

The following examples are designed to build fluency with Laplace transform properties. Each example requires the systematic use of table lookup and one or more transform properties.

Linearity and Frequency Shift

Example 4.3.1. Compute $\mathcal{L}[3e^{-2t} - 4\sin(3t) + 5]$.

Solution 4.3.1. Using linearity,

$$\mathcal{L}[3e^{-2t} - 4\sin(3t) + 5] = 3\mathcal{L}[e^{-2t}] - 4\mathcal{L}[\sin(3t)] + 5\mathcal{L}[1] = \frac{3}{s+2} - \frac{12}{s^2+9} + \frac{5}{s}. \tag{4.9}$$

Time Multiplication Property $\mathcal{L}[tf(t)] = -\dfrac{d}{ds}F(s)$.

Example 4.3.2. Compute $\mathcal{L}[te^{-3t}]$.

Solution 4.3.2. Note that

$$\mathcal{L}[e^{-3t}] = \frac{1}{s+3},$$

and thus

$$\mathcal{L}[te^{-3t}] = -\frac{d}{ds}\left(\frac{1}{s+3}\right) = \frac{1}{(s+3)^2}. \tag{4.10}$$

Example 4.3.3. Compute $\mathcal{L}[t^2 e^{-2t}]$.

Solution 4.3.3. Since

$$\mathcal{L}[e^{-2t}] = \frac{1}{s+2},$$

and

$$\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n}{ds^n} F(s),$$

it follows that

$$\mathcal{L}[t^2 e^{-2t}] = \frac{d^2}{ds^2}\left(\frac{1}{s+2}\right) = \frac{2}{(s+2)^3}. \tag{4.11}$$

Inverse Laplace: Pattern Recognition

Example 4.3.4. Compute $\mathcal{L}^{-1}\left[\dfrac{2s+4}{s^2+4s+8}\right]$.

Solution 4.3.4. First, note that

$$s^2+4s+8=(s+2)^2+4.$$

and

$$2s+4=2(s+2).$$

Thus,

$$\mathcal{L}^{-1}\left[\frac{2s+4}{s^2+4s+8}\right]=\mathcal{L}^{-1}\left[\frac{2(s+2)}{(s+2)^2+4}\right]=2e^{-2t}\cos(2t). \tag{4.12}$$

Repeated Poles

Example 4.3.5. Compute $\mathcal{L}^{-1}\left[\dfrac{1}{(s+2)^3}\right]$.

Solution 4.3.5. Using

$$\mathcal{L}[t^n e^{at}]=\frac{n!}{(s-a)^{n+1}},$$

and setting $n=2$ and $a=-2$, yields

$$\mathcal{L}^{-1}\left[\frac{1}{(s+2)^3}\right]=\frac{t^2}{2}e^{-2t}. \tag{4.13}$$

Time Multiplication Property

Example 4.3.6. Compute $\mathcal{L}[t\sin(2t)]$.

Solution 4.3.6. Note that

$$\mathcal{L}[\sin(2t)]=\frac{2}{s^2+4}.$$

and thus, using the time multiplication property,

$$\mathcal{L}[t\sin(2t)]=-\frac{d}{ds}\left(\frac{2}{s^2+4}\right)=\frac{4s}{(s^2+4)^2}. \tag{4.14}$$

4.4 Guidelines for Using the Laplace Transform

The Laplace transform is a powerful analytical tool for linear systems. However, it is important to understand how it is *actually used* in practice.

4.4.1 The Definition Is Rarely Used Directly

The definition

$$\mathcal{L}[f(t)]=\int_0^\infty f(t)e^{-st}\,dt$$

is fundamental from a theoretical standpoint. However, in engineering practice, we **almost never compute Laplace transforms by evaluating this integral directly**. Instead, transforms are computed by

- Recognizing known transform pairs from a table,
- Using algebraic properties (linearity, shifting, scaling, differentiation),
- Reducing complex expressions to combinations of simpler, known forms.

4.4.2 Laplace Transform as Pattern Recognition

The Laplace transform should be viewed as a structured pattern-matching procedure. For example:

$$\frac{s-a}{(s-a)^2+b^2} = \mathcal{L}[e^{at}\cos(bt)], \quad \frac{b}{(s-a)^2+b^2} = \mathcal{L}[e^{at}\sin(bt)].$$

The goal is not to compute integrals, but to manipulate expressions into recognizable forms.

4.4.3 Use Properties Systematically

When computing transforms

1. Break the signal into simpler pieces using linearity.
2. Apply frequency or time shifts where appropriate.
3. Use differentiation in s when factors of t appear.
4. Reduce the result to standard table forms.

When computing inverse transforms

1. Factor denominators whenever possible.
2. Complete the square for quadratic terms.
3. Rewrite numerators to match sine/cosine patterns.
4. Use partial fractions when necessary.
5. Apply known inverse pairs.

4.4.4 The Laplace Transform Is an Algebraic Tool

In system analysis, the primary purpose of the Laplace transform is to

- Convert differential equations into algebraic equations,
- Transform convolution into multiplication,
- Simplify system analysis using rational functions in s.

Thus, the Laplace transform is best understood as an **algebraic operator on signals**, rather than as an integral to evaluate.

4.4.5 Common Mistakes

- Attempting to compute transforms from the integral definition.
- Forgetting the exponential factor e^{-sT} in time shifts.
- Forgetting to shift $s \to s-a$ for $e^{at}f(t)$.
- Failing to complete the square in quadratic denominators.
- Not rewriting numerators to match standard forms.

Key Takeaway: The Laplace transform is not about integration. It is about recognizing structure and systematically applying properties.

4.5 Transfer Functions

Consider the linear ordinary differential equation

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_0 y = b_m u^{(m)} + b_{m-1} u^{(m-1)} + \ldots + b_0 u. \tag{4.15}$$

Assuming that the initial conditions $y^{(i)}(0) = 0$ and $u^{(j)}(0) = 0$, applying Laplace transform to each term yields

$$a_n s^n Y(s) + a_{n-1} s^{n-1} Y(s) + \ldots + a_0 Y(s) = b_m s^m U(s) + b_{m-1} s^{m-1} U(s) + \ldots + b_0 U(s), \tag{4.16}$$

which can be written as

$$Y(s) = G_{yu}(s) U(s), \tag{4.17}$$

where

$$G_{yu}(s) \triangleq \frac{b_m s^m + b_{m-1} s^{m-1} + \ldots + b_0}{a_n s^n + a_{n-1} s^{n-1} + \ldots + a_0} \tag{4.18}$$

is the **transfer function** from $u(t)$ to $y(t)$. The degree of the denominator polynomial is called the order of the transfer function.

Definition 4.5.1. The **transfer function from the input** u **to output** y, denoted by $G_{yu}(s)$, is the ratio of the Laplace transforms of the output signal and the input signal.

Definition 4.5.2. The **relative degree** of $G_{yu}(s)$ is defined as $n - m$.

1. If $n - m > 0$, then $G_{yu}(s)$ is strictly proper.
2. If $n - m = 0$, then $G_{yu}(s)$ is exactly proper.
3. If $n - m < 0$, then $G_{yu}(s)$ is improper.

Fact 4.5.1. Let $G_{yu}(s)$ be the transfer function of a system. Let $U(s)$ be the Laplace transform of the input signal $u(t)$. Then, the Laplace transform of the output $y(t)$ satisfies

$$Y(s) = G_{yu}(s) U(s). \tag{4.19}$$

Fact 4.5.2. Let $G_{yu}(s)$ be a strictly proper transfer function. Then,

$$G_{yu}(s) = \frac{k(s - z_1) \ldots (s - z_{n_z})}{(s - p_1) \ldots (s - p_{n_p})}, \tag{4.20}$$

where z_i are the **zeros** of the transfer function and p_i are the **poles** of transfer function. The zeros satisfy $G_{yu}(z_i) = 0$, while the poles satisfy $|G_{yu}(p_i)| = \infty$.

Fact 4.5.3. Consider an LTI system with transfer function $G(s)$. Then, $G(s)$ is the Laplace transform of the system's impulse response.

Proof. Recall that, under zero initial conditions,

$$Y(s) = G(s) U(s). \tag{4.21}$$

Taking the inverse Laplace transform and using the convolution property yields

$$y(t) = \mathcal{L}^{-1}[G(s) U(s)] = g(t) * u(t), \tag{4.22}$$

where $g(t) \triangleq \mathcal{L}^{-1}[G(s)]$.

From the state-space realization $G \sim (A, B, C, 0)$ and zero initial condition $x_0 = 0$, the output satisfies

$$y(t) = \int_0^t Ce^{A(t-\tau)}B\, u(\tau)\, d\tau = \left(Ce^{At}B\right) * u(t). \tag{4.23}$$

Comparing (4.22) and (4.23) yields

$$g(t) = Ce^{At}B. \tag{4.24}$$

Next, recall that, if $u(t) = \delta(t)$, then

$$y(t) = \int_0^t Ce^{A(t-\tau)}B\, \delta(\tau)\, d\tau = Ce^{At}B \tag{4.25}$$

is the impulse response of the system. Therefore, the transfer function $G(s)$ is the Laplace transform of the impulse response $g(t)$.

Alternatively, if the input is an impulse, $u(t) = \delta(t)$, then $U(s) = 1$ and

$$Y(s) = G(s)U(s) = G(s). \tag{4.26}$$

Hence, the output $y(t)$ is the impulse response, and its Laplace transform is $G(s)$. □

Example 4.5.1. Effect of initial conditions. Consider the DO

$$m\ddot{q}(t) + c\dot{q}(t) + kq(t) = f(t), \tag{4.27}$$

where $q(t)$ is the displacement and $f(t)$ is the applied force input. Unlike the derivation of the transfer function, we do *not* assume zero initial conditions in this example. Instead, we allow $q(0) \neq 0$ and $\dot{q}(0) \neq 0$ in order to illustrate how initial conditions appear in the frequency-domain representation.

Taking the Laplace transform of both sides gives

$$\mathcal{L}[m\ddot{q} + c\dot{q} + kq] = \mathcal{L}[f] \implies m\mathcal{L}[\ddot{q}] + c\mathcal{L}[\dot{q}] + k\mathcal{L}[q] = F(s). \tag{4.28}$$

Using the Laplace transform properties

$$\mathcal{L}[\dot{q}] = sQ(s) - q(0), \tag{4.29}$$
$$\mathcal{L}[\ddot{q}] = s^2Q(s) - sq(0) - \dot{q}(0), \tag{4.30}$$

it follows that

$$ms^2Q(s) - msq(0) - m\dot{q}(0) + csQ(s) - cq(0) + kQ(s) = F(s), \tag{4.31}$$

which can be reorganized as

$$Q(s) = G_{qf}(s)\, F(s) + G_{qq_0}(s)\, q(0)\Delta(s) + G_{q\dot{q}_0}(s)\, \dot{q}(0)\Delta(s), \tag{4.32}$$

where

$$G_{qf}(s) \triangleq \frac{1}{ms^2 + cs + k}, \quad G_{qq_0} \triangleq \frac{(ms + c)}{ms^2 + cs + k}, \quad G_{q\dot{q}_0} \triangleq \frac{m}{ms^2 + cs + k}. \tag{4.33}$$

Note that $\Delta(s) = \mathcal{L}[\delta(t)] = 1$.

This expression shows that, when initial conditions are nonzero, the system output $q(t)$ is the linear combination of

1. the forced response due to the input $f(t)$, and
2. the homogeneous response due to the initial displacement $q(0)$ and initial velocity $\dot{q}(0)$.

In particular, even when $f(t) \equiv 0$, the system exhibits nontrivial motion determined by its initial conditions.

4.6 Inverse Laplace Transforms via Partial Fraction Expansion

In this section, we develop practical tools for computing inverse Laplace transforms using partial fraction expansion (PFE). This technique allows us to convert rational transfer functions in the s-domain into sums of simple terms whose inverse Laplace transforms are available in closed form. The partial fraction expansion is then used to obtain the time-domain responses (impulse, step, and harmonic responses) of LTI systems.

Consider a strictly proper transfer function

$$G(s) = \frac{N(s)}{D(s)}. \tag{4.34}$$

If the denominator factors into distinct real poles, that is,

$$D(s) = (s - p_1)(s - p_2)\cdots(s - p_n), \tag{4.35}$$

then $G(s)$ can be rewritten as

$$G(s) = \frac{\alpha_1}{s - p_1} + \frac{\alpha_2}{s - p_2} + \cdots + \frac{\alpha_n}{s - p_n}, \tag{4.36}$$

where each coefficient α_i, also called the residue, is computed using the limit formula

$$\alpha_i = \lim_{s \to p_i} (s - p_i) G(s). \tag{4.37}$$

Example 4.6.1. Consider

$$G(s) = \frac{1}{s^2 + 7s + 12} = \frac{1}{(s+3)(s+4)} = \frac{\alpha_1}{s+3} + \frac{\alpha_2}{s+4}. \tag{4.38}$$

Then,

$$\alpha_1 = \lim_{s \to -3} (s+3) \frac{1}{(s+3)(s+4)} = 1, \qquad \alpha_2 = \lim_{s \to -4} (s+4) \frac{1}{(s+3)(s+4)} = -1, \tag{4.39}$$

and therefore

$$G(s) = \frac{1}{s+3} - \frac{1}{s+4}. \tag{4.40}$$

Taking the inverse Laplace yields

$$g(t) = e^{-3t} - e^{-4t}. \tag{4.41}$$

4.7 Using MATLAB to Compute Partial Fractions

For higher-order systems, manual partial fraction expansion becomes tedious. MATLAB provides a built-in routine to compute residues and poles directly.

Let

$$G(s) = \frac{b(s)}{a(s)} = \frac{b_m s^m + \cdots + b_0}{a_n s^n + \cdots + a_0}, \tag{4.42}$$

where n is the system order and $n - m$ is the relative degree. The partial fraction expansion is

$$G(s) = \sum_{i=1}^{n} \frac{r_i}{s - p_i} + k(s), \tag{4.43}$$

where r_i are residues, p_i are poles, and if the transfer function is improper, that is, $n < m$, then $k(s)$ is a polynomial term of degree $m - n$.

In MATLAB:

$$[r, p, k] = \texttt{residue}(b, a), \tag{4.44}$$

with coefficient vectors

$$b = [b_m \; b_{m-1} \; \cdots \; b_0], \qquad a = [a_n \; a_{n-1} \; \cdots \; a_0], \tag{4.45}$$

returns the residues r_i, the corresponding poles p_i, and the coefficients of polynomial $k(s)$. See code 4.1.

4.8 Partial Fraction Expansion: All Pole Cases

Partial fraction expansion (PFE) provides a systematic method for expressing a rational transfer function as a sum of simpler terms whose inverse Laplace transforms are available in closed form. This enables direct computation of time-domain responses of LTI systems from their transfer functions.

Consider a strictly proper transfer function

$$G(s) = \frac{N(s)}{D(s)} = \frac{b_m s^m + \cdots + b_0}{a_n s^n + \cdots + a_0}, \qquad m < n, \tag{4.46}$$

and suppose that the denominator $D(s)$ factors into real and complex roots. The structure of the partial fraction expansion depends on the type and multiplicity of the poles.

4.8.1 Case I: Distinct Real Poles

Suppose $G(s)$ has n distinct real poles $p_1, \ldots, p_n$, so that

$$D(s) = (s - p_1)(s - p_2)\cdots(s - p_n), \tag{4.47}$$

that is, $p_i \neq p_j$ for $i \neq j$. Then $G(s)$ can be written as

$$G(s) = \sum_{i=1}^{n} \frac{r_i}{s - p_i}, \tag{4.48}$$

where each residue r_i is given by

$$r_i = \lim_{s \to p_i} (s - p_i) G(s). \tag{4.49}$$

The corresponding time-domain contribution of each fraction is

$$\mathcal{L}^{-1}\left\{ \frac{r_i}{s - p_i} \right\} = r_i e^{p_i t}. \tag{4.50}$$

4.8.2 Case II: Repeated Real Poles

Suppose $G(s)$ has a real pole p of multiplicity m, so that

$$D(s) = (s - p)^m \, D_r(s), \tag{4.51}$$

where $D_r(s)$ contains no additional factors of $(s - p)$. Then the partial fraction expansion includes the chain

$$\frac{N(s)}{(s - p)^m} = \frac{r_1}{s - p} + \frac{r_2}{(s - p)^2} + \cdots + \frac{r_m}{(s - p)^m}. \tag{4.52}$$

The corresponding time-domain contributions of each fraction is

$$\mathcal{L}^{-1}\left\{ \frac{r_k}{(s - p)^k} \right\} = \frac{r_k}{(k - 1)!} t^{k-1} e^{pt}, \tag{4.53}$$

where $k = 1, 2, \ldots, m$.

4.8.3 Case III: Distinct Complex Conjugate Poles

Suppose $G(s)$ has a complex conjugate pole pair $p_{1,2} = \alpha \pm j\beta$. Then the denominator contains the real quadratic factor $(s - \alpha)^2 + \beta^2$. The corresponding real partial fraction form is

$$\frac{A(s - \alpha) + B}{(s - \alpha)^2 + \beta^2}. \tag{4.54}$$

The corresponding time-domain contributions are

$$\mathcal{L}^{-1}\left\{ \frac{A(s - \alpha)}{(s - \alpha)^2 + \beta^2} \right\} = A e^{\alpha t} \cos(\beta t), \qquad \mathcal{L}^{-1}\left\{ \frac{B}{(s - \alpha)^2 + \beta^2} \right\} = \frac{B}{\beta} e^{\alpha t} \sin(\beta t). \tag{4.55}$$

4.8.4 Case IV: Repeated Complex Conjugate Poles

Suppose $G(s)$ has a complex conjugate pole pair $p_{1,2} = \alpha \pm j\beta$, with multiplicity $\ell > 1$. Then the denominator contains the factor $\left[(s-\alpha)^2 + \beta^2\right]^\ell$, and the corresponding partial fraction expansion includes the chain

$$\sum_{k=1}^{\ell} \frac{A_k(s-\alpha) + B_k}{\left[(s-\alpha)^2 + \beta^2\right]^k}. \tag{4.56}$$

The corresponding time-domain contributions are polynomially weighted oscillatory terms of the form

$$t^{k-1}e^{\alpha t}\cos(\beta t), \qquad t^{k-1}e^{\alpha t}\sin(\beta t). \tag{4.57}$$

4.8.5 General Mixed Case

In general, if $D(s)$ factors as

$$D(s) = \prod_{i=1}^{n_r}(s-p_i)^{m_i} \prod_{k=1}^{n_c}\left[(s-\alpha_k)^2 + \beta_k^2\right]^{\ell_k}, \tag{4.58}$$

then the partial fraction expansion is obtained by summing the corresponding terms from all applicable cases above.

Example 4.8.1. (Mixed real and complex poles). Consider

$$G(s) = \frac{1}{(s+1)(s^2+2s+2)}. \tag{4.59}$$

Then,

$$G(s) = \frac{N_1}{s+1} + \frac{N_2 s + N_3}{(s+1)^2 + 1}, \tag{4.60}$$

where

$$N_1 = 1, \qquad N_2 = -1, \qquad N_3 = -1. \tag{4.61}$$

Hence,

$$G(s) = \frac{1}{s+1} - \frac{s+1}{(s+1)^2 + 1}. \tag{4.62}$$

Therefore, the impulse response is

$$g(t) = \mathcal{L}^{-1}[G(s)] = e^{-t} - e^{-t}\cos t. \tag{4.63}$$

Example 4.8.2. (Repeated real pole with complex conjugate pair) Consider

$$G(s) = \frac{1}{(s-1)^2(s^2+4)}. \tag{4.64}$$

Then,

$$G(s) = \frac{N_1}{s-1} + \frac{N_2}{(s-1)^2} + \frac{N_3 s + N_4}{s^2+4}. \tag{4.65}$$

Rather than solving for the coefficients by hand, we can use MATLAB's built-in `residue` command to compute the residues and poles of $G(s)$, as shown below.

Listing 4.1: MATLAB code to compute residues and poles for $G(s) = \dfrac{1}{(s-1)^2(s^2+4)}$ using `residue`

```
% G(s) = 1 / ((s-1)^2 (s^2 + 4))

a = conv([1 -2 1], [1 0 4]); % (s-1)^2 (s^2 + 4)
b = 1; % numerator

[r, p, k] = residue(b, a);

r % residues
p % poles
```

The output of `residue` returns residues associated with each pole. In this example, MATLAB returns

- two residues corresponding to the repeated real pole at $s = 1$, and
- a complex conjugate pair of residues corresponding to the poles at $s = \pm 2j$.

Combining complex residues into real coefficients. For real-coefficient transfer functions, complex poles and residues always appear in conjugate pairs, which can be combined as

$$\frac{A}{s-j2} + \frac{\overline{A}}{s+j2} = \frac{N_3 s + N_4}{s^2+4}, \tag{4.66}$$

where N_3 and N_4 are real coefficients obtained by algebraic combination of the complex-conjugate residues.

In this case,

$$N_1 = -\frac{2}{25}, \qquad N_2 = \frac{1}{5}, \qquad N_3 = \frac{2}{25}, \qquad N_4 = -\frac{3}{25}, \tag{4.67}$$

and thus

$$G(s) = \frac{N_1}{s-1} + \frac{N_2}{(s-1)^2} + \frac{N_3 s + N_4}{s^2+4} = \frac{N_1}{s-1} + \frac{N_2}{(s-1)^2} + N_3\frac{s}{s^2+4} + N_4\frac{1}{s^2+4}. \tag{4.68}$$

Taking inverse Laplace term-by-term and using standard transform pairs, that is,

$$\mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} = e^t, \quad \mathcal{L}^{-1}\left\{\frac{1}{(s-1)^2}\right\} = te^t, \quad \mathcal{L}^{-1}\left\{\frac{s}{s^2+4}\right\} = \cos(2t), \quad \mathcal{L}^{-1}\left\{\frac{1}{s^2+4}\right\} = \frac{1}{2}\sin(2t),$$

yields

$$g(t) = -\frac{2}{25}e^t + \frac{1}{5}te^t + \frac{2}{25}\cos(2t) - \frac{3}{50}\sin(2t). \tag{4.69}$$

4.9 Limit Theorems

4.9.1 Initial Value Theorem

The Initial Value Theorem provides a simple way to compute the value of a time-domain signal at $t = 0^+$ directly from its Laplace transform.

Theorem 4.9.1. If $F(s) = \mathcal{L}\{f(t)\}$ and the limit exists, then

$$\lim_{t\to 0^+} f(t) = \lim_{s\to\infty} sF(s). \tag{4.70}$$

Example 4.9.1. Consider the DO

$$G(s) = \frac{1}{ms^2 + cs + k}. \tag{4.71}$$

Note that $Q(s) = G(s)U(s)$. Consider an impulse input $u(t) = \delta(t)$. Then,

$$Q(s) = \frac{1}{ms^2 + cs + k} \implies q(0^+) = \lim_{s\to\infty} \frac{s}{ms^2 + cs + k} = 0. \tag{4.72}$$

Next, consider the velocity output. Since $v(t) = \dot{q}(t)$, it follows that $V(s) = sQ(s)$ and thus $V(s) = sG(s)U(s)$. For an impulse input $u(t) = \delta(t)$,

$$V(s) = \frac{s}{ms^2 + cs + k} \implies q(0^+) = \lim_{s\to\infty} \frac{s^2}{ms^2 + cs + k} = \frac{1}{m}. \tag{4.73}$$

Remark 4.9.1. An impulsive input does *not* produce an instantaneous jump in displacement, that is, $q(0^+) = 0$, but it does produce an instantaneous jump in velocity, that is, $v(0^+) = 1/m$. This reflects the physical fact that an impulse imparts momentum to a mass (causing an immediate change in velocity), while the displacement of a mechanical system cannot change instantaneously.

4.9.2 Initial Slope Theorem

The Initial Slope Theorem provides a simple way to compute the initial time-derivative (slope) of a time-domain signal at $t = 0^+$ directly from its Laplace transform.

Theorem 4.9.2. If $F(s) = \mathcal{L}\{f(t)\}$ and the limits exist, then

$$\lim_{t\to 0^+} \dot{f}(t) = \lim_{s\to\infty} s^2 F(s) - sf(0^+). \tag{4.74}$$

Example 4.9.2. Consider the DO

$$G(s) = \frac{1}{ms^2 + cs + k}. \tag{4.75}$$

Note that $Q(s) = G(s)U(s)$. Consider an impulse input $u(t) = \delta(t)$. Then,

$$Q(s) = \frac{1}{ms^2 + cs + k}. \tag{4.76}$$

Since $q(0^+) = 0$ (from the Initial Value Theorem), the Initial Slope Theorem gives

$$\dot{q}(0^+) = \lim_{s\to\infty} s^2 Q(s) - sq(0^+) \tag{4.77}$$

$$= \lim_{s\to\infty} \frac{s^2}{ms^2 + cs + k} = \frac{1}{m}. \tag{4.78}$$

Next, consider the acceleration output $a(t) = \ddot{q}(t)$. Since $A(s) = s^2 Q(s)$ (for zero initial conditions),

$$A(s) = \frac{s^2}{ms^2 + cs + k}. \tag{4.79}$$

Thus, the Initial Value Theorem yields

$$a(0^+) = \lim_{s\to\infty} sA(s) = \lim_{s\to\infty} \frac{s^3}{ms^2 + cs + k} = \infty, \tag{4.80}$$

which reflects the impulsive nature of the applied force.

Remark 4.9.2. The Initial Slope Theorem formalizes the physical interpretation of impulsive inputs in mechanical systems. While the displacement $q(t)$ remains continuous at $t = 0^+$, its slope $\dot{q}(t)$ exhibits a jump determined by the applied impulse. In contrast, the acceleration is unbounded at $t = 0^+$ due to the idealized impulse input, highlighting the limitations of impulsive models in physical realizations.

4.9.3 Final Value Theorem

The Final Value Theorem provides a simple way to compute the steady-state value of a time-domain signal as $t \to \infty$ directly from its Laplace transform.

Theorem 4.9.3. If $F(s) = \mathcal{L}\{f(t)\}$ and all poles of $sF(s)$ lie in the open left half-plane (except possibly a simple pole at $s = 0$), then

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s). \tag{4.81}$$

Example 4.9.3. Consider the DO

$$G(s) = \frac{1}{ms^2 + cs + k}. \tag{4.82}$$

Note that $Q(s) = G(s)U(s)$. Consider a step input $u(t) = 1(t)$ so that $U(s) = 1/s$. Then

$$Q(s) = G(s)U(s) = \frac{1}{s(ms^2 + cs + k)}. \tag{4.83}$$

Applying the Final Value Theorem,

$$\lim_{t\to\infty} q(t) = \lim_{s\to 0} sQ(s) = \lim_{s\to 0} \frac{1}{ms^2 + cs + k} = \frac{1}{k}. \tag{4.84}$$

Remark 4.9.3. The Final Value Theorem is valid only if the system is asymptotically stable and the steady-state limit exists. In particular, the theorem does not apply if $G(s)$ has poles in the right half-plane or repeated poles on the imaginary axis, in which case the time-domain response does not converge to a finite steady-state value.

4.10 Relationship Between State-Space and Transfer Functions

Consider

$$\dot{x}(t) = Ax(t) + Bu(t), \tag{4.85}$$
$$y(t) = Cx(t) + Du(t). \tag{4.86}$$

Taking Laplace transforms yields

$$sX(s) - x_0 = AX(s) + BU(s) \implies X(s) = (sI - A)^{-1}x_0 + (sI - A)^{-1}BU(s), \tag{4.87}$$

and thus

$$Y(s) = C(sI - A)^{-1}x_0 + \big[C(sI - A)^{-1}B + D\big]U(s). \tag{4.88}$$

Therefore, under zero initial conditions ($x_0 = 0$), the transfer function from u to y is

$$G_{yu}(s) = C(sI - A)^{-1}B + D. \tag{4.89}$$

Definition 4.10.1 (State-Space Realization)**.** A quadruple (A, B, C, D) is called a **state-space realization** of the transfer function $G(s)$ if

$$G(s) = C(sI - A)^{-1}B + D. \tag{4.90}$$

We write $G(s) \sim (A, B, C, D)$.

Fact 4.10.1. The state space realization of a transfer function $G(s)$ is not unique.

Proof. Let (A, B, C, D) be a realization of $G(s)$. Then,

$$G(s) = C(sI - A)^{-1}B + D. \tag{4.91}$$

Consider the state-space model

$$\dot{x} = Ax + Bu, \tag{4.92}$$
$$y = Cx + Du, \tag{4.93}$$

where $x \in \mathbb{R}^n$, $u \in \mathbb{R}^m$, and $y \in \mathbb{R}^p$. Define $z \triangleq Vx \in \mathbb{R}^n$, where $V \in \mathbb{R}^{n\times n}$ is a nonsingular matrix. Then,

$$\dot{z} = V\dot{x} = VAx + VBu = VAV^{-1}z + VBu, \tag{4.94}$$
$$y = Cx + Du = CV^{-1}z + Du, \tag{4.95}$$

or equivalently,

$$\dot{z} = \bar{A}z + \bar{B}u, \tag{4.96}$$
$$y = \bar{C}z + \bar{D}u, \tag{4.97}$$

where

$$\bar{A} \triangleq VAV^{-1}, \quad \bar{B} \triangleq VB, \quad \bar{C} \triangleq CV^{-1}, \quad \bar{D} \triangleq D. \tag{4.98}$$

Next, note that

$$\begin{aligned}\bar{C}(sI - \bar{A})^{-1}\bar{B} + \bar{D} &= CV^{-1}(sI - VAV^{-1})^{-1}VB + D\\ &= CV^{-1}V(sI - A)^{-1}V^{-1}VB + D\\ &= C(sI - A)^{-1}B + D\\ &= G(s),\end{aligned} \tag{4.99}$$

which implies that $(\bar{A}, \bar{B}, \bar{C}, \bar{D})$ is also a realization of $G(s)$, proving non-uniqueness. □

Fact 4.10.2. There are an infinite number of realizations of a transfer function $G(s)$.

Fact 4.10.3. Let $G \sim (A, B, C, D)$. The poles of a transfer function $G(s)$ are a subset of the eigenvalues of A.

Proof. Note that

$$G(s) = \frac{N(s)}{D(s)} = C(sI - A)^{-1}B + D = \frac{C\,\mathrm{adj}(sI - A)B}{\det(sI - A)}. \tag{4.100}$$

Hence, the only possible singularities of $G(s)$ occur at the roots of $\det(sI - A)$, which are precisely the eigenvalues of A. However, algebraic cancellations between the numerator and denominator may occur. In particular, if $N(s)$ and $D(s)$ share common factors, the corresponding singularities are canceled and do not appear as poles of $G(s)$. Therefore, the poles of $G(s)$ form a subset of the eigenvalues of A.

□

Example 4.10.1. Consider

$$G(s) = \frac{s+1}{(s+1)(s+2)} = \frac{1}{s+2}. \tag{4.101}$$

Note that the transfer function has a single pole at $s = -2$; the factor $(s + 1)$ cancels and does not appear as a pole of $G(s)$. A realization of $G(s)$ is

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 \end{bmatrix}, \quad D = 0. \tag{4.102}$$

Note that

$$G(s) = C(sI - A)^{-1}B = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{s+1} & 0 \\ 0 & \frac{1}{s+2} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{s+2}. \tag{4.103}$$

The eigenvalues of A are $\{-1, -2\}$, but $G(s)$ has only the pole -2. The eigenvalue -1 does not appear as a pole because of the pole-zero cancellation, which confirms that the poles of $G(s)$ are a subset of the eigenvalues of A.

Chapter 5

Stability

5.1 Stability of Linear State Space System

Consider the linear time-invariant system

$$\dot{x} = Ax, \tag{5.1}$$

where $x \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n \times n}$.

Definition 5.1.1 (Stability of the Linear System)**.** The linear system (5.1) is

1. **asymptotically stable** if $\lim_{t\to\infty} \|x(t)\| = 0$ for all initial conditions $x(0)$.

2. **semistable** if for every initial condition $x(0)$, the limit $\lim_{t\to\infty} x(t)$ exists and is finite (possibly nonzero).

3. **Lyapunov stable** if for each $\epsilon > 0$ there exists $\delta > 0$ such that $\|x(0)\| < \delta$ implies that $\|x(t)\| < \epsilon$ for all $t \geq 0$.

4. **unstable** if it is not Lyapunov stable.

Figure 5.1 shows the relationship between various types of stability properties.

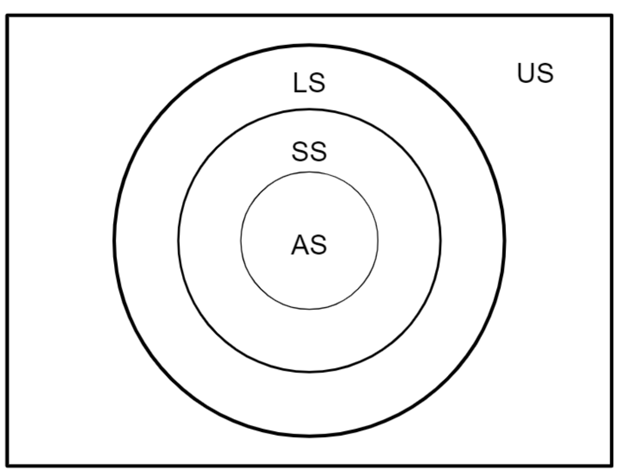

Figure 5.1: Relationship between various types of stability properties.

5.1.1 Stability of Linear Systems via Eigenvalues

Consider the linear time-invariant system (5.1). The stability properties (5.1) are characterized by the eigenvalues of A.

Fact 5.1.1 (Eigenvalue Characterization of Stability)**.** Let $\lambda_1, \ldots, \lambda_n$ denote the eigenvalues of A.

1. The system is **asymptotically stable** if and only if all eigenvalues are in the open left half plane (OLHP).
2. The system is **semistable** if and only if the eigenvalue at the origin in semisimple and all other eigenvalues are in the OLHP.
3. The system is **Lyapunov stable** if and only if all eigenvalues are in the closed left half plane (CLHP), and any eigenvalues on the imaginary axis are semisimple.
4. The system is **unstable** if and only if at least one eigenvalue lies in the open right half plane (ORHP), or there exists a nonsimple eigenvalue on the imaginary axis.

Review Appendix C to review simple, semisimple, and nonsimple eigenvalues and Jordan decomposition.

5.2 Stability of Transfer Functions

The stability of a transfer function is characterized by its time-domain response to an impulse input. Recall that the impulse response of a transfer function $G(s)$ is given by

$$Y(s) = G(s), \tag{5.2}$$

so that $y(t) = \mathcal{L}^{-1}\{G(s)\}$.

Definition 5.2.1 (Stability of a Transfer Function)**.** Consider a transfer function $G(s)$. Let $y_{\rm i}(t)$ denote its impulse response. The transfer function $G(s)$ is

1. **asymptotically stable** if $\lim_{t\to\infty} |y_{\rm i}(t)| = 0$.
2. **semistable** if the limit $\lim_{t\to\infty} y_{\rm i}(t)$ exists and is finite (possibly nonzero).
3. **Lyapunov stable** if $|y_{\rm i}(t)| \le c$, where c is a constant.
4. **unstable** if it is not Lyapunov stable.

Remark 5.2.1. If $G(s)$ is asymptotically stable, then it is semistable and Lyapunov stable. However, a semistable system is not necessarily asymptotically stable.

Remark 5.2.2. If $G(s)$ is semistable, then it is Lyapunov stable.

5.2.1 Stability of Transfer Functions via Poles

Let $(p_1, p_2, \ldots, p_k)$ be the poles of $G(s)$ with multiplicities $(m_1, m_2, \ldots, m_k)$. It follows from the partial fraction expansion of the transfer function that

$$y(t) = \sum_{i=1}^{k} \sum_{j=1}^{m_i} \alpha_{ij} t^{j-1} e^{p_i t}. \tag{5.3}$$

Fact 5.2.1. If all $p_i \in$ OLHP, then $G(s)$ is asymptotically stable.

Fact 5.2.2. If some poles $p_i \in$ IA with multiplicity one and the rest are in the OLHP, then $G(s)$ is Lyapunov stable.

Fact 5.2.3. If at most one pole $p_i = 0$ and the rest are in the OLHP, then $G(s)$ is semi-stable.

Fact 5.2.4. If at least one $p_i \in$ ORHP or any pole on the imaginary axis has multiplicity greater than one, then $G(s)$ is unstable.

5.3 BIBO Stability

In many applications, the primary concern is whether bounded inputs produce bounded outputs. This leads to the notion of *bounded-input bounded-output* (BIBO) stability.

Definition 5.3.1. A system with input $u(t)$ and output $y(t)$ is said to be *BIBO stable* if every bounded input produces a bounded output. More precisely, if, for all $t \geq 0$,

$$|u(t)| \leq M_u < \infty, \tag{5.4}$$

then there exists a constant M_y such that, for all $t \geq 0$,

$$|y(t)| \leq M_y < \infty. \tag{5.5}$$

Thus, a BIBO-stable system cannot generate an unbounded output in response to a bounded input.

Fact 5.3.1. Consider an LTI system with the transfer function $G(s)$ expressed in coprime form, that is, with all pole–zero cancellations removed. Then, the system is BIBO stable if and only if all poles of $G(s)$ lie in the open left half-plane.

Fact 5.3.2. Consider an LTI system with a *minimal realization* (A, B, C, D). Then the system is BIBO stable if and only if all eigenvalues of A lie in the open left half-plane.

Example 5.3.1. Consider

$$G(s) = \frac{s-2}{(s-2)(s+3)} = \frac{1}{s+3}. \tag{5.6}$$

The transfer function has a single pole at -3 and therefore is BIBO stable. However, a nonminimal realization of this transfer function may contain an eigenvalue at 2. Thus, this realization is unstable despite being BIBO stable.

Key Takeaway. BIBO stability does not imply internal stability. Pole-zero cancellations can obscure unstable internal modes in the transfer function.

Chapter 6

Realization

6.1 Realization of Transfer Functions

In practice, a transfer function must be implemented using a dynamical system. A *realization* of a transfer function $G(s)$ is any state-space system

$$\dot{x}(t) = Ax(t) + Bu(t), \tag{6.1}$$
$$y(t) = Cx(t) + Du(t), \tag{6.2}$$

that satisfies

$$G(s) = C(sI - A)^{-1}B + D. \tag{6.3}$$

Thus, realizing a transfer function means finding matrices A, B, C, D such that the above equation holds.

The following fact provides a necessary condition for the existence of a state space form, that is, the matrices A, B, C, and D to realize a transfer function $G(s)$.

Fact 6.1.1 (Relative degree and realizability)**.** Let $G(s)$ be a (real) rational transfer function. If $G(s)$ has *negative relative degree* (equivalently, $G(s)$ is *improper*), then $G(s)$ does not admit a finite-dimensional (LTI) state-space realization of the form

$$G(s) = C(sI - A)^{-1}B + D, \tag{6.4}$$

with constant matrices A, B, C, D.

Proof. Write $G(s) = \frac{N(s)}{D(s)}$, where $N(s)$ and $D(s)$ are coprime polynomials. Negative relative degree means $\deg N(s) > \deg D(s)$, hence

$$\lim_{s\to\infty} G(s) = \infty. \tag{6.5}$$

Now suppose, for contradiction, that $G(s)$ has a finite-dimensional realization

$$G(s) = C(sI - A)^{-1}B + D. \tag{6.6}$$

Since

$$(sI - A)^{-1} = \frac{1}{s}\Big(I - \frac{A}{s}\Big)^{-1} = \frac{1}{s}\Big(I + \frac{A}{s} + \frac{A^2}{s^2} + \cdots\Big), \tag{6.7}$$

it follows that $(sI - A)^{-1} \to 0$ as $s \to \infty$, and therefore

$$\lim_{s\to\infty} C(sI - A)^{-1}B = 0. \tag{6.8}$$

Consequently,

$$\lim_{s\to\infty} G(s) = \lim_{s\to\infty} \left(C(sI - A)^{-1}B + D\right) = D, \tag{6.9}$$

which is finite. This contradicts $\lim_{s\to\infty} G(s) = \infty$. Hence, no such finite-dimensional realization exists. $\square$

A convenient method for obtaining a state-space realization of a transfer function is to first convert the transfer function into a differential equation and then define appropriate state variables.

Example 6.1.1. Consider the transfer function

$$G(s) = \frac{1}{s+3}. \tag{6.10}$$

The corresponding differential equation is

$$\dot{y} = -3y + u. \tag{6.11}$$

Defining $x(t) = y(t)$ yields

$$\dot{x} = -3x + u. \tag{6.12}$$

Thus, a realization is

$$A = -3, \quad B = 1, \quad C = 1, \quad D = 0. \tag{6.13}$$

Example 6.1.2. Consider the transfer function

$$G(s) = \frac{1}{s^2 + 4s + 3}. \tag{6.14}$$

The corresponding differential equation is

$$\ddot{y} + 4\dot{y} + 3y = u. \tag{6.15}$$

Defining

$$x_1 = y, \qquad x_2 = \dot{y}. \tag{6.16}$$

yields

$$\dot{x}_1 = x_2, \tag{6.17}$$
$$\dot{x}_2 = -3x_1 - 4x_2 + u. \tag{6.18}$$

Thus, a realization is

$$A = \begin{bmatrix} 0 & 1 \\ -3 & -4 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = 0. \tag{6.19}$$

Example 6.1.3. Consider the transfer function

$$G(s) = \frac{b_2 s^2 + b_1 s + b_0}{s^3 + a_2 s^2 + a_1 s + a_0}. \tag{6.20}$$

This transfer function corresponds to the ODE

$$\dddot{y} + a_2\ddot{y} + a_1\dot{y} + a_0 y = b_2\ddot{u} + b_1\dot{u} + b_0 u. \tag{6.21}$$

Repeating the steps from the previous example does not produce the state space form because of the presence of time derivatives of the input. To circumvent this problem, factor the transfer function $G(s)$ as $G(s) = G_1(s)G_2(s)$, where

$$G_1(s) \triangleq b_2 s^2 + b_1 s + b_0, \quad G_2(s) \triangleq \frac{1}{s^3 + a_2 s^2 + a_1 s + a_0}. \tag{6.22}$$

Hence, $Y(s) = G_1(s)Z(s) = G_1(s)G_2(s)U(s)$, which implies that

$$\dddot{z} + a_2\ddot{z} + a_1\dot{z} + a_0 z = u, \tag{6.23}$$

$$y = b_2\ddot{z} + b_1\dot{z} + b_0 z. \tag{6.24}$$

Next, defining $x_1 \triangleq z$, $x_2 \triangleq \dot{z}$, $x_3 \triangleq \ddot{z}$, and $x \triangleq [x_1 \quad x_2 \quad x_3]^{\rm T}$, it follows from (6.23) and (6.24) that

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u, \tag{6.25}$$

$$y = \begin{bmatrix} b_0 & b_1 & b_2 \end{bmatrix} x. \tag{6.26}$$

Note that (6.25)-(6.26) is the state-space form of $G(s)$.

@403 students: Ignore the example below.

Example 6.1.4. Consider the differential equation

$$\dot{y} + a_0 y = b_2\ddot{u} + b_1\dot{u} + b_0 u. \tag{6.27}$$

Note that

$$Y(s) = \frac{b_2 s^2 + b_1 s + b_0}{s + a_0} U(s). \tag{6.28}$$

Define

$$x \triangleq \begin{bmatrix} y \\ u \\ \dot{u} \end{bmatrix}, \quad v \triangleq \ddot{u}. \tag{6.29}$$

Then,

$$\dot{x} = \begin{bmatrix} \dot{y} \\ \dot{u} \\ \ddot{u} \end{bmatrix} = \begin{bmatrix} b_2\ddot{u} + b_1\dot{u} + b_0 u - a_0 y \\ x_3 \\ v \end{bmatrix} = \begin{bmatrix} b_2 v + b_1 x_3 + b_0 x_2 - a_0 x_1 \\ x_3 \\ v \end{bmatrix} = Ax + Bv, \tag{6.30}$$

where

$$A \triangleq \begin{bmatrix} -a_0 & b_0 & b_1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad B \triangleq \begin{bmatrix} b_2 \\ 0 \\ 1 \end{bmatrix} \tag{6.31}$$

Note that

$$y = Cx, \tag{6.32}$$

where

$$C \triangleq \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}. \tag{6.33}$$

Note that

$$G_{yv} = C(sI - A)^{-1}B = \frac{b_2 s^2 + b_1 s + b0}{s^2(s + a_0)} \tag{6.34}$$

Since

$$G_{vu} = s^2, \tag{6.35}$$

it follows that

$$G_{yu} = G_{yv}G_{vu} = \frac{b_2 s^2 + b_1 s + b_0}{s + a_0} \tag{6.36}$$

Chapter 7

Control

7.1 Basic Connections

Many control systems are constructed by interconnecting simpler subsystems. Before analyzing feedback systems, it is useful to understand the basic building blocks used in block diagrams and how these subsystems combine to form more complex systems.

Two types of operations arise when constructing block diagrams. First, signals can be manipulated using simple static operations. Signals may be added, subtracted, or scaled. Signal addition or subtraction is implemented using a *summing junction*, while signal scaling is implemented using a *gain*.

Second, dynamical systems can be interconnected to form larger systems. Two common interconnections are *series* and *parallel* connections. In a series connection, the output of one system becomes the input to the next, and the overall transfer function is obtained by multiplying the transfer functions of the individual systems. In a parallel connection, the same input is applied to multiple systems and their outputs are added, so that the overall transfer function is the sum of the individual transfer functions.

These basic operations allow the input–output behavior of interconnected systems to be computed directly from the transfer functions of the individual subsystems.

Definition 7.1.1. A **summing junction** is a system whose output is the sum of all inputs.

Note that the summing junction is a *static* system. Consider the summing junction shown in Figure 7.1. It follows from Figure 7.1 that $y(t) = u_1(t) + u_2(t) - u_3(t)$. Note that the input signal u_3 is multiplied by the minus sign shown near the input junction.

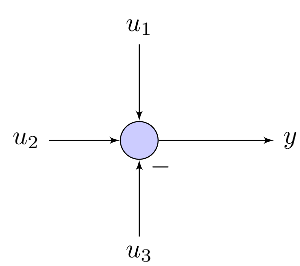

Figure 7.1: Summing junction.

Definition 7.1.2. A **gain** is a system whose output is the scaled input.

Figure 7.2 shows a gain. Note that $y = Ku$, where K may be a scalar, vector, or a matrix. The dimensions of K depend on the dimensions of u and y. Note that the gain is a *static* system.

Definition 7.1.3. Two systems are said to be **connected in series** if the output of the first system is the input to the second system.

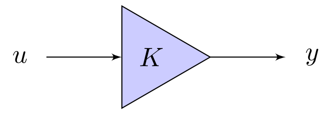

Figure 7.2: Gain. $y = Ku$.

Figure 7.3 shows two systems $G_1(s)$ and $G_2(2)$ connected in series. It follows from Figure 7.3 that

$$W(s) = G_1(s)U(s), \tag{7.1}$$
$$Y(s) = G_2(s)W(s), \tag{7.2}$$

which implies

$$Y(s) = G_2(s)G_1(s)U(s), \tag{7.3}$$

and thus the transfer function $G_{yu}(s)$ from u to y is

$$G_{yu}(s) = G_2(s)G_1(s). \tag{7.4}$$

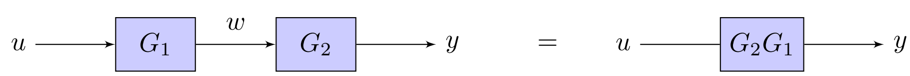

Figure 7.3: Series connection.

Definition 7.1.4. Two systems are said to be **connected in parallel** if the input to both systems is the same and the outputs of both systems are added.

Figure 7.4 shows two systems $G_1(s)$ and $G_2(2)$ connected in parallel. It follows from Figure 7.4 that

$$Y_1(s) = G_1(s)U(s), \tag{7.5}$$
$$Y_2(s) = G_2(s)U(s), \tag{7.6}$$
$$Y(s) = Y_1(s) + Y_2(s), \tag{7.7}$$

which implies

$$Y(s) = (G_1(s) + G_2(s))\, U(s), \tag{7.8}$$

and thus the transfer function $G_{yu}(s)$ from u to y is

$$G_{yu}(s) = G_1(s) + G_2(s). \tag{7.9}$$

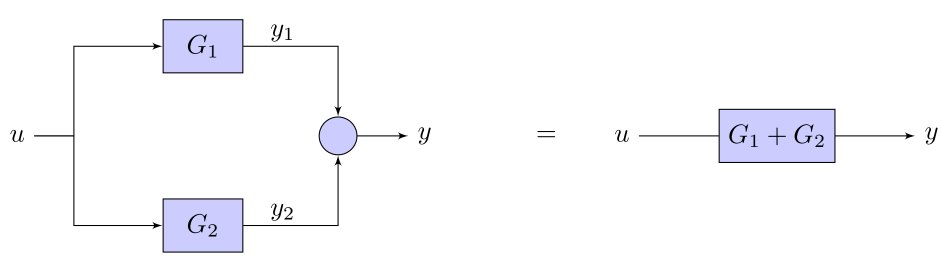

Figure 7.4: Parallel connection.

7.2 Naive Open-Loop Control via Pole–Zero Cancellation

Consider the unstable plant

$$G(s) = \frac{1}{s-2}. \tag{7.10}$$

The objective is to design a controller $G_c(s)$ that ensures that the output of the transfer function $G(s)$ follows the reference signal $r(t)$.

A tempting idea is to design a controller that cancels the plant's unstable pole. Let

$$G_c(s) = s - 2. \tag{7.11}$$

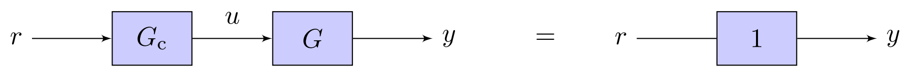

Figure 7.5: Series connection with pole–zero cancellation, where G_c cancels the unstable poles of G.

Note that

$$U(s) = G_c(s)R(s). \tag{7.12}$$

Then,

$$G_c(s)G(s) = (s-2)\frac{1}{s-2} = 1. \tag{7.13}$$

Consequently, the transfer function from r to y is 1 and thus

$$Y(s) = R(s), \tag{7.14}$$

which implies that

$$y(t) = r(t). \tag{7.15}$$

At first glance, this appears to achieve *perfect tracking*. However, this design is fundamentally flawed.

The purpose of this example is to emphasize that pole-zero cancellation is problematic for the following reasons.

1. **Initial conditions reveal the hidden unstable mode.** Transfer functions implicitly assume zero initial conditions. When the plant has nonzero initial conditions, the system's unstable mode reappears. Recall Example 4.5.1, which shows the inclusion of initial conditions in the transfer function.

   In our example, the plant dynamics satisfy

   $$\dot{y} - 2y = u(t). \tag{7.16}$$

   Taking the Laplace transform yields

   $$Y(s) = \frac{1}{s-2}U(s) + \frac{1}{s-2}y(0)\Delta(s). \tag{7.17}$$

   Using (7.12) implies that

   $$Y(s) = R(s) + \frac{1}{s-2}y(0)\Delta(s), \tag{7.18}$$

   which implies that

   $$y(t) = r(t) + e^{2t}y(0). \tag{7.19}$$

   which grows exponentially when $y(0) \neq 0$.

   Thus, although the transfer-function cancellation appears to remove the unstable pole, the unstable mode remains within the system's internal dynamics. Pole–zero cancellation does *not* remove unstable internal dynamics.

2. **Exact pole knowledge is required.** Perfect cancellation requires the unstable pole location to be known exactly. In practice, the plant model is always uncertain. Even a small modeling error can alter the pole location and prevent exact cancellation. As a result, the unstable pole remains in the closed-loop system. Thus, pole-zero cancellation is highly sensitive to modeling uncertainty.

3. **Effect on the Control Signal.** The control input generated by the controller is

$$U(s) = sR(s) - 2R(s). \tag{7.20}$$

Taking the inverse Laplace transform yields

$$u(t) = \dot{r}(t) - 2r(t). \tag{7.21}$$

Thus, the controller differentiates the reference signal. If the reference signal contains high-frequency noise, then $\dot{r}(t)$ can become very large. As a result, the control signal may become excessively large. In general, controllers that differentiate the reference signal tend to amplify high-frequency noise and are therefore highly undesirable in practice.

4. **The controller is not physically realizable.** It follows from Proposition 6.1.1 that if the relative degree of a transfer function $G(s)$ is negative, then a finite-dimensional state-space realization does not exist. Since the relative degree of $G_c(s)$ is -1, it does not admit a state-space realization.

7.3 Why Open-Loop Control Fails?

A natural question is whether feedback is necessary at all. In particular, suppose the plant model $G(s)$ is known. Why not compute an input signal $u(t)$ that produces the desired output $y(t) = r(t)$ directly? This idea corresponds to *open-loop control*. In open-loop control, the control input is computed solely from the reference signal and does not depend on the measured output.

To illustrate this idea, suppose the plant satisfies

$$Y(s) = G(s)U(s). \tag{7.22}$$

If $G(s)$ is known and invertible, then one may attempt to choose

$$U(s) = G(s)^{-1}R(s). \tag{7.23}$$

Substituting into the plant equation yields

$$Y(s) = G(s)G(s)^{-1}R(s) = R(s), \tag{7.24}$$

which appears to achieve perfect tracking.

Although this approach appears attractive, it is fundamentally flawed for several reasons.

1. **Sensitivity to modeling errors.** Open-loop control relies entirely on the accuracy of the plant model. If the actual plant differs even slightly from the model, the computed control input will be incorrect, and the desired output will not be achieved. In practice, all physical systems contain modeling errors due to parameter uncertainty, unmodeled dynamics, and nonlinear effects. Without feedback, there is no mechanism to correct these errors.

2. **Disturbances cannot be rejected.** Real systems are subject to disturbances. Suppose the plant is affected by an additive disturbance $d(t)$,

$$Y(s) = G(s)U(s) + D(s). \tag{7.25}$$

Even if the control input is chosen as

$$U(s) = G(s)^{-1}R(s), \tag{7.26}$$

the output becomes

$$Y(s) = R(s) + D(s). \tag{7.27}$$

Thus, disturbances directly affect the output and cannot be corrected in an open-loop architecture.

3. **Initial conditions cannot be corrected.** Open-loop control does not use measurements of the system output. If the system begins with an unknown initial condition, the controller has no mechanism to compensate for the resulting error. Consequently, the output may deviate significantly from the desired trajectory even if the model of the plant is perfectly known.

These limitations illustrate a fundamental principle of control systems: *feedback is required to achieve robustness to uncertainty, disturbances, and unknown initial conditions.* In contrast to open-loop control, feedback control continuously measures the system output and adjusts the control input to reduce the tracking error.

7.4 Feedback Control System

A **feedback control system** is an interconnected system in which the input to the plant is generated by a controller whose input depends on the plant's output. The controller processes the measured output and produces a control signal that influences the plant's behavior. The subsystem that generates the plant input based on the plant output is called the **feedback controller**. Feedback control systems are designed to achieve one or more of the following objectives.

1. Plant stabilization.
2. Command following or reference tracking.
3. Disturbance or noise rejection.
4. Satisfaction of desired transient-response specifications.

Let $G(s)$ denote the plant transfer function and $G_{\rm c}(s)$ denote the controller transfer function. The plant input is denoted by $u(t)$, and the plant output is denoted by $y(t)$. The controller generates the control signal $u(t)$ based on the tracking error.

The **reference signal** $r(t)$ represents the desired value that the plant output should follow. It is also called the **command signal**. The **error signal** $e(t)$ is defined as the difference between the reference signal and the plant output,

$$e(t) \triangleq r(t) - y(t). \tag{7.28}$$

Definition 7.4.1. The **sensitivity transfer function** $S(s)$ of a feedback control system is the transfer function from the reference signal $r(t)$ to the error signal $e(t)$,

$$S(s) \triangleq G_{er}(s). \tag{7.29}$$

Definition 7.4.2. The **transmissibility transfer function** $T(s)$ (also called the **complementary sensitivity**) is the transfer function from the reference signal $r(t)$ to the plant output $y(t)$,

$$T(s) \triangleq G_{yr}(s). \tag{7.30}$$

7.4.1 Basic Servo Loop

The **basic servo loop** (BSL) is the simplest feedback control architecture used for plant stabilization and reference tracking. The basic servo loop consists of a controller $G_{\rm c}(s)$ connected in series with the plant $G(s)$. The plant output $y(t)$ is compared with the reference signal $r(t)$ to generate the error given by (7.28). The error signal is applied to the controller $G_{\rm c}(s)$, whose output $u(t)$ is the input to the plant $G(s)$. A basic servo loop is shown in Figure 7.6.

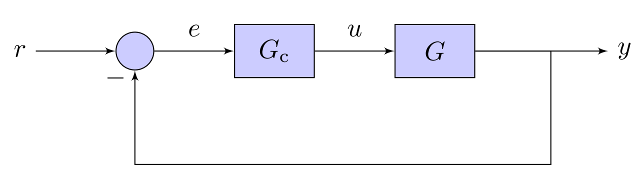

Figure 7.6: Basic servo loop.

Taking the Laplace transform of the error (7.28) yields

$$E(s) = R(s) - Y(s). \tag{7.31}$$

Since the controller $G_c(s)$ and the plant $G(s)$ are connected in series,

$$Y(s) = G_c(s)G(s)E(s). \tag{7.32}$$

Substituting (7.32) into (7.31) yields

$$\begin{aligned} E(s) &= R(s) - G_c(s)G(s)E(s) \\ \implies (1 + G_c(s)G(s))\, E(s) &= R(s) \\ \implies E(s) &= \frac{1}{1 + G_c(s)G(s)} R(s). \end{aligned} \tag{7.33}$$

Furthermore, substituting (7.33) into (7.32) gives

$$Y(s) = \frac{G_c(s)G(s)}{1 + G_c(s)G(s)} R(s). \tag{7.34}$$

Therefore, the transfer functions from the reference signal to the error and output are

$$S(s) = G_{er}(s) = \frac{1}{1 + G_c(s)G(s)}, \qquad T(s) = G_{yr}(s) = \frac{G_c(s)G(s)}{1 + G_c(s)G(s)}. \tag{7.35}$$

In the basic servo loop, the controller $G_c(s)$ and the plant $G(s)$ appear in series inside the feedback loop. It is therefore convenient to define the **loop transfer function**.

Definition 7.4.3. The **loop transfer function** is defined as

$$L(s) \triangleq G_c(s)G(s). \tag{7.36}$$

Using this definition, the closed-loop transfer functions derived previously can be written in compact form as

$$T(s) = \frac{L(s)}{1 + L(s)}, \qquad S(s) = \frac{1}{1 + L(s)}, \tag{7.37}$$

where $T(s)$ is the transmissibility transfer function and $S(s)$ is the sensitivity transfer function. Note that

$$S(s) + T(s) = 1. \tag{7.38}$$

This relation shows that the sensitivity and transmissibility are fundamentally coupled: improving performance with respect to one transfer function necessarily affects the other.

7.4.2 Pole-Zero Structure of the Sensitivity Function

Let the controller and plant be written as rational transfer functions

$$G_c(s) = \frac{N_c(s)}{D_c(s)}, \qquad G(s) = \frac{N(s)}{D(s)}. \tag{7.39}$$

The loop transfer function is therefore

$$L(s) = \frac{N_c(s)N(s)}{D_c(s)D(s)}. \tag{7.40}$$

Substituting this expression into the definition of the sensitivity function yields

$$S(s) = \frac{1}{1 + L(s)} = \frac{1}{1 + \dfrac{N_c(s)N(s)}{D_c(s)D(s)}} = \frac{D_c(s)D(s)}{D_c(s)D(s) + N_c(s)N(s)}. \tag{7.41}$$

This expression reveals two important structural properties.

- The numerator $D_c(s)D(s)$ contains the poles of the controller and the plant, which correspond to the poles of the **open-loop system**.
- The denominator $D_c(s)D(s) + N_c(s)N(s)$ determines the poles of the **closed-loop system**.

Thus, the sensitivity function provides direct insight into how the open-loop dynamics influence the closed-loop behavior of the feedback system.

7.5 Simple Controllers for Command Following

Example 7.5.1 (Stabilization of an Unstable Plant)**.** Consider the basic servo loop where the plant and controller are

$$G(s) = \frac{1}{s-2},\ G_c(s) = 4. \tag{7.42}$$

Since the controller and plant are connected in series, the loop transfer function is

$$L(s) = G_c(s)G(s) = \frac{4}{s-2}. \tag{7.43}$$

The sensitivity transfer function is therefore

$$S(s) = \frac{1}{1+L(s)} = \frac{1}{1+\frac{4}{s-2}} = \frac{s-2}{s+2}. \tag{7.44}$$

Thus,

$$E(s) = S(s)R(s) = \frac{s-2}{s+2}R(s). \tag{7.45}$$

Next, consider a unit-step command

$$r(t) = 1(t) \implies R(s) = \frac{1}{s}. \tag{7.46}$$

Using the final value theorem,

$$e_\infty = \lim_{t\to\infty} e(t) = \lim_{s\to 0} sE(s) = \lim_{s\to 0} s\left(\frac{s-2}{s+2}\frac{1}{s}\right) = -1 \neq 0. \tag{7.47}$$

Note that although the plant is stabilized, the command is **not asymptotically followed**.

Example 7.5.2 (Control of a Double Integrator)**.** Consider the plant

$$G(s) = \frac{1}{ms^2}. \tag{7.48}$$

Proportional Control. First, consider proportional control

$$G_c(s) = K. \tag{7.49}$$

The loop transfer function is

$$L(s) = G_c(s)G(s) = \frac{K}{ms^2}. \tag{7.50}$$

The sensitivity and transmissibility transfer functions are therefore

$$S(s) = \frac{1}{1+L(s)} = \frac{1}{1+\frac{K}{ms^2}} = \frac{ms^2}{ms^2+K}, \tag{7.51}$$

$$T(s) = \frac{L(s)}{1+L(s)} = \frac{K}{ms^2+K}. \tag{7.52}$$

Note that $S(s)$ and $T(s)$ have poles on the imaginary axis. Hence, the closed-loop system is not asymptotically stable. In fact, the step response of the system is

$$Y(s) = T(s)R(s) = \frac{K}{ms^2 + K}\frac{1}{s}. \tag{7.53}$$

Taking the inverse Laplace transform yields

$$y(t) = 1 - \cos\left(\sqrt{\frac{K}{m}}\,t\right), \tag{7.54}$$

which oscillates indefinitely and therefore does not converge to a finite steady-state value. Thus, although proportional control produces bounded oscillatory motion, it does not yield asymptotic tracking of the step command and is not sufficient to achieve asymptotic step tracking for the double-integrator plant.

Proportional–Derivative Control. Next, consider a proportional–derivative controller

$$G_{\rm c}(s) = K_p + K_d s. \tag{7.55}$$

The loop transfer function is

$$L(s) = \frac{K_p + K_d s}{ms^2}. \tag{7.56}$$

The sensitivity and transmissibility transfer functions are therefore

$$S(s) = \frac{1}{1 + L(s)} = \frac{ms^2}{ms^2 + K_d s + K_p}, \tag{7.57}$$

$$T(s) = \frac{K_d s + K_p}{ms^2 + K_d s + K_p}. \tag{7.58}$$

The step response of the system is

$$Y(s) = T(s)R(s) = \frac{K_d s + K_p}{ms^2 + K_d s + K_p}\frac{1}{s}. \tag{7.59}$$

It follows from the final value theorem that $y_\infty = 1$.

Realizable Proportional–Derivative Control. However, the controller

$$G_{\rm c}(s) = K_p + K_d s \tag{7.60}$$

is not physically realizable, that is, an equivalent state-space realization does not exist, because it is an improper transfer function. In particular, the derivative term $K_d s$ requires unbounded gain as $s \to \infty$ and therefore amplifies high-frequency measurement noise.

A common practical implementation introduces a pole in the derivative term. Specifically, the controller is implemented as

$$G_{\rm c}(s) = K_p + \frac{K_d s}{s + p}, \tag{7.61}$$

where $p > 0$ is chosen sufficiently large.

Combining the terms gives

$$G_{\rm c}(s) = \frac{K_p(s + p) + K_d s}{s + p} = \frac{(K_p + K_d)s + K_p p}{s + p}. \tag{7.62}$$

The loop transfer function is

$$L(s) = G_c(s)G(s) = \frac{(K_p + K_d)s + K_p p}{s+p}\frac{1}{ms^2}. \tag{7.63}$$

The sensitivity function is therefore

$$S(s) = \frac{1}{1+L(s)} = \frac{1}{1 + \frac{(K_p+K_d)s+K_p p}{s+p}\frac{1}{ms^2}} = \frac{ms^2(s+p)}{ms^2(s+p) + (K_p+K_d)s + K_p p}. \tag{7.64}$$

Next, consider a unit-step command

$$r(t) = 1(t) \implies R(s) = \frac{1}{s}. \tag{7.65}$$

Since the denominator polynomial

$$ms^2(s+p) + (K_p + K_d)s + K_p p \tag{7.66}$$

has all of its roots in the OLHP if $m > 0$, $K_p > 0$, $K_d > 0$, and $p > 0$, the final value theorem can be applied to compute the asymptotic value of the output. Note that

$$E(s) = S(s)R(s) = S(s)\frac{1}{s}, \tag{7.67}$$

and thus

$$e_\infty = 0, \qquad y_\infty = 1. \tag{7.68}$$

7.6 Steady-State Tracking in the Basic Servo Loop

For the four canonical systems, the transfer function can be written as $G(s) = 1/D(s)$, where

$$D_{\mathrm{URB}}(s) \triangleq ms^2, \tag{7.69}$$

$$D_{\mathrm{DRB}}(s) \triangleq ms^2 + cs, \tag{7.70}$$

$$D_{\mathrm{UO}}(s) \triangleq ms^2 + k, \tag{7.71}$$

$$D_{\mathrm{DO}}(s) \triangleq ms^2 + cs + k. \tag{7.72}$$

For a controller $G_c(s)$, the sensitivity transfer function in the basic servo loop is

$$S(s) = \frac{D(s)}{D(s) + G_c(s)}, \tag{7.73}$$

where the denominator reflects the closed-loop characteristic polynomial. The table below summarizes the resulting asymptotic step-tracking behavior for different controller structures.

Table 7.1: Asymptotic step response in the basic servo loop for several controller structures.

Controller $G_c(s)$	Sensitivity transfer function $S(s)$	URB $D(s) = ms^2$	DRB $D(s) = ms^2 + cs$	UO $D(s) = ms^2 + k$	DO $D(s) = ms^2 + cs + k$
K_p	$\dfrac{D(s)}{D(s)+K_p}$	FVT not applicable	0	$\dfrac{k}{k+K_p}$	$\dfrac{k}{k+K_p}$
$K_d s$	$\dfrac{D(s)}{D(s)+K_d s}$	0	$\dfrac{c}{c+K_d}$	1	1
$K_p + K_d s$	$\dfrac{D(s)}{D(s)+K_d s+K_p}$	0	0	$\dfrac{k}{k+K_p}$	$\dfrac{k}{k+K_p}$
$\dfrac{K_i}{s}$	$\dfrac{D(s)s}{D(s)s+K_i}$	0	0	0	0
$K_p + \dfrac{K_i}{s}$	$\dfrac{D(s)s}{D(s)s+K_p s+K_i}$	0	0	0	0
$K_p + \dfrac{K_i}{s} + K_d s$	$\dfrac{D(s)s}{D(s)s+K_d s^2+K_p s+K_i}$	0	0	0	0

Table 7.1 shows how the steady-state tracking performance depends on both the plant dynamics and the structure of the controller.

7.7 System Type (Not Important)

The steady-state error behavior can be understood in terms of the *type* of the loop transfer function, defined as the number of poles at the origin in $G_c(s)G(s)$.

- **Type 0:** No poles at $s = 0$
- **Type 1:** One pole at $s = 0$
- **Type 2:** Two poles at $s = 0$

Step Tracking and Type. For a step input, the steady-state error depends on the system type as follows.

- Step response of Type 0 systems has nonzero steady-state error.
- Step response of Type 1 or higher systems has zero steady-state error.

The results in Table 7.1 are consistent with this classification.

- The UO and DO systems with proportional or PD control are **Type 0**, and therefore exhibit nonzero steady-state error.
- The URB and DRB systems already contain an integrator in the plant, making them **Type 1**, which explains why proportional control alone achieves zero steady-state error.
- Controllers with integral action (I, PI, PID) introduce an additional pole at the origin, increasing the system type and guaranteeing zero steady-state error for all plant classes.

7.8 Internal Model Principle

The internal model principle provides necessary conditions for the tracking error in a feedback control system to converge to zero. Roughly speaking, the principle states that the feedback loop must contain a *model of the external signal* (reference or disturbance) in order to achieve asymptotic rejection or tracking.

7.8.1 Basic Servo Loop and Error Dynamics

Consider the basic servo loop. The error signal satisfies

$$E(s) = S(s)R(s), \tag{7.74}$$

where the sensitivity transfer function is

$$S(s) = \frac{1}{1 + G_c(s)G(s)}. \tag{7.75}$$

Let

$$G_c(s) = \frac{N_c(s)}{D_c(s)}, \quad G(s) = \frac{N(s)}{D(s)}. \tag{7.76}$$

Then,

$$E(s) = \frac{D_c(s)D(s)}{D_c(s)D(s) + N_c(s)N(s)} R(s), \tag{7.77}$$

where the closed-loop characteristic polynomial is

$$D_{cl}(s) \triangleq D_c(s)D(s) + N_c(s)N(s). \tag{7.78}$$

Assume that the controller is designed such that all roots of

$$D_{cl}(s) = 0 \tag{7.79}$$

are in the open left-half plane (OLHP), so that the closed-loop system is asymptotically stable.

7.8.2 Steady-State Error via Final Value Theorem

The steady-state error is given by

$$e_\infty = \lim_{t\to\infty} e(t) = \lim_{s\to 0} sE(s). \tag{7.80}$$

We now analyze different reference inputs.

Step Command. Let

$$r(t) = 1(t) \implies R(s) = \frac{1}{s}. \tag{7.81}$$

Then,

$$e_\infty = \lim_{s\to 0} s \cdot \frac{D_c(s)D(s)}{D_{cl}(s)} \cdot \frac{1}{s} = \lim_{s\to 0} \frac{D_c(s)D(s)}{D_{cl}(s)}. \tag{7.82}$$

Thus,

$$e_\infty = 0 \quad \iff \quad D_c(0)D(0) = 0. \tag{7.83}$$

This implies

- Either the plant or the controller must have a pole at $s = 0$.
- If $D(0) \neq 0$, then the controller must include an integrator.

Remark 7.8.1. A step input has a pole at $s = 0$, so the loop must contain a pole at $s = 0$ (an integrator) to track it.

Ramp Command. Let

$$r(t) = t\,1(t) \implies R(s) = \frac{1}{s^2}. \tag{7.84}$$

Then,

$$e_\infty = \lim_{s\to 0} s \cdot \frac{D_\mathrm{c}(s)D(s)}{D_\mathrm{cl}(s)} \cdot \frac{1}{s^2} = \lim_{s\to 0} \frac{D_\mathrm{c}(s)D(s)}{D_\mathrm{cl}(s)s}. \tag{7.85}$$

Thus,

$$e_\infty = 0 \quad \iff \quad D_\mathrm{c}(s)D(s) \text{ has at least two zeros at } s = 0. \tag{7.86}$$

This implies

- A ramp input has a double pole at $s = 0$.
- The loop must contain two integrators.

Remark 7.8.2. A ramp input has two poles at $s = 0$, so the loop must contain two poles at $s = 0$ (two integrators) to track it.

Harmonic Command. Let

$$r(t) = \sin(\omega t) \implies R(s) = \frac{\omega}{s^2 + \omega^2}. \tag{7.87}$$

Then,

$$e_\infty = \lim_{s\to 0} s \cdot \frac{D_\mathrm{c}(s)D(s)}{D_\mathrm{cl}(s)} \cdot \frac{\omega}{s^2 + \omega^2}. \tag{7.88}$$

For the final value theorem to be valid, $E(s)$ must not have poles on the imaginary axis. Thus, the poles at $\pm\jmath\omega$ must be canceled by the numerator, which implies

$$D_\mathrm{c}(s)D(s) = 0 \quad \text{has roots at} \quad s = \pm\jmath\omega. \tag{7.89}$$

This implies

- A sinusoidal input has poles at $\pm\jmath\omega$.
- The loop must contain a model of this frequency.

Remark 7.8.3. A harmonic input has two imaginary poles at $s = \pm\jmath\omega$, so the loop must contain two poles at $s = \pm\jmath\omega$ (two integrators) to track it.

7.8.3 Important Remark

The internal model principle provides *necessary but not sufficient* conditions.

- Including the required internal model does not guarantee stability.
- The closed-loop characteristic polynomial must still be stable.
- The controller must embed the dynamics of the signal it is trying to track or reject.

7.9 Time-Domain Specifications

Consider an asymptotically stable system.

Definition 7.1. The **time constant** τ is the time required for the free response of the system to decay to $e^{-1} \approx 37\%$ of its initial value.

Definition 7.2. The **rise time** T_r is the time required for the step response to increase from 10% to 90% of its final value.

Definition 7.3. The **settling time** T_s is the time required for the step response to enter and remain within a specified tolerance band (typically $\pm 2\%$) of its final value.

Definition 7.4. The **natural frequency** ω_n characterizes the oscillatory behavior of the system and corresponds to the imaginary part of the poles in the undamped case.

Definition 7.5. The **natural period** T_n is the time between successive peaks of an oscillatory response and is given by

$$T_\mathrm{n} = \frac{2\pi}{\omega_\mathrm{n}}. \tag{7.90}$$

If the response is non-oscillatory, then $T_\mathrm{n} = \infty$.

Definition 7.6. The **exponential decay rate** σ is the real part of the system poles. For a mode of the form $e^{(\sigma + \jmath\omega)t}$, $\sigma < 0$ determines the rate of decay.

Definition 7.7. The **damping ratio** ζ is defined as

$$\zeta = -\frac{\sigma}{\omega_\mathrm{n}}, \tag{7.91}$$

where σ and ω_n are obtained from the pole locations.

Definition 7.8. The **peak time** T_p is the time at which the step response reaches its first maximum. If the response is non-oscillatory, then $T_\mathrm{p} = \infty$.

Definition 7.9. The **percent overshoot** %OS is defined as

$$\%\mathrm{OS} = \frac{y_\mathrm{p} - y_\infty}{y_\infty} \times 100. \tag{7.92}$$

Pole-Based Interpretation. For a system with poles at

$$s = \sigma \pm \jmath\omega, \tag{7.93}$$

the response contains terms of the form $e^{\sigma t}\cos(\omega t), \quad e^{\sigma t}\sin(\omega t)$. Thus,

- σ determines the rate of decay (transient speed),
- ω determines the oscillation frequency,
- ζ determines the relative dominance of decay vs oscillation.

7.9.1 System Specifications for First-Order Systems

Consider a first-order system

$$G(s) = \frac{1}{s + \lambda}, \tag{7.94}$$

where $\lambda > 0$ ensures stability.

The impulse response is

$$y_\delta(t) = e^{-\lambda t}, \tag{7.95}$$

and the step response is

$$y(t) = \frac{1}{\lambda}(1 - e^{-\lambda t}). \tag{7.96}$$

Fact 7.9.1. For a first-order system,

$$\tau = \frac{1}{\lambda}, \tag{7.97}$$

$$T_\mathrm{r} \approx \frac{2.2}{\lambda}, \tag{7.98}$$

$$T_\mathrm{s} \approx \frac{4}{\lambda}, \tag{7.99}$$

$$\omega_\mathrm{n} = 0, \tag{7.100}$$

$$T_\mathrm{p} = \infty, \tag{7.101}$$

$$\%\mathrm{OS} = 0. \tag{7.102}$$

Remark. First-order systems do not exhibit oscillatory behavior. All transient dynamics are governed by a single real pole.

7.9.2 System Specifications for Second-Order Systems

Consider the standard second-order system

$$G(s) = \frac{\omega_\mathrm{n}^2}{s^2 + 2\zeta\omega_\mathrm{n} s + \omega_\mathrm{n}^2}. \tag{7.103}$$

The poles are located at

$$s = -\zeta\omega_\mathrm{n} \pm \jmath\omega_\mathrm{d}, \tag{7.104}$$

where

$$\omega_\mathrm{d} \triangleq \omega_\mathrm{n}\sqrt{1-\zeta^2}. \tag{7.105}$$

The step response is

$$y(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_\mathrm{n} t}\cos(\omega_\mathrm{d} t - \phi), \tag{7.106}$$

with

$$\phi = \arctan\left(\frac{\zeta}{\sqrt{1-\zeta^2}}\right). \tag{7.107}$$

Fact 7.9.2. For an underdamped second-order system $(0 < \zeta < 1)$,

$$T_\mathrm{p} = \frac{\pi}{\omega_\mathrm{d}}, \tag{7.108}$$

$$\%\mathrm{OS} = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100, \tag{7.109}$$

$$T_\mathrm{s} \approx \frac{4}{\zeta\omega_\mathrm{n}}. \tag{7.110}$$

Remark. Unlike the first-order case, second-order systems exhibit a trade-off between speed of response and oscillatory behavior, governed by the damping ratio ζ.

Important Observation.

- Increasing ω_n speeds up the response.
- Increasing ζ reduces oscillations but may slow the response.

Chapter 8

Complex Plane Methods for Control Analysis and Design

The behavior of dynamical systems is fundamentally determined by the location of their poles and zeros in the complex plane. In this chapter, we develop tools that exploit this structure to analyze and design control systems. In particular, we study how the system's response evolves as parameters vary (root locus), how it responds to sinusoidal inputs across frequencies (frequency response), and how closed-loop stability can be assessed using contour-based arguments (Nyquist criterion). These perspectives provide complementary insights. The root locus describes how closed-loop poles move in the complex plane, the frequency response characterizes the input-output behavior along the imaginary axis, and the Nyquist criterion relates the open-loop frequency response to closed-loop stability. Together, they form a unified framework for understanding stability, performance, and robustness, and for relating complex-plane behavior to time-domain specifications.

8.1 Root Locus

The root locus of a transfer function is used to analyze how the closed-loop poles of a feedback system change as a proportional gain is varied. In a basic servo loop, the gain K is varied from 0 to ∞. The resulting trajectories of the closed-loop poles allow us:

1. to assess robustness to parametric variations,
2. to compute the gain margin,
3. to understand why some plants are difficult to stabilize.

8.1.1 Closed-Loop Poles and Definition

Consider the basic feedback loop. The closed-loop poles are the poles of the sensitivity function

$$S(s) = \frac{1}{1 + L(s)}, \tag{8.1}$$

where the loop transfer function is

$$L(s) \triangleq G_{\mathrm{c}}(s)G(s). \tag{8.2}$$

The **root locus** is the set of all $s \in \mathbb{C}$ such that

$$L(s) = -1. \tag{8.3}$$

Writing $L(s)$ in zero-pole-gain form,

$$L(s) = K\frac{(s-z_1)\cdots(s-z_m)}{(s-p_1)\cdots(s-p_n)}, \tag{8.4}$$

we obtain

$$S(s) = \frac{(s-p_1)\cdots(s-p_n)}{(s-p_1)\cdots(s-p_n)+K(s-z_1)\cdots(s-z_m)}, \tag{8.5}$$

which implies that

1. The open-loop poles of $L(s)$ are the zeros of $S(s)$,
2. The gain K determines the closed-loop pole locations,
3. As $K \to 0$, the poles of $S(s)$ approach the poles of $L(s)$,
4. As $K \to \infty$, the poles of $S(s)$ approach the zeros of $L(s)$.

Thus, the closed-loop poles move from the open-loop poles to the open-loop zeros as K increases.

8.1.2 Root Locus Properties

The root locus satisfies the following properties.

1. The root locus is symmetric about the real axis.
2. Each open-loop zero attracts one branch of the root locus.
3. If $n > m$, then $n - m$ branches go to infinity.
4. The asymptotes intersect the real axis at

$$\alpha \triangleq \frac{1}{n-m}\left(\sum_{i=1}^{n} p_i - \sum_{i=1}^{m} z_i\right). \tag{8.6}$$

5. The asymptote angles are

$$\phi_i = \frac{180^\circ + (i-1)360^\circ}{n-m}, \quad i \in \{1, \ldots, n-m\}. \tag{8.7}$$

6. The root locus lies on real-axis segments to the left of an odd number of real poles and zeros.
7. The departure angle from the pole p_k is

$$\psi_k^{\mathrm{d}} = \sum_{i=1}^{m} \angle(p_k - z_i) - \sum_{j=1, j\neq k}^{n} \angle(p_k - p_j) \pm 180^\circ. \tag{8.8}$$

8. The arrival angle at the zero z_k is

$$\psi_k^{\mathrm{a}} = \sum_{i=1, i\neq k}^{m} \angle(z_k - z_i) - \sum_{j=1}^{n} \angle(z_k - p_j) \pm 180^\circ. \tag{8.9}$$

   If a pole or zero has multiplicity ℓ, the corresponding angle is divided by ℓ.

These properties allow for a qualitative sketch of the root locus without explicitly computing the closed-loop poles.

8.1.3 Closed-Loop Stability using Root Locus

The root locus represents all possible closed-loop poles. If any portion of the root locus lies in the open right-half plane, then the system becomes unstable for some value of K.

To determine the imaginary-axis crossing, let $s = \jmath\omega$. Then,

$$KG(\jmath\omega) = -1. \tag{8.10}$$

The pair (K, ω) satisfying this equation gives the crossing point and the corresponding gain.

8.1.4 Gain Margin

The gain margin is the maximum multiplicative increase in gain before instability. Since

$$S(s) = \frac{1}{1 + KL(s)}, \tag{8.11}$$

at instability we have

$$1 + K_{\max} L(\jmath\omega) = 0. \tag{8.12}$$

The gain margin is

$$\mathrm{GM} \triangleq 20 \log_{10} \left(\frac{K_{\max}}{K} \right) \text{ dB}. \tag{8.13}$$

Example 8.1.1. Consider the system

$$G(s) = \frac{s^4 + 10s^3 + 30s^2 + 10s - 51}{s^5 + 15s^4 + 86s^3 + 248s^2 + 368s + 192}. \tag{8.14}$$

Figure 8.1 shows the root locus of $G(s)$. Note that the root locus is symmetric about the real axis, four closed-loop poles go towards the four open-loop zeros of the system, and one excess pole goes to infinity along the asymptote. The center of the root locus is -1.5 and its angle from the real axis is 180 degrees. Finally, note that the root locus includes the real axis to the left of the odd-numbered real poles and real zeros.

The locus crosses into the open right-half plane at $s = 0$. Thus,

$$1 + KG(0) = 0 \quad \Rightarrow \quad K = \frac{192}{51}. \tag{8.15}$$

The gain margin is thus $\mathrm{GM} = 11.51$ dB.

8.1.5 Dominant Pole Approximation

Consider a higher-order system with multiple closed-loop poles. If a pair of poles lies significantly closer to the imaginary axis than the remaining poles, then the system response is dominated by this pair. In this case, the system can be approximated by a second-order system whose dynamics are determined by the dominant poles. This approximation allows us to

1. analyze transient response using second-order system formulas,
2. relate pole locations to time-domain specifications such as rise time and settling time,
3. simplify controller design.

The approximation is valid when the non-dominant poles are sufficiently far into the left-half plane.

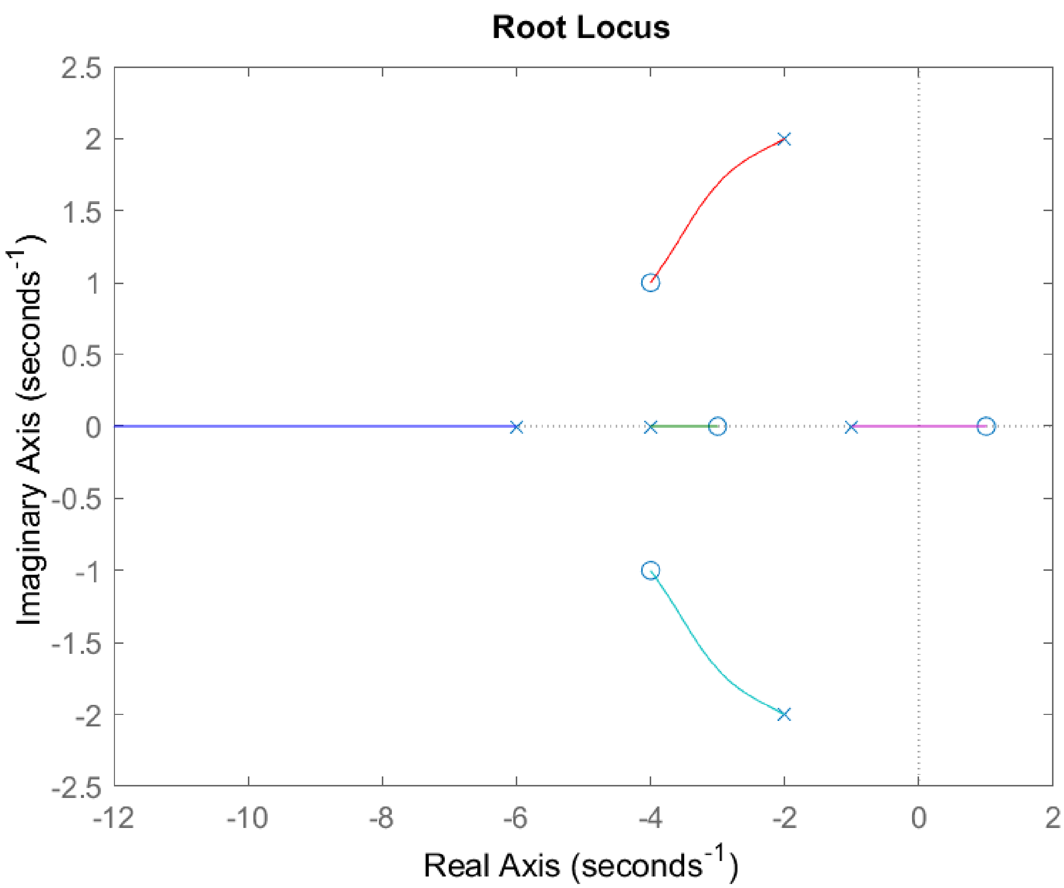

Figure 8.1: Root locus of $G(s)$ given by (**??**).

8.1.6 Design Interpretation

The root locus provides a graphical tool for selecting the gain K such that the closed-loop system satisfies the desired performance specifications. In particular, the gain K can be chosen such that

1. all closed-loop poles lie in the open left-half plane (stability),
2. the dominant poles are placed to achieve the desired transient response,
3. the system maintains adequate robustness to gain variations.

Thus, the root locus bridges the relationship between system parameters and closed-loop performance.

8.2 Frequency Response

8.2.1 Motivating Example: Human Perception of Frequency

The frequency response describes how physical systems respond to inputs at different frequencies. As an example, consider the human auditory system. The human ear can perceive sound roughly in the frequency range of 20 Hz to 20,000 Hz. Within this range, different frequencies are perceived differently: low-frequency sounds are heard as bass, while high-frequency sounds are perceived as treble. Frequencies outside this range are effectively attenuated by the auditory system and are not perceived.

This behavior is the frequency response of the auditory system. The ear amplifies certain frequencies, attenuates others, and introduces phase shifts in the processing of signals. This behavior arises because the auditory system is a dynamical system whose response depends on the input's frequency content.

This motivates the study of frequency response in engineering systems. Given a system described by a transfer function $G(s)$, we are interested in understanding how the system responds to inputs of different frequencies. In particular, we ask: how does the magnitude and phase of the output depend on the frequency of the input signal?

8.2.2 Preliminaries

Fact 8.2.1 (Euler's Identity). For any $\theta \in \mathbb{R}$,

$$e^{\jmath\theta} = \cos\theta + \jmath\sin\theta. \tag{8.16}$$

Consequently,

$$\Re(e^{\jmath\theta}) = \cos\theta, \quad \Im(e^{\jmath\theta}) = \sin\theta, \tag{8.17}$$

where $\Re(z)$ is the real part of z and $\Im(z)$ is the imagninary part of z.

Fact 8.2.2 (Trigonometric Functions in Complex Form). The sine and cosine functions can be written as

$$\cos\theta = \frac{1}{2}\left(e^{\jmath\theta} + e^{-\jmath\theta}\right), \quad \sin\theta = \frac{1}{2\jmath}\left(e^{\jmath\theta} - e^{-\jmath\theta}\right). \tag{8.18}$$

Conversely,

$$\cos(\omega t) = \Re(e^{\jmath\omega t}), \quad \sin(\omega t) = \Re(-\jmath e^{\jmath\omega t}). \tag{8.19}$$

Fact 8.2.3 (Amplitude–Phase Representation). Any sinusoid can be written in amplitude–phase form as

$$\cos(\omega t + \phi) = \Re\left(e^{\jmath\phi}e^{\jmath\omega t}\right). \tag{8.20}$$

Fact 8.2.4 (Polar Representation of Complex Numbers). Any complex number $z \in \mathbb{C}$ can be expressed in polar form as

$$z = |z|e^{\jmath\angle z}, \tag{8.21}$$

where the magnitude $|z|$ and phase $\angle z$ are defined as

$$|z| \triangleq \sqrt{(\Re\{z\})^2 + (\Im\{z\})^2}, \quad \angle z \triangleq \tan^{-1}\left(\frac{\Im\{z\}}{\Re\{z\}}\right). \tag{8.22}$$

The phase can equivalently be defined as $\angle z \triangleq \text{atan2}(\Im(z), \Re(z)$, which correctly accounts for the quadrant.

Definition 8.2.1 (Eigenfunctions of a Transfer Function). Let $G(s)$ denote the transfer function of a linear time-invariant system. The input $u(t)$ and output $y(t)$ satisfy

$$Y(s) = G(s)\,U(s), \tag{8.23}$$

where $U(s)$ and $Y(s)$ are the Laplace transforms of $u(t)$ and $y(t)$, respectively. In the time domain, this defines a mapping $u(t) \longrightarrow y(t)$. A function $u(t)$ is called an eigenfunction of the system if the corresponding output satisfies

$$y(t) = \lambda\, u(t), \tag{8.24}$$

where $\lambda \in \mathbb{C}$ is the corresponding eigenvalue.

Fact 8.2.5 (Complex Exponentials are Eigenfunctions)**.** For an asymptotically stable transfer function $G(s)$, the complex exponential $e^{\jmath\omega t}$ is an eigenfunction. The corresponding eigenvalue is $G(\jmath\omega)$, that is, if $u(t) = e^{\jmath\omega t}$, then

$$y(t) = G(\jmath\omega)\, e^{\jmath\omega t}. \tag{8.25}$$

Proof. Consider the input $u(t) = e^{\jmath\omega t}$. Its Laplace transform is

$$U(s) = \frac{1}{s - \jmath\omega},$$

for $\Re(s) > 0$. The output in the Laplace domain is

$$Y(s) = G(s)\, U(s) = \frac{G(s)}{s - \jmath\omega}.$$

Note that the term $\frac{1}{s-\jmath\omega}$ has a pole at $s = \jmath\omega$, which corresponds to the time-domain signal $e^{\jmath\omega t}$.

Since $G(s)$ is finite at $s = \jmath\omega$, it follows that

$$G(s) = G(\jmath\omega) + (s - \jmath\omega)\, G_1(s),$$

where $G_1(s)$ is finite at $s = \jmath\omega$. Since $G(s)$ is asymptotically stable, $G_1(s)$ is also asymptotically stable. Substituting into $Y(s)$ gives

$$Y(s) = \frac{G(\jmath\omega)}{s - \jmath\omega} + G_1(s).$$

The inverse Laplace transform yields

$$y(t) = G(\jmath\omega)\, e^{\jmath\omega t} + y_{\mathrm{tr}}(t),$$

where $y_{\mathrm{tr}}(t)$ is the transient response corresponding to $G_1(s)$. Since the system is stable, the transient term $y_{\mathrm{tr}}(t)$ decays to zero as $t \to \infty$. Therefore, the steady-state response is

$$y_{\mathrm{ss}}(t) = G(\jmath\omega)\, e^{\jmath\omega t}.$$

Thus, the input is scaled by $G(\jmath\omega)$ without a change in functional form, and hence $e^{\jmath\omega t}$ is an eigenfunction with eigenvalue $G(\jmath\omega)$. □

Fact 8.2.6. For any $z \in \mathbb{C}$ and $\omega \in \mathbb{R}$, the complex number $\jmath\omega + z$ can be written in polar form as

$$\jmath\omega + z = r_z e^{\jmath\theta_z}, \tag{8.26}$$

where $r_z = |\jmath\omega + z|$, and $\theta_z = \arg(\jmath\omega + z)$. Geometrically, r_z and θ_z are the magnitude and angle of the vector from the point $-z$ to the point $\jmath\omega$ in the complex plane, as shown in Figure 8.2.

8.2.3 Harmonic Response

Consider an asymptotically stable LTI system with transfer function $G(s)$. Consider the input

$$u(t) = \sin(\omega t). \tag{8.27}$$

It follows from Fact (8.2.1) that

$$u(t) = \Re\left\{-\jmath e^{\jmath\omega t}\right\}. \tag{8.28}$$

Define the complex input

$$\tilde{u}(t) \triangleq -\jmath e^{\jmath\omega t}, \tag{8.29}$$

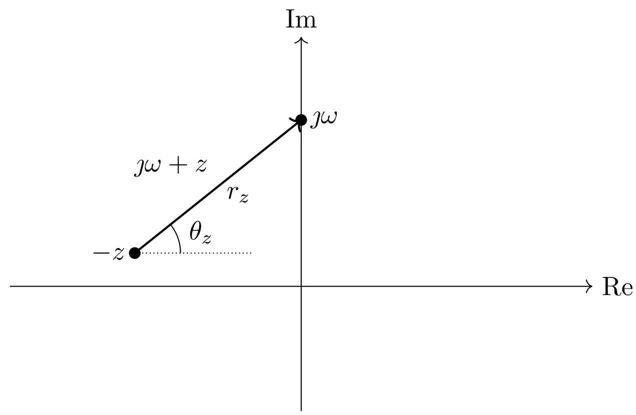

Figure 8.2: Geometric interpretation of $\jmath\omega + z$ in the complex plane. The vector from $-z$ to $\jmath\omega$ is $\jmath\omega + z$, whose magnitude is r_z and whose angle is θ_z.

which implies that $u(t) = \Re\{\tilde{u}(t)\}$. It follows from Fact 8.2.5 that the steady-state response to $\tilde{u}(t)$ is

$$\tilde{y}_{\rm ss}(t) = G(\jmath\omega)\,(-\jmath)e^{\jmath\omega t}. \tag{8.30}$$

Writing $G(\jmath\omega) = M(\omega)e^{\jmath\phi(\omega)}$, where $M(\omega) \triangleq |G(\jmath\omega)|$, and $\phi(\omega) \triangleq \angle G(\jmath\omega)$, it follows that

$$\tilde{y}_{\rm ss}(t) = M(\omega)e^{\jmath(\omega t + \phi(\omega) - \pi/2)}, \tag{8.31}$$

whose real part is

$$y_{\rm ss}(t) = \Re\left\{\tilde{y}_{\rm ss}(t)\right\} = M(\omega)\cos\left(\omega t + \phi(\omega) - \frac{\pi}{2}\right) = M(\omega)\sin(\omega t + \phi(\omega)). \tag{8.32}$$

Therefore, the steady-state response of a stable LTI system to the input $u(t) = \sin(\omega t)$ is a sinusoid at the same frequency, whose amplitude is $|G(\jmath\omega)|$ and whose phase shift is $\angle G(\jmath\omega)$.

8.2.4 Evaluating Frequency Response

As shown in Section 8.2.3, the steady-state response of a transfer function $G(s)$ to a harmonic input $\sin(\omega t)$ is

$$y_{\rm ss}(t) = M(\omega)\sin(\omega t + \phi(\omega)), \tag{8.33}$$

where $M(\omega) = |G(\jmath\omega)|$, and $\phi(\omega) = \angle G(\jmath\omega)$. Thus, the steady-state magnitude and phase shift at frequency ω can be obtained by evaluating the transfer function at $s = \jmath\omega$. In this section, we develop a graphical method to evaluate $G(\jmath\omega)$.

Writing the transfer function as

$$G(s) = K\frac{(s+z_1)(s+z_2)\cdots(s+z_m)}{(s+p_1)(s+p_2)\cdots(s+p_n)}, \tag{8.34}$$

where $-z_1, \ldots, -z_m \in \mathbb{C}$ are the zeros and $-p_1, \ldots, -p_n \in \mathbb{C}$ are the poles of $G(s)$, we evaluate at $s = \jmath\omega$ to obtain

$$G(\jmath\omega) = K\frac{(\jmath\omega+z_1)(\jmath\omega+z_2)\cdots(\jmath\omega+z_m)}{(\jmath\omega+p_1)(\jmath\omega+p_2)\cdots(\jmath\omega+p_n)}. \tag{8.35}$$

It follows from Fact 8.2.6 that each term $\jmath\omega + x$ can be expressed in polar form as

$$\jmath\omega + x = r_x e^{\jmath\theta_x}, \tag{8.36}$$

where r_x is the magnitude and θ_x is the angle of the vector from $-x$ to $\jmath\omega$ in the complex plane. Consequently,

$$G(\jmath\omega) = K \frac{\Pi_{i=1}^{m} r_{z_i}}{\Pi_{j=1}^{n} r_{p_j}} \; e^{\jmath\left(\sum_{i=1}^{m} \theta_{z_i} - \sum_{j=1}^{n} \theta_{p_j}\right)}. \tag{8.37}$$

and thus

$$M(\omega) = |K| \frac{\Pi_{i=1}^{m} r_{z_i}}{\Pi_{j=1}^{n} r_{p_j}}, \tag{8.38}$$

$$\phi(\omega) = \angle K + \sum_{i=1}^{m} \theta_{z_i} - \sum_{j=1}^{n} \theta_{p_j}. \tag{8.39}$$

Definition 8.2.2. The **DC gain** of $G(s)$ is the steady-state gain at zero frequency, defined as

$$\text{DC gain} \triangleq G(0) = K \frac{\Pi_{i=1}^{m} z_i}{\Pi_{j=1}^{n} p_j}. \tag{8.40}$$

The corresponding phase at $\omega = 0$ is either $0°$ or $180°$, depending on the sign of $G(0)$.

8.2.5 Bode Plot

The frequency response of a system is given by evaluating the transfer function at $s = \jmath\omega$. Note that

$$G(\jmath\omega) = M(\omega) e^{\jmath\phi(\omega)}, \tag{8.41}$$

where $M(\omega) = |G(\jmath\omega)|$ is the magnitude and $\phi(\omega) = \angle G(\jmath\omega)$ is the phase. The Bode plot provides a convenient way to visualize $M(\omega)$ and $\phi(\omega)$ as functions of frequency.

Definition 8.2.3. The **Bode plot** consists of

- the **magnitude plot**, which shows $20 \log_{10} M(\omega)$ versus $\log \omega$, and
- the **phase plot**, which shows $\phi(\omega)$ versus $\log \omega$.

The magnitude is expressed in decibels (dB), the frequency ω is expressed in rad/s on a logarithmic scale, and the phase is expressed in degrees.

Geometric Interpretation

Consider a transfer function written in zero-pole-gain form, that is,

$$G(s) = K \frac{\prod_i (s - z_i)}{\prod_j (s - p_j)}. \tag{8.42}$$

Evaluating at $s = \jmath\omega$ yields

$$G(\jmath\omega) = K \frac{\prod_i (\jmath\omega - z_i)}{\prod_j (\jmath\omega - p_j)}. \tag{8.43}$$

Thus,

- the magnitude $M(\omega)$ is the product of distances from the zeros divided by the product of distances from the poles,
- the phase $\phi(\omega)$ is the sum of angles from the zeros minus the sum of angles from the poles.

Logarithmic Representation

Using the logarithmic property $\log(ab) = \log(a) + \log(b)$, the logarithm of (8.43) yields

$$20\log_{10} M(\omega) = 20\log_{10}|K| + \sum_i 20\log_{10}|\jmath\omega - z_i| - \sum_j 20\log_{10}|\jmath\omega - p_j|. \tag{8.44}$$

Remark 8.2.1. The Bode magnitude plot converts multiplication into addition. This decomposition allows the Bode plot to be constructed by summing the contributions of simple building blocks.

Basic Building Blocks

We now study the Bode plots of a few basic systems.

Example 8.2.1. Consider the system $G(s) = K$, where K is a constant gain. The frequency response is $G(\jmath\omega) = K$, and thus

$$M(\omega) = |K|, \tag{8.45}$$

$$\phi(\omega) = \begin{cases} 0, & K > 0, \\ 180, & K < 0. \end{cases} \tag{8.46}$$

Note that the magnitude plot is constant at $20\log_{10}|K|$, and the phase plot is constant.

Example 8.2.2. Consider the system $G(s) = s$. Then, $M(\omega) = \omega$ and $\phi(\omega) = 90°$. The magnitude plot has a slope of +20 dB/decade for all ω.

Example 8.2.3. Consider the system $G(s) = \frac{1}{s}$. Then, $M(\omega) = \frac{1}{\omega}$ and $\phi(\omega) = -90°$. The magnitude plot has a slope of −20 dB/decade for all ω.

Example 8.2.4. Consider the system $G(s) = s+z$, where $z > 0$. The frequency response is $G(\jmath\omega) = \jmath\omega + z$, and thus

$$M(\omega) = \sqrt{\omega^2 + z^2}, \tag{8.47}$$

$$\phi(\omega) = \tan^{-1}\left(\frac{\omega}{z}\right). \tag{8.48}$$

Note that

1. For $\omega \ll z$, $M(\omega) \approx z$, and thus the magnitude plot $20\log_{10} M(\omega)$ has slope 0 dB/decade.
2. For $\omega \gg z$, $M(\omega) \approx \omega$, and thus the magnitude plot $20\log_{10} M(\omega)$ has slope +20 dB/decade.
3. For $\omega \ll z$, $\phi(\omega) \approx 0$, whereas for $\omega \gg z$, $\phi(\omega) \approx 90$ degrees.

Example 8.2.5. Consider the system

$$G(s) = \frac{1}{s+p}, \tag{8.49}$$

where $p > 0$. The frequency response is $G(\jmath\omega) = \frac{1}{\jmath\omega + p}$, and thus

$$M(\omega) = \frac{1}{\sqrt{\omega^2 + p^2}}, \tag{8.50}$$

$$\phi(\omega) = -\tan^{-1}\left(\frac{\omega}{p}\right). \tag{8.51}$$

Note that

1. For $\omega \ll p$, $M(\omega) \approx \frac{1}{p}$, and thus the magnitude plot $20\log_{10} M(\omega)$ has slope 0 dB/decade.
2. For $\omega \gg p$, $M(\omega) \approx \frac{1}{\omega}$, and thus the magnitude plot $20\log_{10} M(\omega)$ has slope −20 dB/decade.
3. For $\omega \ll p$, $\phi(\omega) \approx 0$, whereas for $\omega \gg p$, $\phi(\omega) \approx -90$ degrees.

Break Frequency

Definition 8.2.4. The **break frequency** is a frequency associated with a pole or zero of the transfer function, used to identify where the slope of the asymptotic Bode magnitude plot changes. For a real pole or zero at $s = -a$ with $a > 0$,

$$\omega_b = a. \tag{8.52}$$

For a complex conjugate pair at $s = -\sigma \pm \jmath\omega_d$,

$$\omega_b = \sqrt{\sigma^2 + \omega_d^2}. \tag{8.53}$$

Remark 8.2.2. The break frequency serves as a convenient reference point for constructing asymptotic Bode plots. It is introduced for ease of analysis and does not represent an intrinsic feature of the system response.

Asymptotic Approximation

Instead of evaluating exact expressions, the Bode plot can be approximated using asymptotic lines.

Definition 8.2.5. The **low-frequency asymptote** is the asymptotic behavior as $\omega \to 0$.

Definition 8.2.6. The **high-frequency asymptote** is the asymptotic behavior as $\omega \to \infty$.

Bode Plot Approximation Rules

Fact 8.2.7. Each real zero contributes +20 dB/decade to the magnitude slope and approximately $+90^\circ$ phase shift.

Fact 8.2.8. Each real pole contributes −20 dB/decade to the magnitude slope and approximately -90° phase shift.

Fact 8.2.9. Each complex conjugate zero pair contributes +40 dB/decade and approximately $+180^\circ$ phase shift.

Fact 8.2.10. Each complex conjugate pole pair contributes −40 dB/decade and approximately -180° phase shift.

Fact 8.2.11. A constant gain shifts the magnitude plot vertically by $20 \log_{10} |K|$.

Fact 8.2.12. Each phase transition occurs gradually over approximately two decades centered at the break frequency.

Procedure for Sketching a Bode Plot

The following step-by-step procedure is useful for constructing an approximate Bode plot by hand.

1. Factor the transfer function into standard Bode form:

$$G(s) = K \frac{\prod_i (s + z_i)}{\prod_j (s + p_j)}, \tag{8.54}$$

   where K is the constant gain, z_i are zeros, and p_j are poles.

2. Identify and mark all break frequencies $\omega_\mathrm{b} = z_i$, and $\omega_\mathrm{b} = p_j$ on the logarithmic frequency axis. For a complex conjugate pole (or zero) pair of the form

$$s^2 + 2\zeta\omega_n s + \omega_n^2, \tag{8.55}$$

   the break frequency is $\omega_\mathrm{b} = \omega_n$.

3. Determine the DC gain of the transfer functions, that is, $G(0)$.
4. Construct the magnitude plot from left to right. Begin with the initial slope, then modify the slope at each break frequency as follows:
   - Add +20 dB/decade for each real zero,
   - subtract 20 dB/decade for each real pole,
   - add +40 dB/decade for each complex conjugate zero pair,
   - subtract 40 dB/decade for each complex conjugate pole pair.
5. Construct the phase plot from left to right.
   - each real zero contributes approximately $+90°$,
   - each real pole contributes approximately $-90°$,
   - each complex conjugate zero pair contributes approximately $+180°$,
   - each complex conjugate pole pair contributes approximately $-180°$,
   - each contribution changes gradually over approximately two decades centered at its break frequency.

Remark 8.2.3. A convenient way to construct a Bode plot is by superposition. In particular, the Bode plot can be approximated by summing the individual contributions of each pole, zero, and gain factor.

Example 8.2.6. Consider the system

$$G(s) = K\frac{s+z}{s+p}, \tag{8.56}$$

where $p > 0$ and $z > 0$. The frequency response is

$$G(\jmath\omega) = K\frac{\jmath\omega+z}{\jmath\omega+p}, \tag{8.57}$$

and thus

$$20\log_{10} M(\omega) = 20\log_{10}|K| + 20\log_{10}\sqrt{\omega^2+z^2} - 20\log_{10}\sqrt{\omega^2+p^2}, \tag{8.58}$$

$$\phi(\omega) = \tan^{-1}\left(\frac{\omega}{z}\right) - \tan^{-1}\left(\frac{\omega}{p}\right). \tag{8.59}$$

Note that the behavior depends on the relative magnitude of z and p.

1. **Case 1: $z < p$.**
   (a) For $\omega \ll z$, $M(\omega) \approx \dfrac{|K|z}{p}$, and the magnitude plot has slope 0 dB/decade.
   (b) For $z \ll \omega \ll p$, $M(\omega) \approx \dfrac{|K|\omega}{p}$, and the magnitude plot has slope +20 dB/decade.
   (c) For $\omega \gg p$, $M(\omega) \approx |K|$, and the magnitude plot has slope 0 dB/decade.
   (d) For $\omega \ll z$, $\phi(\omega) \approx 0$, and for $\omega \gg p$, $\phi(\omega) \approx 0$. Between the break frequencies, the phase increases from 0 to 90 degrees (due to the zero at $-z$) and then decreases from 90 back to 0 degrees (due to the pole at $-p$).
2. **Case 2: $p < z$.**
   (a) For $\omega \ll p$, $M(\omega) \approx \dfrac{|K|z}{p}$, and the magnitude plot has slope 0 dB/decade.

(b) For $p \ll \omega \ll z$, $M(\omega) \approx \dfrac{|K|z}{\omega}$, and the magnitude plot has slope -20 dB/decade.

(c) For $\omega \gg z$, $M(\omega) \approx |K|$, and the magnitude plot has slope 0 dB/decade.

(d) For $\omega \ll p$, $\phi(\omega) \approx 0$, and for $\omega \gg z$, $\phi(\omega) \approx 0$. Between the break frequencies, the phase decreases from 0 to -90 degrees (due to the pole at $-p$) and then increases from -90 back to 0 degrees (due to the zero at $-z$).

3. The break frequencies are $\omega = z$ and $\omega = p$.

Note that the Bode magnitude and the phase plot are thus the sum of the Bode magnitude and the phase plots in Examples 8.2.1, 8.2.4, and 8.2.5.

8.2.6 Bode plot of a second-order system with Complex Poles

Consider the system

$$G(s) = \frac{\omega_{\rm n}^2}{s^2 + 2\zeta\omega_{\rm n} s + \omega_n^2}. \tag{8.60}$$

Note that

$$G(\jmath\omega) = \frac{\omega_{\rm n}^2}{\omega_n^2 - \omega^2 + 2\zeta\omega_{\rm n}\omega\jmath}, \tag{8.61}$$

and thus

$$M(\omega) = \frac{\omega_{\rm n}^2}{\sqrt{(\omega_n^2 - \omega^2)^2 + 4\zeta^2\omega_{\rm n}^2\omega^2}}, \tag{8.62}$$

$$\phi(\omega) = -\tan^{-1}\frac{2\zeta\omega_{\rm n}\omega}{\omega_n^2 - \omega^2} \tag{8.63}$$

For $\omega << \omega_{\rm n}$, $M(\omega) \approx 1$, and thus the low-frequency aysmptote is $\log M(\omega) = 0$. For $\omega >> \omega_{\rm n}$, $M(\omega) \approx \omega_{\rm n}^2/\omega^2$, and thus the slope of the high-frequency asymptote is -40 dB/decade. Finally, for $\omega = \omega_{\rm n}$,

$$M(\omega_{\rm n}) = \frac{1}{2\zeta}. \tag{8.64}$$

Definition 8.2.7. The **peak magnitude** is the maximum value of the frequency response magnitude, that is,

$$M_{\rm p} = \max_{\omega \geq 0} M(\omega). \tag{8.65}$$

Definition 8.2.8. The **peak frequency** is the frequency at which the peak magnitude is attained, that is,

$$\omega_{\rm p} = \arg\max_{\omega \geq 0} M(\omega). \tag{8.66}$$

For the second-order system (8.60), the peak frequency is obtained by setting

$$\frac{{\rm d}M(\omega)}{{\rm d}\omega} = -\frac{1}{2}\frac{\omega_{\rm n}^2\left(-4\omega(\omega_n^2 - \omega^2) + 8\zeta^2\omega_{\rm n}^2\omega\right)}{\left((\omega_n^2 - \omega^2)^2 + 4\zeta^2\omega_{\rm n}^2\omega^2\right)^{3/2}} = 0, \tag{8.67}$$

which yields

$$\omega^3 + \omega\omega_n^2(2\zeta^2 - 1) = 0. \tag{8.68}$$

The solutions of (8.68) are $\omega = 0$ and $\omega = \omega_n\sqrt{1-2\zeta^2}$. Thus, a nonzero peak frequency exists only if

$$1 - 2\zeta^2 > 0, \tag{8.69}$$

that is, when

$$\zeta < \frac{1}{\sqrt{2}}. \tag{8.70}$$

For $\zeta < 1/\sqrt{2}$, substituting $\omega = \omega_\mathrm{p} = \omega_n\sqrt{1-2\zeta^2}$ into $M(\omega)$ yields the peak magnitude

$$M_\mathrm{p} = \frac{1}{2\zeta\sqrt{1-\zeta^2}}. \tag{8.71}$$

Note that $M_\mathrm{p} \to \infty$ as $\zeta \to 0$. This corresponds to resonance in the system, where the response to inputs near the natural frequency becomes unbounded in the absence of damping. For $\zeta \geq 1/\sqrt{2}$, the magnitude does not exhibit a nonzero peak, and thus $\omega_\mathrm{p} = 0$, and $M_\mathrm{p} = M(0) = 1$.

Thus, the presence of a peak in the magnitude response is determined entirely by the damping ratio ζ.

8.2.7 Stability Margins from Bode Plot

Consider the basic servo loop with loop transfer function

$$L(s) = G_\mathrm{c}(s)G(s). \tag{8.72}$$

The sensitivity transfer function is

$$S(s) = \frac{1}{1 + L(s)}. \tag{8.73}$$

The poles of the closed-loop system are the zeros of $1 + L(s)$.

Fact 8.2.13. The closed-loop system has a pole at $s = s_0$ if

$$1 + L(s_0) = 0. \tag{8.74}$$

In particular, if for some frequency ω,

$$1 + L(\jmath\omega) = 0, \tag{8.75}$$

then the closed-loop system has a pole on the imaginary axis.

Equivalently, a pole on the imaginary axis occurs when

$$L(\jmath\omega) = -1 = e^{-\jmath\pi}, \tag{8.76}$$

that is,

- the magnitude is $|L(\jmath\omega)| = 1$ (0 dB), and
- the phase is $-180°$.

Thus, if the Bode plot crosses 0 dB at the same frequency where the phase reaches $-180°$, the closed-loop system has poles on the imaginary axis.

Gain Margin

Definition 8.2.9. The **gain crossover frequency**, denoted $\omega_{\rm gc}$, is the frequency at which

$$|L(\jmath\omega_{\rm gc})| = 1. \tag{8.77}$$

Definition 8.2.10. The **phase crossover frequency**, denoted $\omega_{\rm pc}$, is the frequency at which

$$\angle L(\jmath\omega_{\rm pc}) = -180^\circ. \tag{8.78}$$

Definition 8.2.11. The **gain margin** is the amount of gain increase required to bring the closed-loop poles of the system to the imaginary axis, that is,

$$\mathrm{GM} = \frac{1}{|L(\jmath\omega_{\rm pc})|}. \tag{8.79}$$

In decibels,

$$\mathrm{GM}_{\rm dB} = -20\log_{10}|L(\jmath\omega_{\rm pc})|. \tag{8.80}$$

Phase Margin

Definition 8.2.12. The **phase margin** is the additional phase lag required at the gain crossover frequency to bring the closed-loop poles of the system to the imaginary axis, that is,

$$\mathrm{PM} = 180^\circ + \angle L(\jmath\omega_{\rm gc}). \tag{8.81}$$

Remark 8.2.4. Larger gain margin and phase margin generally indicate greater robustness to modeling uncertainty and parameter variations.

8.2.8 Time Delay as a Pure Phase Shift

Consider a system with a pure time delay,

$$y(t) = u(t-T) \tag{8.82}$$

Taking the Laplace transform yields

$$Y(s) = e^{-sT}U(s), \tag{8.83}$$

so the transfer function is

$$G(s) = e^{-sT} \tag{8.84}$$

Evaluating on the imaginary axis yields

$$G(j\omega) = e^{-j\omega T} \tag{8.85}$$

The frequency response of the transfer function of the time-delay satisfies

$$|G(j\omega)| = 1 = 0 \text{ dB} \tag{8.86}$$

$$\angle G(j\omega) = -\omega T \tag{8.87}$$

Thus, a pure time delay does not affect the magnitude but introduces a phase lag that increases linearly with frequency.

Implications for Control

- A time delay introduces a phase lag of $-\omega T$, which decreases without bound as $\omega \to \infty$.
- Consequently, the phase will necessarily cross all multiples of -180°.
- For a high-gain-stable (HGS) system, the phase does not cross -180°, and the gain margin is infinite.
- Introducing any nonzero time delay guarantees a -180° phase crossing.
- Therefore, any nonzero time delay renders the gain margin finite.

8.2.9 Nyquist Plot

Like the Bode plot, the Nyquist plot represents the frequency response of a transfer function, but directly in the complex plane. The Nyquist plot of a transfer function $L(s)$ is the parametric curve of $L(j\omega)$ for $\omega \in (-\infty, \infty)$. As the frequency ω varies over the entire imaginary axis, the complex number $L(j\omega)$ traces a curve in the complex plane.

Relation to Bode Plot. The Nyquist and Bode plots encode the same frequency response information in different forms. The Bode plot shows

$$20 \log_{10} |L(j\omega)| \text{ and } \angle L(j\omega)$$

as functions of $\log_{10} \omega$ for $\omega \in (0, \infty)$, whereas the Nyquist plot represents

$$L(j\omega) = |L(j\omega)| e^{j\angle L(j\omega)}$$

directly as a curve in the complex plane.

Key distinction. The Bode plot describes the frequency response only for positive frequencies, whereas the Nyquist plot captures the response over the *entire imaginary axis*. As a result, the Nyquist plot contains complete information about the frequency response in a single curve.

Fact 8.2.14. If $L(s)$ has real coefficients, then $L(-j\omega) = \overline{L(j\omega)}$. Thus, the Nyquist plot is symmetric about the real axis. In practice, it is sufficient to compute the curve for $\omega \geq 0$ and reflect it.

Asymptotic Behavior

Consider the zero-pole-gain form

$$L(s) = \frac{K(s - z_1) \cdots (s - z_m)}{(s - p_1) \cdots (s - p_n)}.$$

Low-frequency behavior. As $\omega \to 0$,

$$L(j\omega) \to L(0) = \frac{K(-z_1) \cdots (-z_m)}{(-p_1) \cdots (-p_n)},$$

which determines the starting point of the Nyquist plot.

High-frequency behavior ($\omega \to \infty$) The behavior is determined by the relative degree $n - m$.

- If $n > m$, then $L(j\omega) \to 0$.
- If $n = m$, then $L(j\omega) \to K$.
- If $n < m$, then $|L(j\omega)| \to \infty$.

Procedure for Sketching

1. Compute $L(0)$ (starting point).
2. Determine the high-frequency limit.
3. Identify poles and zeros.
4. Determine how the phase evolves with frequency.
5. Sketch the curve for $\omega \geq 0$.
6. Reflect about the real axis to obtain the full plot.

Gain and Phase Margin from Nyquist Plot

The Nyquist plot provides a direct geometric interpretation of gain and phase margins.

Key observation. The critical point is $-1+0j$. Gain and phase margins quantify how close the Nyquist plot comes to this point.

Gain Margin. The gain margin is determined at the frequency where the Nyquist plot intersects the negative real axis, i.e., where

$$\angle L(j\omega) = -180^\circ.$$

Let this frequency be ω_{pc} (phase crossover frequency). At this point, the Nyquist plot lies on the real axis at $L(j\omega_{pc}) \in \mathbb{R}_{<0}$. The gain margin is given by

$$\mathrm{GM} = \frac{1}{|L(j\omega_{pc})|}.$$

At $\omega = \omega_{pc}$, the Nyquist plot lies on the negative real axis. The gain margin is the factor by which the plot must be radially scaled so that it passes through the point -1.

Phase Margin. The phase margin is determined at the frequency where the Nyquist plot intersects the unit circle, i.e., where

$$|L(j\omega)| = 1.$$

Let this frequency be ω_{gc} (gain crossover frequency). At this point,

$$L(j\omega_{gc}) = e^{j\angle L(j\omega_{gc})}.$$

The phase margin is given by

$$\mathrm{PM} = 180^\circ + \angle L(j\omega_{gc}).$$

At $\omega = \omega_{gc}$, the Nyquist plot lies on the unit circle. The phase margin is the angular separation between the point $L(j\omega_{gc})$ and the negative real axis.

Fact 8.2.15. Nyquist Stability Criterion. Let P be the number of unstable poles of $G(s)$ and Z be the number of unstable poles of $S(s)$. Let N denote the encirclements of the complex number $-1+0\jmath$ by the Nyquist plot of $G(s)$. Note that a clockwise (CW) encirclement is counted as positive, whereas a counterclockwise (CCW) encirclement is counted as negative. Then,

$$Z = N + P. \tag{8.88}$$

Example 8.2.7. Figure 8.3 shows the Bode plot of

$$G(s) = \frac{15}{(s+1)(s+2)(s+3)}. \tag{8.89}$$

Note that $N = 0$ and $P = 0$, and thus $Z = 0$. Furthermore, the closed-loop poles in the basic servo loop are at $-4.6, -0.7 \pm 2.02\jmath$.

Example 8.2.8. Figure 8.4 shows the Bode plot of

$$G(s) = \frac{15}{(s-1)(s+3)(s+4)}. \tag{8.90}$$

Note that $N = -1$ and $P = 1$, and thus $Z = 0$. Furthermore, the closed-loop poles in the basic servo loop are at $-5.14, -0.43 \pm 0.63\jmath$.

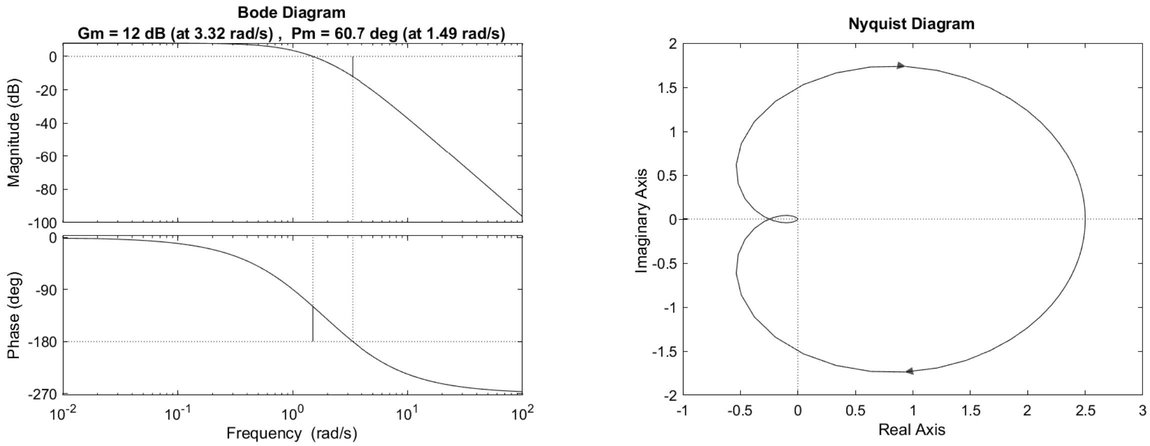

Figure 8.3: Bode and Nyquist plot for Example 8.2.7.

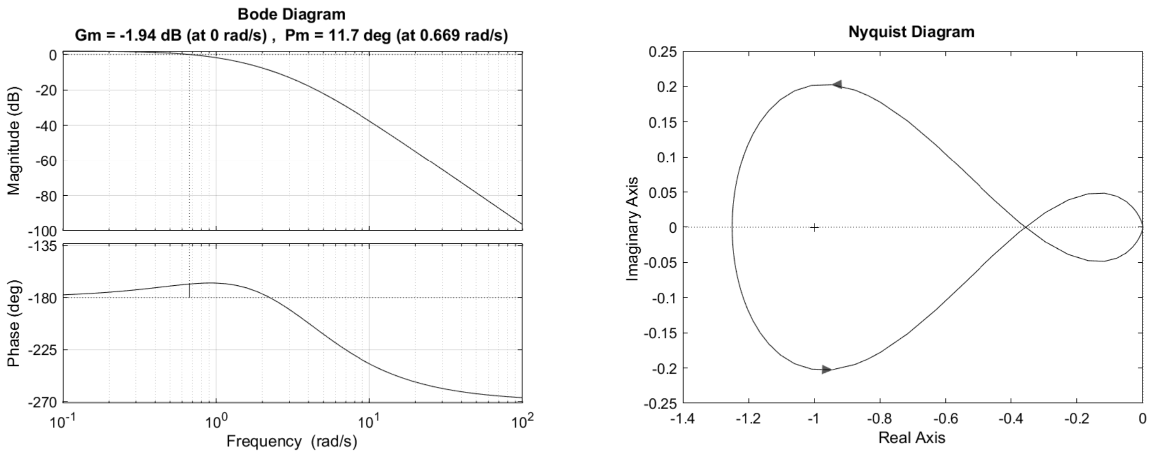

Figure 8.4: Bode and Nyquist plot for Example 8.2.8.

Example 8.2.9. Figure 8.5 shows the Bode plot of

$$G(s) = \frac{(s+1)(s+2)(s+3)}{20}. \tag{8.91}$$

Note that $N = -1$ and $P = 0$, and thus $Z = -1$. Furthermore, the closed-loop poles in the basic servo loop are at $-4.8, -0.58 \pm 2.24\jmath$.

Note that the sign of the gain margin and the phase margin does not provide any information about the stability of the closed-loop system.

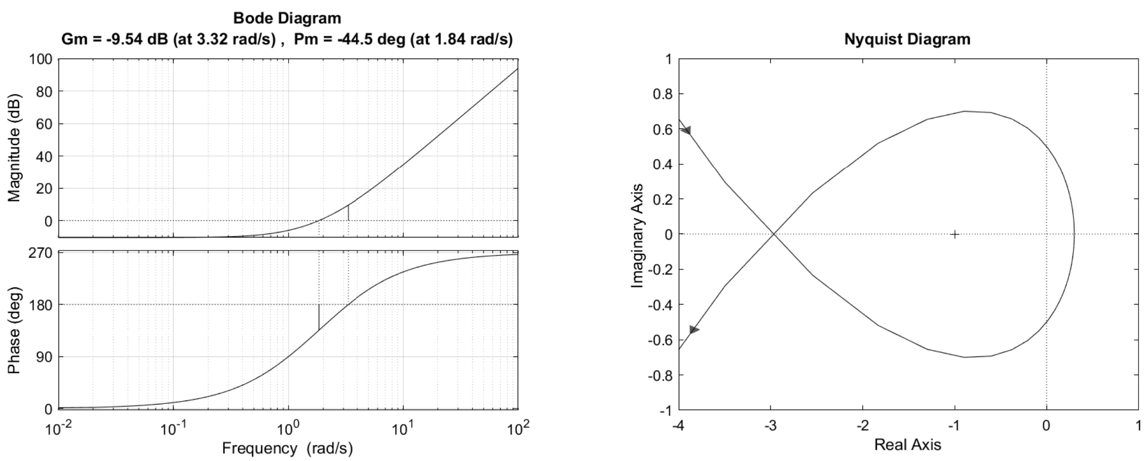

Figure 8.5: Bode and Nyquist plot for Example 8.2.9.

Chapter M

Midterm Review

M.1 Midterm 01 Review

TBD.

M.2 Midterm 02 Review

M.2.1 Introduction

These review notes summarize the key concepts from Chapters 7 and 8. The goal is to understand the relationships between time-domain specifications, pole locations, and frequency-domain representations. A unifying idea throughout is

$$1 + L(s) = 0,$$

which determines the closed-loop poles of a feedback system.

M.2.2 Second-Order System Fundamentals

Consider the standard second-order system

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2},$$

whose poles are given by

$$s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}.$$

Interpretation of Poles

- The real part determines the rate of exponential decay.
- The imaginary part determines oscillation frequency.
- Complex conjugate poles imply oscillatory behavior.

Key idea: The system behavior is completely determined by the location of its poles in the complex plane.

Time-Domain Specifications

For underdamped systems ($0 < \zeta < 1$),

$$T_s \approx \frac{4}{\zeta\omega_n}, \tag{M.1}$$

$$T_p = \frac{\pi}{\omega_d}, \quad \omega_d = \omega_n\sqrt{1-\zeta^2}, \tag{M.2}$$

$$\%OS = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100. \tag{M.3}$$

Tradeoffs

- Increasing ω_n speeds up the response (reduces T_s and T_p).
- Increasing ζ reduces oscillations (reduces $\%OS$).
- Increasing ζ also reduces ω_d, which increases T_p.

These effects are coupled, leading to a tradeoff between speed and oscillation.

M.2.3 Specifications in the Complex Plane

Time-domain specifications can be interpreted geometrically.

- Damping ratio ζ corresponds to lines at constant angle from the origin.
- Settling time corresponds to a vertical boundary in the left-half plane.
- Peak time T_p corresponds to horizontal contours, since $T_p = \pi/\omega_d$ and ω_d is the imaginary part of the pole location.

Thus, design requirements define a region in the complex plane where closed-loop poles must lie.

M.2.4 Root Locus

The root locus describes how closed-loop poles move as gain varies. Closed-loop poles satisfy

$$1 + L(s) = 0 \quad \Longleftrightarrow \quad L(s) = -1.$$

Key Properties

- The root locus begins at open-loop poles.
- It ends at open-loop zeros (or infinity).
- It is symmetric about the real axis.
- Real-axis segments depend on pole-zero configuration.

Interpretation

The root locus provides insight into

- Stability
- Speed of response
- Oscillatory behavior

Changing gain moves poles continuously along these trajectories.

M.2.5 Frequency Response and Bode Plots

The frequency response is obtained by evaluating the transfer function at

$$s = j\omega.$$

A Bode plot consists of

- Magnitude (in dB)
- Phase (in degrees)

Key Frequencies

$$|L(j\omega_{gc})| = 1 \quad \text{(gain crossover frequency)} \tag{M.4}$$
$$\angle L(j\omega_{pc}) = -180^\circ \quad \text{(phase crossover frequency)} \tag{M.5}$$

Stability Insight

A system is has closed loop poles on the imaginary axis when

- $|L(j\omega)| = 1$,
- $\angle L(j\omega) = -180^\circ$,

which corresponds to

$$L(j\omega) = -1.$$

M.2.6 Stability Margins

Gain Margin

$$GM = \frac{1}{|L(j\omega_{pc})|}.$$

Represents how much the gain can increase before instability.

Phase Margin

$$PM = 180^\circ + \angle L(j\omega_{gc}).$$

Represents how much additional phase lag can be tolerated.

Interpretation

- Larger margins indicate greater robustness.
- Small margins indicate proximity to crossing of the imaginary axis.

M.2.7 Frequency-Domain and Time-Domain Connection

There is a strong relationship between frequency-domain and time-domain behavior:

- Phase margin is related to damping ratio.
- Damping ratio determines overshoot.

Thus, frequency response can be used to estimate time-domain performance.

M.2.8 Nyquist Criterion

The Nyquist plot represents $L(j\omega)$ in the complex plane.

Nyquist Stability Criterion

$$Z = N + P$$

where:

- P = number of unstable open-loop poles
- N = number of encirclements of -1
- Z = number of unstable closed-loop poles

Key Idea

Stability is determined by how the Nyquist plot encircles the point -1.

Important Notes

- The direction of encirclement matters.
- Open-loop unstable poles must be accounted for.
- Changing gain scales the Nyquist plot.

M.2.9 Effect of Gain

Across all methods

- Root locus: poles move in the complex plane.
- Bode plot: magnitude shifts.
- Nyquist plot: the plot scales radially.

Thus, gain affects both stability and performance.

M.2.10 Unifying Perspective

All methods are different ways of analyzing

$$1 + L(s) = 0.$$

- Root locus: analyze solutions in the complex plane.
- Bode plot: analyze magnitude and phase.
- Nyquist plot: analyze encirclements of -1.

M.2.11 Final Remarks

- Closed-loop pole locations determine system behavior.
- Time-domain and frequency-domain analyses are closely related.
- Root locus, Bode plots, and Nyquist plots provide complementary perspectives.

Chapter 10

State Space Control

10.1 Motivation

Classical control design based on transfer functions has been extremely successful for low-order, single-input single-output (SISO) systems. Techniques such as root locus, Bode plots, and Nyquist analysis provide valuable insight into stability and frequency-domain behavior. However, transfer-function-based design has several fundamental limitations.

- **Limited applicability to MIMO systems:** Transfer functions primarily describe input-output behavior. For multi-input multi-output (MIMO) systems, interactions between states are not explicitly captured, making systematic controller design difficult.
- **Lack of internal structure:** Transfer functions do not explicitly represent the internal state of the system. As a result, they provide limited insight into how energy, dynamics, or coupling evolve within the system.
- **Difficulty with higher-order systems:** Designing controllers for higher-order plants using classical methods can become non-intuitive, especially when multiple modes interact.
- **Limited design flexibility:** Classical methods often rely on graphical tuning or trial-and-error. There is no direct way to arbitrarily shape the closed-loop dynamics.
- **No systematic notion of optimality:** While performance metrics such as gain margin or bandwidth can be assessed, there is no unified framework to define and compute an *optimal* controller.

These limitations motivated the development of the **state-space framework** in the 1960s, which represents systems in terms of their internal dynamics:

$$\dot{x} = Ax + Bu, \tag{10.1}$$
$$y = Cx + Du, \tag{10.2}$$

where $x \in \mathbb{R}^n$, $u \in \mathbb{R}^m$, and $y \in \mathbb{R}^p$.

The state-space representation offers several key advantages.

- **Natural handling of MIMO systems:** The matrix formulation directly captures multiple inputs, outputs, and their interactions.
- **Explicit representation of internal dynamics:** The state vector x encodes the system's internal behavior, enabling deeper analysis and control design.
- **Systematic design via eigenvalues:** Stability and transient response are determined by the eigenvalues of A. Through feedback, these eigenvalues can be directly modified.

- **Full-state feedback design:** If the system is controllable, we can arbitrarily assign the closed-loop dynamics using state feedback.
- **Foundation for modern control methods:** State-space provides the basis for optimal control (LQR), estimation (Kalman filtering), robust control, and nonlinear control.

In summary, transfer-function methods analyze input-output behavior, while state-space methods enable direct shaping of the system dynamics. In this section, we begin with **full-state feedback**, which illustrates how feedback can be used to directly modify the system matrix and achieve desired closed-loop behavior.

10.2 Full-State Feedback Control

Consider the linear system (10.1), (10.2). For fullstate feedback, assume that x is available for feedback. Note that this assumption is often unrealistic in practice, but it provides a foundational starting point for control design. State estimation will be introduced later to relax this assumption.

10.2.1 Full-State Feedback Law

Consider the control law

$$u = Kx, \tag{10.3}$$

where $K \in \mathbb{R}^{m\times n}$. Substituting into the system dynamics (10.1) yields the closed-loop dynamics

$$\dot{x} = (A + BK)x. \tag{10.4}$$

The closed-loop dynamics matrix $A+BK$ determines the closed-loop behavior. By choosing the gain matrix K, we can shape the system dynamics.

10.2.2 Controllability

Definition 10.2.1. The system $\dot{x} = Ax + Bu$ is **controllable** if, for any initial state $x(0)$ and any desired final state x_f, there exists an input $u(t)$ defined over a finite time interval $[0, T]$ that transfers the system from $x(0)$ to x_f.

Controllability means that the input $u(t)$ has sufficient authority to drive the system to any state in the state space. Controllability of a system can be verified with an algebraic test.

Definition 10.2.2. The pair (A, B) is **controllable** if

$$\operatorname{rank}\,[B \ \ AB \ \ A^2B \ \ \cdots \ \ A^{n-1}B] = n. \tag{10.5}$$

Fact 10.2.1. If (A, B) is controllable, then the eigenvalues of $A + BK$ can be arbitrarily assigned.

Using full-state feedback, we can assign any desired closed-loop dynamics. This is the central advantage of the state-space approach.

In practice, the gain K is computed using pole placement techniques. The MATLAB routine `place` provides a convenient tool to determine K such that the eigenvalues of $A + BK$ are assigned to desired locations.

In particular, the gain

$$K = \texttt{place}(A, B, p), \tag{10.6}$$

places the closed loop eigenvalues of $A + BK$ at $p \in \mathbb{C}^n$.

10.2.3 Hand Computation of State Feedback Gain

Example 10.2.1. Consider

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \tag{10.7}$$

Note that

$$\mathcal{C} = [B \; AB] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \tag{10.8}$$

Since $\operatorname{rank}(\mathcal{C}) = 2$, the system is controllable. Next, let

$$K = \begin{bmatrix} k_1 & k_2 \end{bmatrix}. \tag{10.9}$$

Then

$$A + BK = \begin{bmatrix} 0 & 1 \\ k_1 & k_2 \end{bmatrix}. \tag{10.10}$$

The characteristic equation is

$$\det(sI - (A + BK)) = s^2 - k_2 s - k_1. \tag{10.11}$$

To place the poles at p_1, p_2, match with

$$(s - p_1)(s - p_2) = s^2 - (p_1 + p_2)s + p_1 p_2, \tag{10.12}$$

which implies that

$$k_2 = p_1 + p_2, \tag{10.13}$$
$$k_1 = -p_1 p_2. \tag{10.14}$$

For example, placing poles at $p_1 = -1$, $p_2 = -1$ gives

$$k_2 = -2, \quad k_1 = -1. \tag{10.15}$$

Example 10.2.2. Consider

$$A = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}. \tag{10.16}$$

Note that

$$\mathcal{C} = [B \; AB] = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}. \tag{10.17}$$

Since $\operatorname{rank}(\mathcal{C}) = 1 < 2$, the system is not controllable. Next, let

$$K = \begin{bmatrix} k_1 & k_2 \end{bmatrix}. \tag{10.18}$$

Then

$$A + BK = \begin{bmatrix} k_1 & 2 + k_2 \\ 0 & 0 \end{bmatrix}. \tag{10.19}$$

The characteristic equation is

$$\det(sI - (A + BK)) = s^2 - k_1 s. \tag{10.20}$$

To place the poles at p_1, p_2, match with

$$(s - p_1)(s - p_2) = s^2 - (p_1 + p_2)s + p_1 p_2, \tag{10.21}$$

which implies that

$$k_1 = p_1 + p_2, \tag{10.22}$$
$$0 = p_1 p_2. \tag{10.23}$$

Thus, one pole must always be at the origin and cannot be moved.

10.2.4 Closed-Loop Transfer Functions

So far, we have focused on shaping the *internal dynamics* of the system using full-state feedback. We now examine how this affects the *input-output behavior*.

Consider the system (10.1)-(10.2). The transfer function from input u to output y is

$$G_{yu}(s) = C(sI - A)^{-1}B + D. \tag{10.24}$$

Consider the control law

$$u = Kx + r, \tag{10.25}$$

where $K \in \mathbb{R}^{m \times n}$ is the full-state feedback gain. Substituting (10.25) into the system dynamics yields the closed-loop system

$$\dot{x} = (A + BK)x + Br, \tag{10.26}$$
$$y = (C + DK)x + Dr. \tag{10.27}$$

Figure 10.1 illustrates the resulting control architecture.

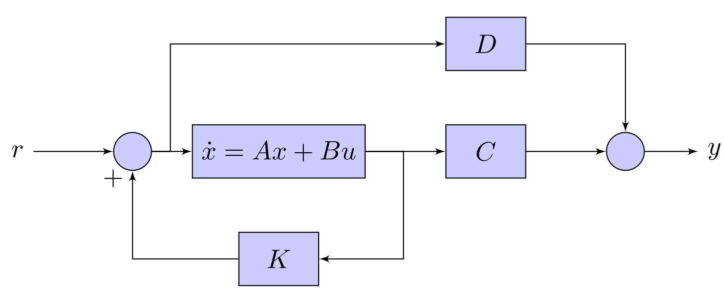

Figure 10.1: Full-state feedback control system.

Note that the transfer function from the reference input r to the output y is

$$G_{yr}(s) = (C + DK)(sI - (A + BK))^{-1}B + D. \tag{10.28}$$

While the feedback gain K determines the poles of the system through $A + BK$, it also influences the input-output behavior through the matrices $(C + DK)$ and B. However, pole placement alone does not guarantee desirable tracking performance, which motivates additional design steps in the following sections.

10.2.5 Integral Control

To ensure that y converges to r, we need to include an integrator in the loop as shown in Figure 10.2.

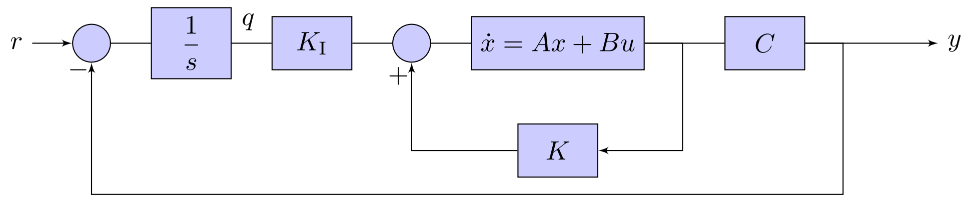

Figure 10.2: Full-state feedback with integral action.

Note that

$$\dot{q} = e = r - y = r - Cx. \tag{10.29}$$

Defining the augmented state

$$x_{\mathrm{a}} = \begin{bmatrix} x \\ q \end{bmatrix}, \tag{10.30}$$

it follows that

$$\dot{x}_{\mathrm{a}} = A_{\mathrm{a}} x_{\mathrm{a}} + B_{\mathrm{a}} u + B_{\mathrm{r}} r, \tag{10.31}$$
$$y = C_{\mathrm{a}} x_{\mathrm{a}}. \tag{10.32}$$

where

$$A_{\mathrm{a}} \triangleq \begin{bmatrix} A & 0 \\ -C & 0 \end{bmatrix}, \quad B_{\mathrm{a}} \triangleq \begin{bmatrix} B \\ 0 \end{bmatrix}, \quad B_{\mathrm{r}} \triangleq \begin{bmatrix} 0 \\ I \end{bmatrix}, \quad C_{\mathrm{a}} \triangleq \begin{bmatrix} C & 0 \end{bmatrix}. \tag{10.33}$$

Consider the control law

$$u = K_{\mathrm{a}} x_{\mathrm{a}}. \tag{10.34}$$

The closed-loop system is thus

$$\dot{x}_{\mathrm{a}} = (A_{\mathrm{a}} + B_{\mathrm{a}} K_{\mathrm{a}}) x_{\mathrm{a}} + B_{\mathrm{r}} r, \tag{10.35}$$
$$y = C_{\mathrm{a}} x_{\mathrm{a}}. \tag{10.36}$$

Assuming $(A_{\mathrm{a}}, B_{\mathrm{a}})$ is controllable, the state feedback gain matrix K_{a} can be chosen such that $(A_{\mathrm{a}} + B_{\mathrm{a}} K_{\mathrm{a}})$ is asymptotically stable. Assuming that the closed-loop is asymptotically stable, it follows that if r is a constant, x_{a} converges. This implies that q converges, and thus $e \to 0$, which implies that $y \to r$.

Remark 10.2.1. $(A_{\mathrm{a}}, B_{\mathrm{a}})$ is not necessarily controllable. Note that

$$A_{\mathrm{a}}^i B_{\mathrm{a}} = \begin{bmatrix} A^i B \\ -CA^{i-1} B \end{bmatrix}, \tag{10.37}$$

and thus

$$\begin{aligned} \mathcal{C}(A_{\mathrm{a}}, B_{\mathrm{a}}) &= \begin{bmatrix} B_{\mathrm{a}} & A_{\mathrm{a}} B_{\mathrm{a}} & \dots & A_{\mathrm{a}}^{n-1} B_{\mathrm{a}} \end{bmatrix} \\ &= \begin{bmatrix} B & AB & A^2 B & \dots & A^{n-1} B \\ 0 & -CB & -CAB & \dots & -CA^{n-2} B \end{bmatrix}. \end{aligned} \tag{10.38}$$

Assume that C is full rank, then $\mathcal{O}(A, C)$ is full rank. In this case, note that

```
A = [0 1; 0 0];
B = [0;1];
C = eye(2);
obsv(A,C)
Aa = [A zeros(2,2); -C zeros(2,2)];
Ba = [B; zeros(2,1)];
rank(ctrb(Aa,Ba))
```

yields rank 3, which implies that $(A_\mathrm{a}, B_\mathrm{a})$ is not controllable.

10.3 Observer

Although a full-state feedback controller guarantees stabilization, it requires the entire state to implement the controller. The entire state is usually not available to implement the FSF controller. However, under certain conditions, it is possible to estimate the state using only the input and output measurements. Consider the system

$$\dot{x} = Ax + Bu, \tag{10.39}$$
$$y = Cx. \tag{10.40}$$

The objective is to estimate x with the measurements of u and y.

Consider the dynamic system

$$\dot{\hat{x}} = A\hat{x} + Bu + F(y - C\hat{x}). \tag{10.41}$$

Defining the state error $e = x - \hat{x}$, it follows that

$$\begin{aligned} \dot{e} &= \dot{x} - \dot{\hat{x}} \\ &= Ax + Bu - A\hat{x} - Bu - F(y - C\hat{x}) \\ &= (A - FC)e. \end{aligned} \tag{10.42}$$

By appropriately choosing the gain F, the eigenvalues of A may be arbitrarily assigned.

Fact 10.3.1. Eigenvalues of A and A^T are the same.

This fact implies that eigenvalues of $A - FC$ and $A^\mathrm{T} - C^\mathrm{T}F^\mathrm{T}$ are the same.

Fact 10.3.2. All eigenvalues of $A^\mathrm{T} - C^\mathrm{T}F^\mathrm{T}$ can be arbitrarily assigned if $(A^\mathrm{T}, C^\mathrm{T})$ is controllable.

Fact 10.3.3. The pair (A, C) is **observable** if and only if

$$\operatorname{rank} \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix} = n. \tag{10.43}$$

Fact 10.3.4. All eigenvalues of $A - FC$ can be arbitrarily assigned if and only if (A, C) is observable.

Fact 10.3.5. If (A, C) is observable, then $(A^\mathrm{T}, C^\mathrm{T})$ is controllable.

10.4 Output Feedback Stabilization

Consider the controller

$$u = K\hat{x}, \tag{10.44}$$

where the observed state $\hat{x}$ is given by (10.41). Note that the controller can be written as

$$\begin{aligned}\dot{\hat{x}} &= Ax + Bu + F(y - C\hat{x}) \\ &= Ax + BK\hat{x} + Fy - FC\hat{x} \\ &= (A + BK - FC)\hat{x} + Fy, &(10.45)\\ u &= K\hat{x}. &(10.46)\end{aligned}$$

and thus

$$G_{\rm c}(s) = K(sI - (A + BK - FC))^{-1}F. \tag{10.47}$$

10.4.1 Closed-loop Transfer Function

Define

$$x_{\rm a} = \begin{bmatrix} x \\ \hat{x} \end{bmatrix}. \tag{10.48}$$

Note that

$$\begin{aligned}\dot{x}_{\rm a} &= \begin{bmatrix} Ax + Bu \\ (A + BK - FC)\hat{x} + Fy \end{bmatrix} \\ &= \begin{bmatrix} Ax + BK\hat{x} \\ (A + BK - FC)\hat{x} + FCx \end{bmatrix} \\ &= \begin{bmatrix} A & BK \\ A + BK - FC & FC \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix}. &(10.49)\end{aligned}$$

Therefore, the stability of $x = 0$ is determined by the eigenvalues of $A_{\rm a}$, where

$$A_{\rm a} \triangleq \begin{bmatrix} A & BK \\ A + BK - FC & FC \end{bmatrix}. \tag{10.50}$$

Fact 10.4.1. The eigenvalues of $A_{\rm a}$ consist of the eigenvalues of $A + BK$ and the eigenvalues of $A - FC$, that is,

$$\mathrm{eig}(A_{\rm a}) = \mathrm{eig}(A + BK) \cup \mathrm{eig}(A - FC). \tag{10.51}$$

$$\begin{aligned}\dot{e} &= \dot{x} - \dot{\hat{x}} \\ &= Ax + Bu - A\hat{x} - Bu - F(y - C\hat{x}) \\ &= (A - FC)e &(10.52)\end{aligned}$$

$$\begin{aligned}\dot{x} &= Ax + Bu \\ &= Ax + BK\hat{x} \\ &= (A + BK)x - BKe &(10.53)\end{aligned}$$

$$\begin{aligned}\begin{bmatrix}\dot{x}\\ \dot{e}\end{bmatrix} &= \begin{bmatrix}(A+BK)x - BKe\\ (A-FC)e\end{bmatrix}\\ &= \begin{bmatrix}A+BK & -BK\\ 0 & A-FC\end{bmatrix}\begin{bmatrix}x\\ e\end{bmatrix}\end{aligned} \tag{10.54}$$

10.5 Dynamic Output Feedback Reference Tracking Controller

Consider the system

$$\dot{x} = Ax + Bu, \tag{10.55}$$
$$y = Cx. \tag{10.56}$$

Consider the controller

$$u = K_x \hat{x} + K_q q = \begin{bmatrix} K_x & K_q \end{bmatrix} \begin{bmatrix} \hat{x} \\ q \end{bmatrix}, \tag{10.57}$$

where

$$\dot{q} = e = r - y = r - Cx, \tag{10.58}$$

and

$$\begin{aligned} \dot{\hat{x}} &= A\hat{x} + Bu + F(y - C\hat{x}) \\ &= A\hat{x} + BK_x\hat{x} + BK_q q + F(y - C\hat{x}) \\ &= (A + BK_x - FC)\hat{x} + BK_q q + Fy \end{aligned} \tag{10.59}$$

The state error dynamics is

$$\dot{e} = (A - FC)e. \tag{10.60}$$

If (A, C) is observable, then eigenvalues of $A - FC$ can be placed anywhere.

Define

$$x_{\mathrm{a}} = \begin{bmatrix} x \\ \hat{x} \end{bmatrix} \tag{10.61}$$

Then,

$$\begin{aligned} \dot{x}_{\mathrm{a}} &= \begin{bmatrix} \dot{x} \\ \dot{\hat{x}} \end{bmatrix} \\ &= \begin{bmatrix} Ax + BK_x\hat{x} + BK_q q \\ (A - FC)\hat{x} + BK_x\hat{x} + BK_q q + FCx \end{bmatrix} \\ &= \begin{bmatrix} A & BK_x \\ FC & A + BK_x - FC \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix} + \begin{bmatrix} I \\ I \end{bmatrix} BK_q q \end{aligned} \tag{10.62}$$

Define

$$x_{\mathrm{b}} = \begin{bmatrix} x_{\mathrm{a}} \\ q \end{bmatrix} \tag{10.63}$$

Then,

$$\begin{aligned}
\dot{x}_{\rm b} &= \begin{bmatrix} \dot{x}_{\rm a} \\ \dot{q} \end{bmatrix} \\
&= \begin{bmatrix} \begin{bmatrix} A & BK_x \\ FC & A+BK_x-FC \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix} + \begin{bmatrix} I \\ I \end{bmatrix} BK_q q \\ r - Cx \end{bmatrix} \\
&= \begin{bmatrix} A & BK_x & BK_q \\ FC & A+BK_x-FC & BK_q \\ -C & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \\ q \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ I \end{bmatrix} r \\
&= \begin{bmatrix} A & 0 & 0 \\ FC & A-FC & 0 \\ -C & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \\ q \end{bmatrix} + \begin{bmatrix} B \\ B \\ 0 \end{bmatrix} u + \begin{bmatrix} 0 \\ 0 \\ I \end{bmatrix} r
\end{aligned} \tag{10.64}$$

Fact 10.5.1. The eigenvalues of $A_{\rm b}$ are

$$\mathrm{eig}(A - FC) \cup \mathrm{eig}(A_a + B_a K_a) \tag{10.65}$$

$$\begin{bmatrix} A & BK_q & BK_x \\ -C & 0 & 0 \\ FC & BK_q & A+BK_x-FC \end{bmatrix} \begin{bmatrix} x \\ q \\ \hat{x} \end{bmatrix} \tag{10.66}$$

$$\begin{aligned}
\dot{x} &= Ax + Bu \\
&= Ax + BK_x\hat{x} + BK_q q \\
&= Ax + BK_x(x - e) + BK_q q \\
&= (A + BK_x)x - BK_x e + BK_q q && (10.67) \\
\dot{e} &= (A - FC)e && (10.68) \\
\dot{q} &= r - Cx && (10.69)
\end{aligned}$$

$$\begin{aligned}
\dot{x}_{\rm b} &= \begin{bmatrix} \dot{x} \\ \dot{q} \\ \dot{e} \end{bmatrix} \\
&= \begin{bmatrix} (A+BK_x)x - BK_x e + BK_q q \\ r - Cx \\ (A - FC)e \end{bmatrix} \\
&= \begin{bmatrix} A+BK_x & BK_q & -BK_x \\ -C & 0 & 0 \\ 0 & 0 & A-FC \end{bmatrix} \begin{bmatrix} x \\ q \\ e \end{bmatrix} + \begin{bmatrix} B \\ B \\ 0 \end{bmatrix} u + \begin{bmatrix} 0 \\ 0 \\ I \end{bmatrix} r
\end{aligned} \tag{10.70}$$

Consider the basic servo loop shown in Figure 10.3. The controller G_c is an output-feedback controller since the output of the system is used to compute the control input u. In this architecture, the controller G_c is designed to simultaneously accomplish stabilization and command following.

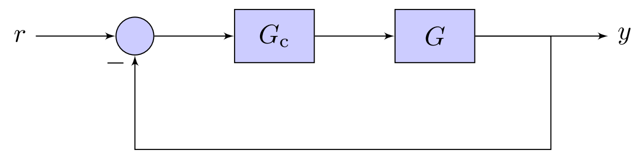

Figure 10.3: Basic servo loop.

A full-state feedback (FSF) controller is used to stabilize the system. Note that the FSF controller only stabilizes the system G, and does not necessarily drive the output to the commanded value. However, FSF can be extended to follow commands by rearranging the system as shown later.

Unlike the transfer-function based approaches, a FSF controller requires the system G to be represented in the state-space form.

Fact 10.5.2. All eigenvalues of $A + BK$ can be arbitrarily assigned if and only if (A, B) is controllable.

Fact 10.5.3. Let $A \in \mathbb{R}^{n\times n}$ and $B \in \mathbb{R}^{n\times m}$. The pair (A, B) is controllable iff

$$\operatorname{rank}\,[B \ AB \ A^2B \ \cdots \ A^{n-1}B] = n. \tag{10.71}$$

If (A, B) is controllable, then the Matlab function `K = place(A,B,p)` can be used to compute the gain K that places the eigenvalues of $A + BK$ at p.

With full-state feedback, the closed-loop transfer function from r to y is

$$G_{ry}(s) = C(sI - A - BK)^{-1}B. \tag{10.72}$$

Assuming $r = 1(t)$, it follows from the FVT that

$$y_\infty = -C(A + BK)^{-1}B. \tag{10.73}$$

The system $G(s)$ in state-space form is

$$\dot{x} = Ax + Bu, \tag{10.74}$$
$$y = Cx. \tag{10.75}$$

With full-state feedback control law $u = Kx+r$, where the gain matrix K is obtained either by the function `place` or `lqr`, the closed-loop system is

$$\dot{x} = (A + BK)x + Br, \tag{10.76}$$
$$y = Cx. \tag{10.77}$$

Figure 10.4 shows the stabilized closed-loop for the control law $u = Kx+r$. The closed-loop transfer function from r to y is

$$T(s) = C(sI - A - BK)^{-1}B. \tag{10.78}$$

For $y_\infty = r$, $T(0) = 1$ However, the control law $u = Kx + r$ may not yield $T(0) = 1$.

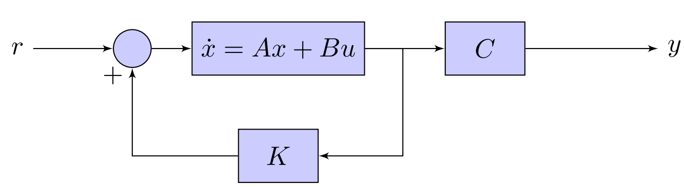

Figure 10.4: Full-state feedback stabilization.

Appendices

Appendix A

Useful Facts

A. Goel

A.1 Leibniz Formula

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{a(t)}^{b(t)} f(t,\tau)\mathrm{d}\tau = f(t,b(t))\frac{\mathrm{d}}{\mathrm{d}t}b(t) - f(t,a(t))\frac{\mathrm{d}}{\mathrm{d}t}a(t) + \int_{a(t)}^{b(t)} \frac{\partial}{\partial t}f(t,\tau)\mathrm{d}\tau. \tag{A.1}$$

A.2 Sum of Sinusoids

Let $g_1(t) = A_1\sin(\omega t + \phi_1)$ and $g_2(t) = A_2\sin(\omega t + \phi_2)$. The sum of the sinusoids $g(t) \triangleq g_1(t) + g_2(t)$ is also a sinusoid, that is, $g(t) = A\sin(\omega t + \phi)$. Using the identity $\sin(A+B) = \sin A\cos B + \cos A\sin B$, it follows that

$$\begin{aligned}
g_1(t) + g_2(t) &= A_1\sin(\omega t)\cos\phi_1 + A_1\cos(\omega t)\sin\phi_1 + A_2\sin(\omega t)\cos\phi_2 + A_2\cos(\omega t)\sin\phi_2 \\
&= (A_1\cos\phi_1 + A_2\cos\phi_2)\sin(\omega t) + (A_1\sin\phi_1 + A_2\sin\phi_2)\cos(\omega t) \\
&= A\cos\phi\sin(\omega t) + A\sin\phi\cos(\omega t) \\
&= A\sin(\omega t + \phi),
\end{aligned}$$

where

$$A\cos\phi = A_1\cos\phi_1 + A_2\cos\phi_2, \tag{A.2}$$

$$A\sin\phi = A_1\sin\phi_1 + A_2\sin\phi_2. \tag{A.3}$$

Simplifying (A.2) and (A.3) yields

$$\begin{aligned}
A &\triangleq \sqrt{(A_1\cos\phi_1 + A_2\cos\phi_2)^2 + (A_1\sin\phi_1 + A_2\sin\phi_2)^2} \\
&= \sqrt{A_1^2 + A_2^2 + 2A_1A_2(\cos\phi_1\cos\phi_2 + \sin\phi_1\sin\phi_2)},
\end{aligned} \tag{A.4}$$

$$\phi \triangleq \arctan\frac{A_1\sin\phi_1 + A_2\sin\phi_2}{A_1\cos\phi_1 + A_2\cos\phi_2}. \tag{A.5}$$

A.3 Integration by Parts

The integration by parts formula is

$$\int_a^b f(t)\frac{\mathrm{d}}{\mathrm{d}t}g(t)\mathrm{d}t = f(b)g(b) - f(a)g(a) - \int_a^b g(t)\frac{\mathrm{d}}{\mathrm{d}t}f(t)\mathrm{d}t. \tag{A.6}$$

Letting $f(t) = e^{-st}$ yields

$$\int_a^b e^{-st}\frac{\mathrm{d}}{\mathrm{d}t}g(t)\mathrm{d}t = e^{-sb}g(b) - e^{-sa}g(a) + s\int_a^b g(t)e^{-st}\mathrm{d}t. \tag{A.7}$$

A.4 Descartes' Rule of Signs

Consider the nth order polynomial

$$p(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots a_0. \tag{A.8}$$

Then, the number of positive roots of $p(x)$ is either equal to the number of sign differences between the consecutive nonzero coefficients of $p(x)$, or is less than it by an even number.

The proof appears in [1].

A.5 Geometric Series

Consider the function

$$S_n = \sum_{i=0}^{n} ar^i. \tag{A.9}$$

Then,

$$S_n = \frac{a(1-r^{n+1})}{1-r}. \tag{A.10}$$

If $|r| < 1$, then

$$S_\infty = \frac{a}{1-r}. \tag{A.11}$$

A.6 Matrix Inversion

Let

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \tag{A.12}$$

where A_{11} and A_{22} are $m \times m$ and $n \times n$ symmetric positive definite matrices, respectively, and A_{12} and A_{21} are matrices of appropriate dimensions. Then,

$$A^{-1} = \begin{bmatrix} (A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1} & -A_{11}^{-1}A_{12}(A_{22} - A_{21}A_{11}^{-1}A_{12})^{-1} \\ -A_{22}^{-1}A_{21}(A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1} & (A_{22} - A_{21}A_{11}^{-1}A_{12})^{-1} \end{bmatrix} \tag{A.13}$$

$$= \begin{bmatrix} (A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1} & -(A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1}A_{12}A_{22}^{-1} \\ -(A_{22} - A_{21}A_{11}^{-1}A_{12})^{-1}A_{21}A_{11}^{-1} & (A_{22} - A_{21}A_{11}^{-1}A_{12})^{-1} \end{bmatrix}. \tag{A.14}$$

Need to check. The second equation may require $A_{12} = A_{21}^{\mathrm{T}}$.

A.7 Taylor Series

Let $f : \mathbb{R} \to \mathbb{R}$. Then,

$$f(x_0+\xi) = f(x_0) + \left.\frac{\partial f}{\partial x}\right|_{x=x_0} \xi + \left.\frac{\partial^2 f}{\partial x^2}\right|_{x=x_0} \xi^2 + \ldots, \tag{A.15}$$

where $x_0, \xi \in \mathbb{R}$.

Let $f : \mathbb{R}^n \to \mathbb{R}$. Then,

$$f(x_0+\xi) = f(x_0) + \left.\frac{\partial f}{\partial x}\right|_{x=x_0} \xi + \xi^{\mathrm{T}} \left.\frac{\partial^2 f}{\partial x^2}\right|_{x=x_0} \xi + \ldots, \tag{A.16}$$

where $x_0, \xi \in \mathbb{R}^n$.

A.8 Euler's Identity

Euler's identity is

$$e^{\jmath x} = \cos x + \jmath \sin x, \tag{A.17}$$

it follows that where $\jmath \triangleq \sqrt{-1}$. Consequently,

$$\cos x = \frac{e^{\jmath x} + e^{-\jmath x}}{2}, \quad \sin x = \frac{e^{\jmath x} - e^{-\jmath x}}{2\jmath}. \tag{A.18}$$

A.9 Roots of Unity

The nth root of unity is

$$\omega_n \triangleq e^{\jmath \frac{2\pi}{n}}. \tag{A.19}$$

Fact A.9.1. $\omega_n^n = 1$.

Proof. $\omega_n^n = e^{\jmath 2\pi} = \cos(2\pi) = 1$. □

Fact A.9.2. For $i \in \mathbb{Z}$, $|\omega_n^i| = 1$.

Proof. $\omega_n^i = e^{\jmath 2\pi i/n} = \cos(2\pi i/n) + \jmath \sin(2\pi i/n)$. Next, note that $|\omega_n^i| = |\cos(2\pi i/n) + \jmath \sin(2\pi i/n)| = 1$. □

Fact A.9.3. For $i \in \mathbb{Z}$, $\omega_n^i = \omega_n^{n-i}$.

A.10 Pade Approximation

A Pade approximant is the *best* approximation of a function near a specific point by a rational function of given order. Given a function f and two integers $m \geq 0$ and $n \geq 1$, the pade approximate of order $[m/n]$ is the rational function

$$R(x) \triangleq \frac{\sum_{j=0}^{m} a_j x^j}{1 + \sum_{k=1}^{n} b_k x^k} \tag{A.20}$$

such that, for $i = 0, \ldots, m+n$,

$$f^{(i)}(0) = R^{(i)}(0). \tag{A.21}$$

Example A.10.1.

$$e^x \approx \frac{1 + \frac{1}{2}x + \frac{1}{9}x^2}{1 - \frac{1}{2}x + \frac{1}{9}x^2}. \tag{A.22}$$

A.11 Convolution Integral

The **convolution integral** is a fundamental operation that arises naturally in the analysis of linear dynamical systems and signal processing. In the context of linear time-invariant (LTI) systems, convolution provides a convenient way to express the response of a system to an arbitrary input in terms of its response to an impulse.

Definition A.11.1. Let $f, g : \mathbb{R} \to \mathbb{R}$ be two functions. The **convolution** of f and g, denoted by $f * g$, is defined as

$$(f * g)(t) \triangleq \int_{-\infty}^{\infty} f(t-\tau)\, g(\tau)\, \mathrm{d}\tau. \tag{A.23}$$

If $g(t) = 0$ for $t < 0$, then

$$(f * g)(t) = \int_{0}^{t} f(t-\tau)\, g(\tau)\, \mathrm{d}\tau. \tag{A.24}$$

This simplification is useful for causal dynamical systems, where $g(t)$ is typically the input signal $u(t)$ which is assumed to be zero for $t < 0$.

Remark A.11.1. Convolution can be interpreted as a weighted accumulation of past input values $g(\tau)$, where the weighting function $f(t-\tau)$ encodes how strongly past inputs influence the present output. Recent inputs typically have a stronger influence than distant past inputs for stable systems.

Remark A.11.2. For the LTI state-space system

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad x(0) = 0, \tag{A.25}$$

the state $x(t)$ is given by

$$x(t) = \int_{0}^{t} e^{A(t-\tau)} Bu(\tau)\, \mathrm{d}\tau. \tag{A.26}$$

This expression has the form of a convolution between the matrix-valued function $e^{At}B$ and the input $u(t)$, and highlights the role of $e^{At}B$ as the **impulse response** of the state dynamics.

Remark A.11.3. Convolution provides a direct connection between time-domain and frequency-domain descriptions of linear systems. In particular, convolution in time corresponds to multiplication in the Laplace or Fourier domain, which underlies many frequency-response and transfer-function methods used in control and signal processing.

Appendix B

ODEs

A. Goel

B.1 Solution of Linear ODE

Consider the linear ordinary differential equation

$$y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_0 y = b_0 u, \tag{B.1}$$

where

$$y^{(n)} \triangleq \frac{\mathrm{d}^n y}{\mathrm{d}t^n}, \tag{B.2}$$

with initial conditions, for $i = 0, 1, 2, \ldots, n-1$,

$$y^{(i)}(0) = y_0^{(i)}. \tag{B.3}$$

The objective is to find a function $y(t)$ that simultaneously satisfies (B.1) and (B.3).

Let $Y(t)$ denote the solution of (B.1). The general solution can be written as the sum of a homogeneous solution and a particular solution,

$$Y(t) = Y_{\mathrm{h}}(t) + Y_{\mathrm{p}}(t), \tag{B.4}$$

where $Y_{\mathrm{h}}(t)$ and $Y_{\mathrm{p}}(t)$ are defined below.

B.1.1 Homogeneous Solution

The homogeneous solution $Y_{\mathrm{h}}(t)$ satisfies (B.1) with zero input, that is, for all $t \geq 0$,

$$Y_{\mathrm{h}}^{(n)} + a_{n-1}Y_{\mathrm{h}}^{(n-1)} + \cdots + a_0 Y_{\mathrm{h}} = 0. \tag{B.5}$$

Assume a solution of the form $y(t) = e^{\lambda t}$. Substituting into (B.5) yields

$$\left(\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_0\right) e^{\lambda t} = 0. \tag{B.6}$$

Since $e^{\lambda t} \neq 0$, the characteristic equation is

$$\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_0 = 0. \tag{B.7}$$

By the Fundamental Theorem of Algebra, (B.7) has n (possibly repeated or complex) roots $\lambda_1, \ldots, \lambda_n$, and hence

$$Y_{\mathrm{h}}(t) = c_1 e^{\lambda_1 t} + \cdots + c_n e^{\lambda_n t}, \tag{B.8}$$

where the constants $c_1, \ldots, c_n$ are determined from the initial conditions (B.3).

B.1.2 Particular Solution

The particular solution $Y_{\mathrm{p}}(t)$ can be obtained using the method of undetermined coefficients. This method applies when the input $u(t)$ is a polynomial, exponential, sinusoid, or finite combinations of these functions. Typical choices for $Y_{\mathrm{p}}(t)$ are summarized in Table B.1.

Input	Particular Solution
Polynomial, $P_n = \sum_{i=0}^{n} \beta_i t^i$	$\sum_{i=0}^{n} d_i t^i$
Exponential, $e^{\alpha t}$	$de^{\alpha t}$
Sinusoid, $\sin(\omega t), \cos(\omega t)$	$d_1 \sin(\omega t) + d_2 \cos(\omega t)$
$P_n e^{\alpha t} \sin(\omega t)$	$P_n e^{\alpha t}(d_1 \sin(\omega t) + d_2 \cos(\omega t))$

Table B.1: Typical Particular Solutions

Example B.1.1. Consider (B.1) with input $u(t) = e^{\alpha t}$. From Table B.1, assume $Y_{\mathrm{p}}(t) = de^{\alpha t}$. Substituting into (B.1) yields

$$d\left(\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_0\right) e^{\alpha t} = e^{\alpha t}. \tag{B.9}$$

Equating coefficients gives

$$d = \frac{1}{\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_0}. \tag{B.10}$$

Example B.1.2. Consider the mass–spring–damper system

$$m\ddot{q} + c\dot{q} + kq = f, \tag{B.11}$$

with initial conditions $q(0) = q_0$ and $\dot{q}(0) = \dot{q}_0$, and input $f(t) = \sin(\omega t)$.

The homogeneous solution is

$$q_{\mathrm{h}}(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}, \tag{B.12}$$

where λ_1 and λ_2 are the roots of

$$m\lambda^2 + c\lambda + k = 0. \tag{B.13}$$

The constants c_1, c_2 are obtained from

$$c_1 + c_2 = q_0, \tag{B.14}$$

$$c_1\lambda_1 + c_2\lambda_2 = \dot{q}_0. \tag{B.15}$$

Writing in matrix form,

$$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ \lambda_1 & \lambda_2 \end{bmatrix}^{-1} \begin{bmatrix} q_0 \\ \dot{q}_0 \end{bmatrix}. \tag{B.16}$$

A particular solution is assumed as

$$q_{\mathrm{p}}(t) = d_1 \sin(\omega t) + d_2 \cos(\omega t). \tag{B.17}$$

Substituting into (B.11) and equating coefficients yields

$$(-m\omega^2 + k)d_1 - c\omega d_2 = 1, \tag{B.18}$$

$$c\omega d_1 + (-m\omega^2 + k)d_2 = 0, \tag{B.19}$$

which can be written compactly as

$$\begin{bmatrix} d_1 \\ d_2 \end{bmatrix} = \begin{bmatrix} -m\omega^2 + k & -c\omega \\ c\omega & -m\omega^2 + k \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix}. \tag{B.20}$$

The complete solution is thus

$$\begin{aligned} q(t) &= q_{\mathrm{h}}(t) + q_{\mathrm{p}}(t) \\ &= \begin{bmatrix} e^{\lambda_1 t} & e^{\lambda_2 t} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ \lambda_1 & \lambda_2 \end{bmatrix}^{-1} \begin{bmatrix} q_0 \\ \dot{q}_0 \end{bmatrix} + \begin{bmatrix} \sin(\omega t) & \cos(\omega t) \end{bmatrix} \begin{bmatrix} -m\omega^2 + k & -c\omega \\ c\omega & -m\omega^2 + k \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix}. \end{aligned} \tag{B.21}$$

Appendix C

Diagonalization

Let $A \in \mathbb{R}^{n\times n}$ (or $\mathbb{C}^{n\times n}$), and let $\lambda \in \mathbb{C}$ be an eigenvalue of A.

Definition C.0.1. The *algebraic multiplicity* of λ, denoted by $\mathrm{AM}(\lambda)$, is its multiplicity as a root of the characteristic polynomial.

Definition C.0.2. The *geometric multiplicity* of λ, denoted by $\mathrm{GM}(\lambda)$, is the dimension of the $\mathcal{N}(\lambda I - A)$.

C.1 Simple, Semisimple, and Nonsimple Matrices

Definition C.1.1. A matrix A is

1. **simple**, if for each eigenvalue λ_i, $\mathrm{AM}(\lambda_i) = 1$.
2. **semisimple**, if for each eigenvalue λ_i, $\mathrm{AM}(\lambda) = \mathrm{GM}(\lambda)$.
3. **nonsimple (defective)** if $\mathrm{AM}(\lambda) > \mathrm{GM}(\lambda)$.

Remark C.1.1. Every simple eigenvalue is semisimple, but the converse need not hold. An eigenvalue may be semisimple with algebraic multiplicity greater than one (for example, $A = \mathrm{diag}(\lambda, \lambda)$). Nonsimple eigenvalues give rise to nontrivial Jordan blocks in the Jordan canonical form of A.

Remark C.1.2. The distinction between semisimple and nonsimple eigenvalues is particularly important for the stability of linear systems. In particular, eigenvalues on the imaginary axis must be semisimple in order for the system $\dot{x} = Ax$ to be Lyapunov stable. If an eigenvalue on the imaginary axis is nonsimple, solutions exhibit polynomial growth in time and the system is unstable.

Example C.1.1 (Examples of Matrices with Simple, Semisimple, and Nonsimple Eigenstructure)**.** Let

$$A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}. \tag{C.1}$$

- The matrix A_1 has two distinct eigenvalues, $\lambda_1 = 1$ and $\lambda_2 = 2$, each with algebraic multiplicity one. Hence, A_1 has only *simple* eigenvalues (and is therefore diagonalizable).
- The matrix A_2 has a repeated eigenvalue $\lambda = 3$ with algebraic multiplicity two. Since $\dim \mathcal{N}(\lambda I - A_2) = 2$, the geometric multiplicity equals the algebraic multiplicity. Thus, A_2 is *diagonalizable* and has a repeated but *semisimple* eigenvalue.
- The matrix A_3 has a repeated eigenvalue $\lambda = 4$ with algebraic multiplicity two, but $\dim \mathcal{N}(\lambda I - A_3) = 1$. Hence, A_3 is *not diagonalizable* and has a *nonsimple (defective)* eigenvalue with a nontrivial Jordan block.

C.2 Diagonalization

Fact C.2.1 (Diagonalizability)**.** Let $A \in \mathbb{C}^{n\times n}$. If all eigenvalues of A are semisimple, then A is diagonalizable. That is, there exists an invertible matrix V such that

$$A = V\Lambda V^{-1}, \tag{C.2}$$

where Λ is diagonal. The matrix V contains the eigenvectors of A and the diagonal matrix Λ contains the corresponding eigenvalues on the diagonal.

Example C.2.1. Reconsider matrix A_2 from Example C.1.1. The eigenvalues of A_2 are $\lambda_1 = \lambda_2 = 3$. Since $3I - A_2 = 0$, the columns of the 2×2 identity matrix I are the eigenvectors. Therefore,

$$A_2 = I \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} I^{-1}. \tag{C.3}$$

Fact C.2.2 (Matrix Exponential of a Semisimple Matrix)**.** Consider a semisimple matrix A. Then,

$$e^{A}t = Ve^{\Lambda t}V^{-1}, \tag{C.4}$$

and

$$e^{\Lambda t} = \begin{bmatrix} e^{\lambda_1 t} & 0 & \cdots & 0 \\ 0 & e^{\lambda_2 t} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & e^{\lambda_n t} \end{bmatrix} \in \mathbb{C}^{n\times n}. \tag{C.5}$$

For diagonalizable matrices, the matrix exponential reduces to exponentiating each eigenvalue independently. In contrast, for nonsimple (defective) matrices, the matrix exponential contains additional polynomial terms arising from the Jordan blocks.

C.3 Jordan Decomposition

Fact C.3.1 (Jordan Decomposition)**.** Let $A \in \mathbb{C}^{n\times n}$. If A has at least one nonsimple (defective) eigenvalue, then A is not diagonalizable. In this case, A can be decomposed as

$$A = TJT^{-1}, \tag{C.6}$$

where J is in Jordan canonical form and contains at least one nontrivial Jordan block and T contains the eigenvectors and the generalized eigenvectors of A. This decomposition is called the Jordan decomposition of A.

Let A have $\ell < n$ distinct eigenvalues and their geometric multiplicity is μ_i. Then,

$$J = \text{blockdiagonal}(J_1, J_2, \ldots, J_\ell), \tag{C.7}$$

where, for $i \in 1, 2, \ldots, \ell$,

$$J_i \triangleq \begin{bmatrix} \lambda_i & 1 & 0 & \cdots & 0 \\ 0 & \lambda_i & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & \lambda_i & 1 \\ 0 & \cdots & 0 & 0 & \lambda_i \end{bmatrix} \in \mathbb{C}^{\mu_i\times\mu_i}, \tag{C.8}$$

Fact C.3.2 (Matrix Exponential of a nonsimple Matrix). Consider a nonsimple matrix A. Then,

$$e^{At} = Te^{Jt}T^{-1}, \tag{C.9}$$

where

$$e^{Jt} = \text{blockdiagonal}(e^{J_1 t}, e^{J_2 t}, \ldots, e^{J_\ell t}), \tag{C.10}$$

where, for $i \in 1, 2, \ldots, \ell$,

$$e^{J_i t} \triangleq \begin{bmatrix} e^{\lambda_i t} & te^{\lambda_i t} & t^2 e^{\lambda_i t}/2 & \cdots & t^{(\mu-1)!} e^{\lambda_i t}/(\mu-1)! \\ 0 & e^{\lambda_i t} & te^{\lambda_i t} & \cdots & t^{(\mu-2)!} e^{\lambda_i t}/(\mu-2)! \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & e^{\lambda_i t} & te^{\lambda_i t} \\ 0 & \cdots & 0 & 0 & e^{\lambda_i t} \end{bmatrix} \in \mathbb{R}^{\mu_i \times \mu_i}. \tag{C.11}$$

Example C.3.1 (A 2×2 Nonsimple Matrix and Its Exponential). Consider the defective (nonsimple) matrix

$$A = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}. \tag{C.12}$$

The eigenvalue λ has algebraic multiplicity 2 and geometric multiplicity 1, so A is not diagonalizable. Furthermore, $J = A$, and thus

$$e^{At} = \begin{bmatrix} e^{\lambda t} & te^{\lambda t} \\ 0 & e^{\lambda t} \end{bmatrix}. \tag{C.13}$$

Appendix D

Discretization

D.1 Discretization of Linear Systems under Zero-Order Hold

Consider the continuous-time linear system

$$\dot{x}(t) = Ax(t) + Bu(t), \tag{D.1}$$

where $x(t) \in \mathbb{R}^n$ and $u(t) \in \mathbb{R}^m$. The solution of the linear system is

$$x(t) = e^{A(t-t_0)}x(t_0) + \int_{t_0}^{t} e^{A(t-\tau)}Bu(\tau)\, d\tau. \tag{D.2}$$

Assume that the input is held constant over each sampling interval, that is, for $t \in [kT_s, (k+1)T_s)$,

$$u(t) = u_k, \tag{D.3}$$

where T_s is the sampling period. Setting $t_0 = kT_s$ and $t = (k+1)T_s$, and using the zero-order hold assumption $u(\tau) = u_k$ yields

$$x_{k+1} = e^{AT_s}x_k + \int_0^{T_s} e^{A\tau}B\, d\tau\, u_k. \tag{D.4}$$

Therefore, the discrete-time system is

$$x_{k+1} = A_{\mathrm{d}}x_k + B_{\mathrm{d}}u_k, \tag{D.5}$$

where

$$A_{\mathrm{d}} \triangleq e^{AT_s}, \qquad B_{\mathrm{d}} \triangleq \int_0^{T_s} e^{A\tau}B\, d\tau. \tag{D.6}$$

The matrix A_{d} is the state transition matrix over one sampling interval, while B_{d} accounts for the effect of the constant input applied during the interval.

D.2 Computation of $(A_{\mathrm{d}}, B_{\mathrm{d}})$ using the Block Matrix Exponential

Direct evaluation of the matrix integral

$$B_{\mathrm{d}} = \int_0^{T_s} e^{A\tau}B\, d\tau \tag{D.7}$$

can be inconvenient. A convenient alternative uses a block matrix exponential.

Define the augmented matrix

$$M \triangleq \begin{bmatrix} A & B \\ 0 & 0 \end{bmatrix}. \tag{D.8}$$

Then, the matrix exponential, using the Taylor series is

$$e^{MT} = \sum_{k=0}^{\infty} \frac{(MT)^k}{k!}. \tag{D.9}$$

Note that, for $k \geq 1$,

$$M^k = \begin{bmatrix} A^k & A^{k-1}B \\ 0 & 0 \end{bmatrix}. \tag{D.10}$$

which implies

$$e^{MT} = \begin{bmatrix} \displaystyle\sum_{k=0}^{\infty} \frac{A^k T^k}{k!} & \displaystyle\sum_{k=1}^{\infty} \frac{A^{k-1} B\, T^k}{k!} \\ 0 & I \end{bmatrix}. \tag{D.11}$$

Next, note that the upper-left block satisfies

$$\sum_{k=0}^{\infty} \frac{A^k T^k}{k!} = e^{AT}, \tag{D.12}$$

and the upper-right block can be rewritten as

$$\sum_{k=1}^{\infty} \frac{A^{k-1} B\, T^k}{k!} = \sum_{j=0}^{\infty} \frac{A^j B\, T^{j+1}}{(j+1)!}. \tag{D.13}$$

Furthermore, it follows from

$$e^{A\tau} = \sum_{j=0}^{\infty} \frac{A^j \tau^j}{j!}, \tag{D.14}$$

that

$$\int_0^T e^{A\tau} B\, d\tau = \sum_{j=0}^{\infty} \frac{A^j B}{j!} \int_0^T \tau^j\, d\tau = \sum_{j=0}^{\infty} \frac{A^j B\, T^{j+1}}{(j+1)!}. \tag{D.15}$$

Finally, substituting (D.12) and (D.15) in (D.11) yields

$$e^{MT} = e^{\begin{bmatrix} A & B \\ 0 & 0 \end{bmatrix} T_s} = \begin{bmatrix} e^{AT} & \int_0^T e^{A\tau} B\, d\tau \\ 0 & I \end{bmatrix} = \begin{bmatrix} A_{\mathrm{d}} & B_{\mathrm{d}} \\ 0 & I \end{bmatrix}. \tag{D.16}$$

This result is known as the **Van Loan discretization formula** and is commonly used to compute the exact zero-order-hold discretization of linear systems.

Bibliography

[1] X. Wang, "A simple proof of descartes's rule of signs," *The American Mathematical Monthly*, vol. 111, no. 6, pp. 525–526, 2004, ISSN: 00029890, 19300972. Accessed: Jun. 3, 2023. [Online]. Available: http://www.jstor.org/stable/4145072.